23. (a) $Q = \frac{X_C}{R} = \frac{1}{\omega_0 CR}$

$= \frac{1}{RC} \times \sqrt{LC} = \frac{1}{R}\sqrt{\frac{L}{C}}$

For better tuning, Q-factor must be high.

$\therefore Q = \frac{\omega_0 L}{R} = \frac{1}{\sqrt{LC}}\left(\frac{L}{R}\right) = \frac{1}{R}\sqrt{\frac{L}{C}}$

R and C should be small and L should be high.

24. (c)

$\tan\phi = \frac{\pi}{6}$

When C is removed,

$\tan\frac{\pi}{3} = \frac{X_C}{R} \Rightarrow X_C = R\sqrt{3}$

When L is removed,

$\tan\frac{\pi}{3} = \frac{X_L}{R} \Rightarrow X_L = R\sqrt{3}$

$\tan\phi = \frac{X_L - X_C}{R}$

$\tan\phi = \frac{\sqrt{3}R - \sqrt{3}R}{R} = 0$

$\cos\phi = 1 \qquad [\because \phi = 0]$

25. (c) Here, $V_1 = V_L$, $V_2 = V_C$

$\therefore V_L = V_C = 300$ V

Circuit is in resonance

$\therefore V_R = V_3 = V = 220$ V

$I = \frac{V}{Z} = \frac{220}{100} = 2.2\text{A} \quad (Z = R)$

26. (d) In series LCR, current is maximum at resonance.

$\therefore$ Resonant frequency $\omega = \frac{1}{\sqrt{LC}}$

$\therefore \omega^2 = \frac{1}{LC}$ or, $L = \frac{1}{\omega^2 C}$

Given $\omega = 1000\ s^{-1}$ and $C = 10\ \mu F$

$\therefore L = \frac{1}{1000 \times 1000 \times 10 \times 10^{-6}} = 0.1\ H = 100\ mH$

27. (a) Quality factor, $Q = \frac{\omega L}{R}$

Since, $\omega^2 = \frac{1}{LC}$

$\therefore$ Quality factor, $Q = \frac{1}{\omega^2 RC}$

28. (b) In the resonant condition, the impedance will be minimum and the current be maximum. This is only possible when $X_L = X_C$.

Therefore $\tan\theta = \frac{X_L - X_C}{R} = 0$ or $\theta = 0$.

29. (d) Total energy remains constant.

$\frac{CV^2}{2} + \frac{1}{2}Li^2 = \text{constant}$

Initially, $i = 0$, $V = V_1$

Total energy $= \frac{1}{2}CV_1^2$

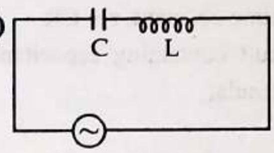
When $V \to V_2$

$\frac{1}{2}C(V_2)^2 + \frac{1}{2}Li^2 = \frac{1}{2}CV_1^2$

$Li^2 = C\left(V_1^2 - V_2^2\right)$

$i^2 = \frac{C}{L}\left(V_1^2 - V_2^2\right) \Rightarrow i = \sqrt{\frac{C}{L}\left(V_1^2 - V_2^2\right)}$

30. (c)



Resonant frequency, $f = \frac{1}{2\pi\sqrt{LC}}$

Now, $L_1 = 2L$, $C_1 = 4C$

New resonant frequency

$f' = \frac{1}{2\pi\sqrt{L_1 C_1}} = \frac{1}{2\pi\sqrt{2L \times 4C}}$

$= \frac{1}{2\sqrt{2}}\left[\frac{1}{2\pi\sqrt{LC}}\right]\left[\because f = \frac{1}{2\pi\sqrt{LC}}\right] = \frac{f}{2\sqrt{2}}$

31. (b) Resonance angular frequency of LCR circuit,

$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5 \times 80 \times 10^{-6}}}$

$\omega_0 = \frac{1}{0.02}$

$\omega_0 = 50$ rad/s

Now, $\Delta\omega = \frac{R}{L} = \frac{40}{5} = 8$ rad / s

Required half power frequency is given as:

$\omega_L = \omega_0 - \frac{\Delta\omega}{2} = 50 - \frac{8}{2} = 46$ rad/s

$\omega_H = \omega_0 + \frac{\Delta\omega}{2} = 50 + \frac{8}{2} = 54$ rad/s

32. (c) Here, $R = 50\Omega$

$X_L = \omega L = 314 \times 20 \times 10^{-3} = 6.28\ \Omega$

$X_C = \frac{1}{\omega C} = \frac{1}{314 \times 100 \times 10^{-6}} = 31.84\ \Omega$

$$Z=\sqrt{R^2+(X_L-X_C)^2}$$

$$Z=\sqrt{(50)^2+(25.56)^2}=56.15\,\Omega$$

$$P=V_rI_r\cos\phi=\frac{V_r^2}{Z^2}\times R=\frac{\left(\frac{10}{\sqrt{2}}\right)^2\times 50}{(56.15)^2}=0.79\text{ W}$$

33. (a) $\tan\phi=\dfrac{X_L-X_C}{R}$

$$\tan\phi=\frac{V_L-V_C}{V_R}=\frac{100-40}{80}\text{ or }\phi=37^\circ$$

Power factor $=\cos\phi=\cos 37^\circ=\dfrac{4}{5}$ or 0.8

34. (a) $X_C=\dfrac{1}{\omega C}=\dfrac{1}{340\times 50\times 10^{-6}}=58.8\,\Omega$

$X_L=\omega L=340\times 20\times 10^{-3}=6.8\,\Omega$

$$Z=\sqrt{R^2+(X_C-X_L)^2}$$

$$=\sqrt{40^2+(58.8-6.8)^2}=\sqrt{4304}\,\Omega$$

$$P=i_{rms}^2R=\left\{\frac{V_{rms}}{Z}\right\}^2R$$

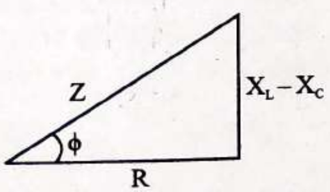
$$=\left\{\frac{10/\sqrt{2}}{\sqrt{4304}}\right\}^2\times 40=\frac{50\times 40}{4304}=0.47\text{ W}$$

So best answer will be (a)

35. (d) $P_{avg}=V_{rms}I_{rms}\cos\phi$

$$=\frac{1}{2}\times\frac{1}{2}\times\cos 60^\circ=\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}=\frac{1}{8}\text{ watt}$$

36. (d)



As we know, in series LCR circuit

$P=VI\cos\phi$

$$=\frac{V^2}{Z}\cos\phi$$

$$=\frac{V^2}{Z}\times\frac{R}{Z}$$

$$P=\frac{V^2R}{\left[R^2+\left(L\omega-\frac{1}{C\omega}\right)^2\right]}\quad\text{Here, }V=\varepsilon$$

37. (d) $e=E_0\sin\omega t$

$i=I_0\sin(\omega t-\phi)$

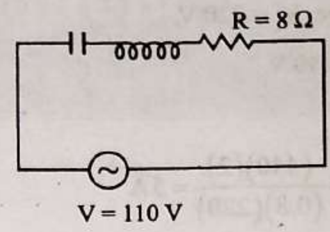
The average power in the circuit,

$P_{avg}=V_{rms}.I_{rms}\cos\phi$

$$=\frac{E_o}{\sqrt{2}}\times\frac{I_o}{\sqrt{2}}\times\cos\phi=\frac{E_oI_o}{2}\cos\phi$$

where ϕ is phase difference between current and voltage.

38. (c)



Here, $X_C=25\,\Omega$, $X_L=31\,\Omega$, $R=8\,\Omega$

$$\text{Power factor}=\frac{R}{Z}=\frac{R}{\sqrt{R\ +(X_L-X_C)}}$$

$$=\frac{8}{\sqrt{64+(31-25)^2}}=\frac{8}{\sqrt{64+36}}=\frac{8}{10}=0.8$$

39. (c) In A.C circuit power loss $P=VI\cos\phi$

$P=VI=I^2R$ ($\because\phi=0$ at resonance)

40. (b) The dissipation of power in an a.c. circuit is

$(P)=V\times I\times\cos\theta$.

Therefore current flowing in the circuit depends upon the phase voltage (V) and current (I) of the a.c. circuit.

41. (b) Current (I) = 5 sin (100t – π/2) and voltage (V) = 200 sin (100 t). Comparing the given equation, with the standard equation, we find that between current and voltage is

$$\phi=\frac{\pi}{2}=90^\circ.$$

Power consumption, $P=I_{rms}V_{rms}\cos\phi$

$$=I_{rms}V_{rms}\cos 90^\circ=0$$

42. (a) As the major portion of the A.C. flows on the surface of the wire. So where a thick wire is required, a number of thin wires are joined together to give an equivalent effect of a thick wire.

Therefore multiple strands are suitable for transporting A.C. Similarly multiple strands can also be used for D.C.

43. (d) For an ideal transformer,

Input power = Output Power

$$\Rightarrow V_pI_p=V_sI_s\Rightarrow I_p=\frac{V_sI_s}{V_p}$$

$$\Rightarrow I_p=\frac{44}{220}\text{ [Given that output power, }V_sI_s=44\text{ W]}$$

$$\Rightarrow I_p=\frac{1}{5}$$

$$\Rightarrow I_p=0.2\text{ A}$$

44. (b) $\eta = \frac{V_S I_S}{V_P I_P} \Rightarrow 0.9 = \frac{V_S(6)}{3\times10^3} \Rightarrow V_S = 450\text{ V}$

As $V_P I_P = 3000$ so, $I_P = \frac{3000}{200}\text{ A} = 15\text{ A}$

45. (a) Given, primary voltage, $V_o = 220$ V,

Secondary voltage, $V_P = 440$ V,

Efficiency, $\eta = 0.8$

$\therefore\ \eta = \frac{V_S I_S}{V_P I_P} = 0.8 \Rightarrow I_P = \frac{(440)(2)}{(0.8)(220)} = 5\text{A}$

46. (b) Efficiency of transformer

$(\eta) = \frac{V_S I_S}{V_P I_P} \times 100$

$(V_S I_S)$ = Power in secondary coil = 100 W

$\eta = \frac{100}{220\times0.5}\times100 = 90.9 \approx 90\%$

47. (a) $N_A = 50$ = Turns in primary coil

$N_B = 1500$ = Turns in secondary coil

Since, magnetic flux linked with the primary coil is ϕ.

Voltage across primary coil $= \frac{d\phi}{dt} = \frac{d(\phi_0 + 4t)}{dt} = 4$ V

As we know that

$\frac{V_A}{V_B} = \frac{N_A}{N_B} \Rightarrow \frac{4}{V_B} = \frac{50}{1500}$

$\Rightarrow V_B = 120$ volts

$\Rightarrow$ Output voltage across secondary coil = 120 V

48. (a) Eddy currents are produced on the metal surface. Laminating the surface will provide insulation and prevent the heat loss due to eddy currents. Since, there is change in magnetic flux

49. (b) Note: Load coil = Secondary coil

$\frac{E_s}{E_p} = \frac{N_s}{N_p} = \frac{I_P}{I_s} \Rightarrow \frac{25}{1} = \frac{I_P}{2}$

Therefore, $I_p = 50$ A

50. (c) Turns on primary winding, $N_p = 500$; Turns on secondary winding, $N_s = 5000$; Primary winding voltage $(E_p) = 20$ V and frequency = 50 Hz.

$\therefore\ \frac{N_s}{N_p} = \frac{E_s}{E_p}$ or $\frac{5000}{500} = \frac{E_s}{20}$

Or $E_s = \frac{5000\times20}{500} = 200\text{ V}$ and frequency remains the same. Therefore secondary winding will have an output of 200 V, 50 Hz.

CHAPTER 8

Electromagnetic Waves

Displacement Current, Ampere Maxwell's Law

1. A 100 Ω resistance and a capacitor of 100 Ω reactance are connected in series across a 220 V source. When the capacitor is 50% charged, the peak value of the displacement current is: *(2016 - II)*

a. 4.4 A
b. $11\sqrt{2}$ A
c. 2.2 A
d. 11 A

Properties and Applications (i.e. Velocity, Amplitude, Energy Density) of Electromagnetic Waves

2. When light propagates through a material medium of relative permittivity ϵ_r and relative permeability μ_r, the velocity of light, v is given by : (c - velocity of light in vacuum) *(2022)*

a. $v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$
b. $v = c$
c. $v = \sqrt{\frac{\mu_r}{\epsilon_r}}$
d. $v = \sqrt{\frac{\epsilon_r}{\mu_r}}$

3. For a plane electromagnetic wave propagating in x-direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively? *(2021)*

a. $-\hat{j}+\hat{k}, -\hat{j}-\hat{k}$
b. $\hat{j}+\hat{k}, -\hat{j}-\hat{k}$
c. $-\hat{j}+\hat{k}, -\hat{j}+\hat{k}$
d. $\hat{j}+\hat{k}, \hat{j}+\hat{k}$

4. Light with an average flux of 20 W/cm² falls on non-reflecting surface at normal incidence having surface area 20 cm². The energy received by the surface during time span of 1 minute is: *(2020)*

a. 12×10^3 J
b. 24×10^3 J
c. 48×10^3 J
d. 10×10^3 J

5. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is : (c = speed of electromagnetic waves) *(2020)*

a. 1 : 1
b. 1 : c
c. 1 : c^2
d. c : 1

6. The magnetic field in an electromagnetic wave is given by, $B_y = 2 \times 10^{-7} \sin(\pi \times 10^3 x + 3\pi \times 10^{11} t)$T
Calculate the wavelength. *(2020-Covid)*

a. 2×10^{-3} m
b. 2×10^3 m
c. $\pi \times 10^{-3}$ m
d. $\pi \times 10^3$ m

7. An em wave is propagating in a medium with a velocity $v = \hat{i}v$. The instantaneous oscillating electric field of this em wave is along +y axis. Then the direction of oscillating magnetic field of the em wave will be along. *(2018)*

a. –y direction
b. +z direction
c. –z direction
d. –x direction

8. In an electromagnetic wave in free space the root mean square value of the electric field is E_{rms} = 6 V/m. The peak value of the magnetic field is: *(2017-Delhi)*

a. 2.83×10^{-8} T
b. 0.70×10^{-8} T
c. 4.23×10^{-8} T
d. 1.41×10^{-8} T

9. Out of the following options which one can be used to produce a propagating electromagnetic wave? *(2016 - I)*

a. A charge moving at constant velocity
b. A stationary charge
c. A charge less particle
d. An accelerating charge

10. Radiation of energy 'E' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is (*C* = velocity of light): *(2015)*

a. $\frac{2E}{C}$
b. $\frac{2E}{C^2}$
c. $\frac{E}{C^2}$
d. $\frac{E}{C}$

11. Light with an energy flux of 25×10^4 W/m² falls on a perfectly reflecting surface at normal incidence. If the surface area is 15 cm², the average force exerted on the surface is: *(2014)*

a. 1.25×10^{-6} N
b. 2.50×10^{-6} N
c. 1.20×10^{-6} N
d. 3.0×10^{-6} N

12. The ratio of amplitude of magnetic field to the amplitude of electric field for an electromagnetic wave propagating in vacuum is equal to: *(2012Mains)*

a. The speed of light in vacuum

b. Reciprocal of speed of light in vacuum

c. The ratio of magnetic permeability to the electric susceptibility of vacuum

d. Unity

13. The dimensions of $(\mu_0\varepsilon_0)^{\frac{1}{2}}$ are: *(2011 Pre)*

a. $\left[L^{-\frac{1}{2}}T^{\frac{1}{2}}\right]$ b. $\left[L^{\frac{1}{2}}T^{\frac{1}{2}}\right]$

c. $[L^{-1}T]$ d. $[LT^{-1}]$

14. The electric and the magnetic field associated with an e.m. wave, propagating along the +z axis, can be represented by: *(2011 Pre)*

a. $\left[\vec{E}=E_0\hat{i},\vec{B}=B_0\hat{j}\right]$ b. $\left[\vec{E}=E_0\hat{k},\vec{B}=B_0\hat{i}\right]$

c. $\left[\vec{E}=E_0\hat{j},\vec{B}=B_0\hat{i}\right]$ d. $\left[\vec{E}=E_0\hat{j},\vec{B}=B_0\hat{k}\right]$

15. The electric field of an electromagnetic wave in free space is given by $E=10\cos\left(10^7t+kx\right)\hat{j}$ v/m where t and x are in second and metres respectively. It can be inferred that

1. The wavelength λ is 188.4 m
2. The wave number k is 0.33 rad/m
3. The wave amplitude is 10 V/m
4. The wave is propagating along +x direction

Where one of the following pairs of statement is correct? *(2010 Mains)*

a. 3 and 4 b. 1 and 2

c. 2 and 3 d. 1 and 3

16. Which of the following statement is false for the properties of electromagnetic waves? *(2010 Pre)*

a. These waves do not require any material medium for propagation

b. Both electric and magnetic field vectors attains the maxima and minima at the same place and same time

c. The energy in electromagnetic wave is divided equally between electric and magnetic vectors

d. Both electric and magnetic field vectors are parallel to each other and perpendicular to the direction of propagation of wave

17. The electric field part of an electromagnetic wave in a medium is represented by $E_x = 0$ *(2009)*

$$E_y = 2.5\cos\left[\left(2\pi\times10^6\frac{\text{rad}}{\text{m}}\right)t-\left(\pi\times10^{-2}\frac{\text{rad}}{\text{s}}\right)x\right];$$

$E_z = 0$. The wave is:

a. Moving along x direction with frequency 10^6 Hz and wave length 100 m.

b. Moving along x direction with frequency 10^6 Hz and wave length 200 m.

c. Moving along – x direction with frequency 10^6 Hz and wave length 200 m.

d. Moving along y direction with frequency $2\pi\times10^6$ Hz and wave length 200 m.

18. The velocity of electromagnetic radiation in a medium of permittivity ε_0 and permeability μ_0 is given by: *(2008)*

a. $\sqrt{\frac{\mu_0}{\varepsilon_0}}$ b. $\sqrt{\frac{\varepsilon_0}{\mu_0}}$

c. $\sqrt{\mu_0\varepsilon_0}$ d. $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$

19. The velocity of electromagnetic wave is parallel to: *(2002)*

a. $\vec{B}\times\vec{E}$ b. $\vec{E}\times\vec{B}$

c. $\vec{E}$ d. $\vec{B}$

20. Frequency of an E.M. waves is 10 MHz then its wavelength is: *(1999)*

a. 30 m b. 300 m

c. 3 m d. None of the above

21. If ε_0 and μ_0 are the electric permittivity and magnetic permeability in a free space, ε and μ are the corresponding quantities in medium, the refractive index of the medium is: *(1994)*

a. $\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}}$ b. $\sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}}$

c. $\sqrt{\frac{\varepsilon_0\mu}{\varepsilon\,\mu_0}}$ d. $\sqrt{\frac{\varepsilon}{\varepsilon_0}}$

22. The frequency of electromagnetic wave, which best suited to observe a particle of radius 3×10^{-4} cm is of the order of *(1991)*

a. 10^{15} b. 10^{14}

c. 10^{13} d. 10^{12}

Electromagnetic Spectrum

23. Match List-I with List-II *(2022)*

List-I (Electromagnetic waves)	List-II (Wavelength)
a. AM radio waves	(i) 10^{-10} m
b. Microwaves	(ii) 10^2 m
c. Infra-red radiations	(iii) 10^{-2} m
d. X-rays	(iv) 10^{-4} m

Choose the correct answer from the options given below:

a. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

b. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

c. (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

d. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

24. The E.M. wave with shortest wavelength among the following is, *(2020-Covid)*

a. X-rays b. Gamma-rays

c. Microwaves d. Ultraviolet rays

25. Which colour of the light has the longest wavelength? *(2019)*

a. Red b. Blue

c. Green d. Violet

26. The energy of the E.M. waves is of the order of 15 keV. To which part of the spectrum does it belong? *(2015 Pre)*

a. Gamma-rays b. X-rays

c. Infra-red rays d. Ultraviolet rays

27. The condition under which a microwave oven heats up a food item containing water molecules most efficiently is: (2013)
a. Infra-red waves produce heating in a microwave oven
b. The frequency of the microwaves must match the resonant frequency of the water molecules
c. The frequency of the microwaves has no relation with natural frequency of water molecules
d. Microwaves are heat waves, so always produce heating

28. If λ_v, λ_x and λ_m represent the wavelengths of visible light, X-rays and microwaves respectively, then: (2005)
a. $\lambda_m > \lambda_x > \lambda_v$
b. $\lambda_v > \lambda_m > \lambda_x$
c. $\lambda_m > \lambda_v > \lambda_x$
d. $\lambda_v > \lambda_x > \lambda_m$

29. Which of the following ray are not electromagnetic waves: (2003)
a. X-rays
b. γ-rays
c. β-rays
d. Heat rays

30. What is the cause of "Green house effect": (2002)
a. Infra-red rays
b. Ultra violet rays
c. X-rays
d. Radio waves

31. Biological importance of Ozone layer is: (2001)
a. It stops ultraviolet rays
b. Ozone layer reduces green house effect
c. Ozone layer reflects radio waves
d. Ozone layer controls O_2/H_2 ratio in atmosphere

32. Which is having minimum wavelength: (2002)
a. X-ray
b. Ultra violet rays
c. γ-rays
d. Cosmic rays

33. The frequency order for γ-rays (b), X-rays (a), UV-rays (c) is (2000)
a. b > a > c
b. a > b > c
c. c > b > a
d. a > c > b

34. Which of the following electromagnetic radiations have the smaller wavelength? (1994)
a. X-rays
b. γ-rays
c. UV waves
d. Microwaves

35. The structure of solids is investigated by using (1992)
a. Cosmic rays
b. X-rays
c. γ-rays
d. Infra-red radiations

36. Which of the following electromagnetic radiations have the longest wavelength? (1989)
a. X-rays
b. γ-rays
c. Microwaves
d. Radio waves

Effects of Dielectrics in Capacitors

37. When light propagates through a material medium of relative permittivity ϵ_r and relative permeability μ_r, the velocity of light, v is given by : (c - velocity of light in vacuum) (2022)
a. $v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$
b. $v = c$
c. $v = \sqrt{\frac{\mu_r}{\epsilon_r}}$
d. $v = \sqrt{\frac{\epsilon_r}{\mu_r}}$

38. A parallel plate capacitor of capacitance 20 μF is being charged by a voltage source whose potential is changing at the rate of 3 V/s. The conduction current through the connecting wires, and the displacement current through the plates of the capacitor, would be, respectively. (2019)
a. Zero, 60 μA
b. 60 μA, 60 μA
c. 60 μA, zero
d. Zero, zero

Signals and Communication

39. A signal emitted by an antenna from a certain point can be received at another point of the surface in the form of (1993)
a. Sky wave
b. Ground wave
c. Sea wave
d. Both (a) and (b)

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	a	a	b	a	a	b	a	d	a	b	b	c	a	d	d	b
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
d	b	a	b	b	a	b	a	b	b	c	c	a	a	d	a	b
35	36	37	38	39												
b	d	a	b	d												

Explanations

1. (c) Since, the maximum value of displacement current is equal to maximum value of capacitor current. Therefore,

$$(i_d)_{max} = (i_c)_{max} = i = \frac{\varepsilon_0}{Z} = \frac{220\sqrt{2}}{\sqrt{100^2+100^2}} = 2.2\text{ A}$$

As we are asked amplitude of displacement current.
So, need not worry about charge on capacitor.

2. (a) Refractive index $n = \sqrt{\epsilon_r u_r}$

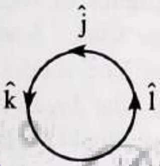
However $v = \frac{c}{n} \Rightarrow v = \left(\frac{c}{\sqrt{\epsilon_r \mu_r}}\right)$

3. (a) Given, Direction of propagation of electromagnetic waves is along x-axis. (i.e., i)

Direction of propagation of electromagnetic wave is given by $\vec{E}\times\vec{B}$.

i.e., $\vec{V} \parallel (\vec{E}\times\vec{B})$

Now, we need to check all options one by one as follows:



(a) $(\hat{j}+\hat{k})\times(\hat{j}+\hat{k}) = 0+\hat{i}+(-\hat{i})+0=0$

(b) $(-\hat{j}+\hat{k})\times(-\hat{j}-\hat{k}) = 0+\hat{i}+\hat{i}-0=2\hat{i}$

(c) $(\hat{j}+\hat{k})\times(-\hat{j}-\hat{k}) = 0-\hat{i}+\hat{i}-0=0$

(d) $(-\hat{j}+\hat{k})\times(-\hat{j}+\hat{k}) = 0-\hat{i}+\hat{i}+0=0$

Here, when electric field $(\vec{E})$ is directed along $(-\hat{j}-\hat{k})$ and magnetic field $(\vec{B})$ is directed along $(-\vec{j}-\vec{k})$, then the propagation of electromagnetic wave will be in $\hat{i}$ direction i.e., in x-axis.

4. (b) Intensity$(I) = \frac{E}{At}$

$\therefore$ Energy$(E) = IAt = \frac{20}{10^{-9}}\times 20\times10^{-9}\times 60 = 24\times10^3\text{ J}$

5. (a) In EMW, electric field and magnetic field have same energy density and same intensities.

6. (a) Wavelength $(\lambda) = \frac{2\pi}{k} = \frac{2\pi}{\pi\times10^3} \Rightarrow \lambda = 2\times10^{-3}\text{m}$

7. (b) Direction $(\vec{v})$ is decided by $\vec{E}\times\vec{B}$

$\vec{V} = \vec{E}\times\vec{B}$

$\hat{i} = \hat{J}\times\hat{B}$

$\hat{B} = \hat{k} = +z$ direction

8. (a) $E_{rms} = \frac{E_0}{\sqrt{2}}$

$E_0 = \sqrt{2}E_{rms}$

$E_0 = \sqrt{2}\times 6\text{ V/m}$

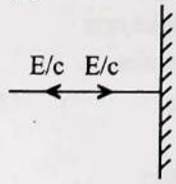
$\frac{E_0}{B_0} = c$

$B_0 = \frac{E_0}{c}\quad \frac{6\sqrt{2}\text{ V/m}}{3\times10^8\text{ m/s}}$

$B_0 = 2.824\times10^{-8}\text{ T}$

9. (d) To generate electromagnetic waves we need accelerating charge particle.

10. (a)



Momentum of light $p = \frac{E}{c}$

So, momentum transferred to the surface

$= p_f - p_i = \frac{2E}{c}$

11. (b) Average force $F_{av} = \frac{\Delta p}{\Delta t} = \frac{2\phi A}{c}$

$= \frac{2\times25\times10^4\times15\times10^{-4}}{3\times10^8}$

$= 2.50\times10^{-6}\text{N}$

12. (b) As $E = cB$

So, required ratio $\frac{B_0}{E_0} = \frac{1}{c}$

13. (c) Speed of light, $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$

$[c] = \left[(\mu_0\varepsilon_0)^{-\frac{1}{2}}\right] = \left[LT^{-1}\right]^{-1}$

14. (a) As we know $\vec{E}\times\vec{B}$ gives the direction of wave propagation and e.m. wave is propagating Along x-axis

So, $\hat{i}\times\hat{j} = \hat{k}$

15. (d) Comparing given eq. with $E = E_0\text{Cos}(\omega t + kx)$

$E_0 = 10\text{ V/m}, \omega = 10^7 \Rightarrow \omega = 2\pi n = \frac{2\pi c}{\lambda}$

$\Rightarrow \lambda = \frac{2\pi c}{\omega} = 188.4\text{ m}$

16. (d) Electric and magnetic field vectors are perpendicular to each other in electromagnetic wave.

17. **(b)** Here, $\omega = 2\pi \times 10^6$ and e.m. wave is moving in x direction as E.F. is varying with x

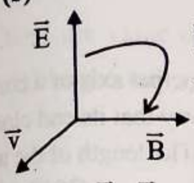
$$\text{Frequency} = \frac{\omega}{2\pi} = \frac{2\pi \times 10^6}{2\pi} = 10^6\,\text{Hz}$$

$$\text{Wavelength} = \frac{2\pi}{\kappa} = \frac{2\pi}{\pi \times 10^{-2}} = 200\text{ m}$$

18. **(d)** According to Maxwell equation

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

19. **(b)**



$\vec{E}\,\hat{j} \times \vec{B}\,\hat{i} = -\vec{v}\,\hat{k}$

velocity is in negative z - axis

$\downarrow$

$(-\hat{k})$

$\vec{E} \times \vec{B}$ parallel to $\vec{v}$

20. **(a)** $\lambda = \frac{c}{v} = \frac{3 \times 10^8}{10 \times 10^6} = 30$ metre

21. **(b)** $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ (free space)

$v = \frac{1}{\sqrt{\mu\varepsilon}}$ (medium) $\therefore\ n = \frac{c}{v} = \sqrt{\frac{\mu\varepsilon}{\mu_0 \varepsilon_0}}$

22. **(b)** The wave length of radiation used should be less than the size of the particle

Size of particle $= \lambda = \frac{c}{\upsilon}$ $\quad c = 3 \times 10^{10}$ cm/s

$3 \times 10^{-4} = \frac{3 \times 10^{10}}{\upsilon}$

or $\upsilon = 10^{14}$ hertz

However, when frequency is higher than this, wavelength is still smaller. Resolution becomes better.

23. **(a)** (a) Wavelength of Radio wave $\approx 10^2$ m (ii)
(b) Wavelength of Microwave $\approx 10^{-2}$ m (iii)
(c) Wavelength of Infrared radiations $\approx 10^{-4}$ m (iv)
(d) Wavelength of X-ray $\approx$ Å $= 10^{-10}$ m (i)
(a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

24. **(b)** Gamma rays has the shortest wavelength among the given options of E.M waves.

25. **(a)** Red has the longest wavelength while violet has smallest wavelength among the given options.

26. **(b)** Wavelength of the ray

$\lambda = \frac{hc}{E} = 0.826$ Å

γ	X-ray	UV
10^{-} A	1Å	3×10^{-7} m

since 10^{-3}nm $\le \lambda \le 1$mm

So, it is X-ray

27. **(b)** In this condition the energy from the waves is transferred efficiently to the K.E. of the molecules.

28. **(c)** $\lambda_m > \lambda_v > \lambda_x$

X-rays has minimum wavelength here and microwave has maximum wavelength.

29. **(c)** β rays are not electromagnetic waves.

30. **(a)** When the electromagnetic radiations from sun pass through the atmosphere, some of them are absorbed by it while other reach the surface of earth. The range of wavelength which reaches earth lies in infrared region.

This part of the radiation from the sun has shorter wavelength and can penetrate through the layer of gases like CO_2 and reach earth surface. But the radiation from the earth being of longer wavelength can escape through this layer. As a result the earth surface gets warm. This is known as green house effect.

31. **(a)** The ozone layer absorbs the harmful ultraviolet rays coming from sun.

32. **(d)** Cosmic rays have minimum wavelength.

33. **(a)** γ-rays $= 3 \times 10^{19}$ Hz
X-rays $= 3 \times 10^{16} - 3 \times 10^{19}$ Hz
UV-rays $= 8 \times 10^{14} - 3 \times 10^{16}$ Hz

34. **(b)** γ-rays have the smaller wavelength.

35. **(b)** X-rays are used for the investigation of structure of solids because its wavelength is of the order of interatomic distances in the solid.

36. **(d)**

Rays	wave length [Range in m]
X–rays	1×10^{-11} to 3×10^{-8}
γ–rays	6×10^{-14} to 1×10^{-11}
Microwaves	10^{-3} to 0.3
Radiowaves	10 to 10^4

37. **(a)** Refractive index $n = \sqrt{\epsilon_r u_r}$

However $v = \frac{c}{n} \Rightarrow v = \left(\frac{c}{\sqrt{\epsilon_r \mu_r}}\right)$

38. **(b)** Given capacitance of capacitor $C = 20\,\mu F = 20 \times 10^{-6}$ F

Rate of change of potential $\left(\frac{dV}{dt}\right) = 3$ V/s

$\frac{dq}{dt} = C\frac{dV}{dt}$

$i_c = 20 \times 10^{-6} \times 3$
$= 60 \times 10^{-6}$ A $= 60\,\mu$A

Also $i_d = i_c = 60\,\mu$A

39. **(d)** Signals from antenna can be transmitted in the form of sky waves and ground waves.

CHAPTER 9

Ray Optics and Optical Instruments

Photometry

1. A lens having focal length f and aperture of diameter d forms an image of intensity I. Aperture of diameter d/2 in central region of lens is covered by a black paper. Focal length of lens and intensity of image now will be respectively: *(2010 Pre)*

a. $\frac{f}{2}$ and $\frac{I}{2}$ b. f and $\frac{I}{4}$

c. $\frac{3f}{4}$ and $\frac{I}{2}$ d. f and $\frac{3I}{4}$

Reflection of Light and Spherical Mirrors

2. An object is placed on the principal axis of a concave mirror at a distance of 1.5 f (f is the focal length). The image will be at, *(2020-Covid)*

a. 1.5f b. – 1.5f

c. 3f d. –3f

3. An object is placed at a distance of 40 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be: *(2018)*

a. 30 cm towards the mirror

b. 36 cm away from the mirror

c. 30 cm away the mirror

d. 36 cm towards the mirror

4. Match the corresponding entries of column-1 with column-2. [where m is the magnification produced by the mirror] *(2016 - I)*

	Column-1		Column-2
(A)	$m = -2$	(1)	Convex mirror
(B)	$m = -\frac{1}{2}$	(2)	Concave mirror
(C)	$m = +2$	(3)	Real image
(D)	$m = \sqrt{2}/\sqrt{3}$	(4)	Virtual image

a. A → 2 and 3; B → 2 and 3; C → 2 and 4; D → 1 and 4

b. A → 1 and 3; B → 1 and 4; C → 1 and 2; D → 3 and 4

c. A → 1 and 4; B → 2 and 3; C → 2 and 4; D → 2 and 3

d. A → 3 and 4; B → 2 and 4; C → 2 and 3; D → 1 and 4

5. A rod of length 10 cm lies along the principal axis of a concave mirror of focal length 10 cm in such a way that its end closer to the pole is 20 cm away from the mirror. The length of the image is: *(2012 Mains)*

a. 10 cm b. 15 cm

c. 2.5 cm d. 5 cm

6. A concave mirror of focal length f_1 is placed at a distance of d from a convex lens of focal length f_2. A beam of light coming from infinity and falling on this convex lens-concave mirror combination returns to infinity. The distance 'd' must equal: *(2012 Pre)*

a. $-2f_1 + f_2$ b. $f_1 + f_2$

c. $-f_1 + f_2$ d. $2f_1 + f_2$

7. A tall man of height 6 feet, want to see his full image. Then required minimum length of the mirror will be: *(2000)*

a. 12 feet b. 3 feet

c. 6 feet d. Any length

Laws of Refraction; Refractive Index

8. The frequency of a light wave in a material is 2×10^{14} Hz and wavelength is 5000Å. The refractive index of material will be: *(2007)*

a. 1.50 b. 3.00

c. 1.33 d. 1.40

9. A ray of light travelling in air haves wavelength λ, frequency n, velocity v and intensity, I. If this ray enters into water than these parameter are λ, n′, v′ and I′ respectively. Which relation is correct: *(2001)*

a. $\lambda = \lambda'$ b. $n = n'$

c. $v = v'$ d. $I = I'$

10. The refractive index of water is 1.33. What will be the speed of light in water? *(1996)*

a. 4×10^8 m/s b. 1.33×10^8 m/s

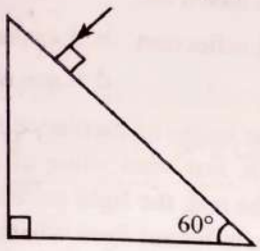
c. 3×10^8 m/s d. 2.25×10^8 m/s

Refraction Through Glass Slab and Lateral Shift

11. A light ray falls on a glass surface of refractive index $\sqrt{3}$, at an angle 60°. The angle between the refracted and reflected rays would be : *(2022)*

a. 120° b. 30°
c. 60° d. 90°

12. Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$. *(2021)*



a. 30° b. 45°
c. 90° d. 60°

13. A thin prism having refracting angle 10° is made of glass of refractive index 1.42. This prism is combined with another thin prism of glass of refractive index 1.7. This combination produces dispersion without deviation. The refracting angle of second prism should be: *(2017-Delhi)*

a. 6° b. 8°
c. 10° d. 4°

14. A microscope is focused on a mark on a piece of paper and then a slab of glass of thickness 3 cm and refractive index 1.5 is placed over the mark. How should the microscope be moved to get the mark in focus again? *(2006)*

a. 1 cm upward b. 0.5 cm downward
c. 1 cm downward d. 0.5 cm upward

15. A beam of light composed of red and green rays is incident obliquely at a point on the face of a rectangular glass slab. When coming out on the opposite parallel face, the red and green rays emerge from: *(2004)*

a. Two points propagating in two different parallel directions
b. One point propagating in two different directions through slab
c. One point propagating in the same direction through slab
d. Two points propagating in two different non parallel directions

16. Light travels through a glass plate of thickness t and having a refractive index μ. If c is the velocity of light in vacuum, the time taken by light to travel this thickness of glass is *(1996)*

a. $\frac{t}{\mu c}$ b. $\frac{\mu t}{c}$
c. $t\mu c$ d. $\frac{tc}{\mu}$

17. Time taken by sunlight to pass through a window of thickness 4 mm whose refractive index is $\frac{3}{2}$ is *(1993)*

a. 2×10^{-4} s b. 2×10^{8} s
c. 2×10^{-11} s d. 2×10^{11} s

Real and Apparent Depths

18. An air bubble in a glass slab with refractive index 1.5 (near normal incidence) is 5 cm deep when viewed from one surface and 3 cm deep when viewed from the opposite surface. The thickness (in cm) of the slab is: *(2016 - II)*

a. 12 b. 16
c. 8 d. 10

19. A bubble in glass slab (μ – 1.5) when viewed from one side appears at 5 cm and 2 cm from other side, then thickness of slab is: *(2000)*

a. 3.75 cm b. 3 cm
c. 10.5 cm d. 2.5 cm

Total Internal Reflection and Critical Angle

20. Two transparent media A and B are separated by a plane boundary. The speed of light in those media are 1.5×10^8 m/s and 2.0×10^8 m/s, respectively. The critical angle for a ray of light for these two media is: *(2022)*

a. $\tan^{-1}(0.750)$ b. $\sin^{-1}(0.500)$
d. $\sin^{-1}(0.750)$ d. $\tan^{-1}(0.500)$

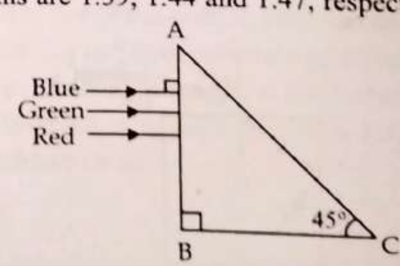
21. If the critical angle for total internal reflection from a medium to vacuum is 45°, then velocity of light in the medium is, *(2020-Covid)*

a. $\frac{3}{\sqrt{2}} \times 10^8$ m/s b. $\sqrt{2} \times 10^8$ m/s
c. 3×10^8 m/s d. 1.5×10^8 m/s

22. In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction? *(2019)*

a. 180°
b. 0°
c. Equal to angle of incidence
d. 90°

23. A beam of light consisting of red, green and blue colors is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47, respectively



The prism will: *(2015 Pre)*

a. Separate the red color part from the green and blue colors
b. Separate the blue color part from the red and green colors
c. Separate all the three colors from one another
d. Not separate the three colors at all

24. Which of the following is not due to total internal reflection? *(2011 Pre)*

a. Working of optical fiber
b. Difference between apparent and real depth of pond
c. Mirage on hot summer days
d. Brilliance of diamond

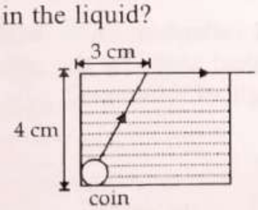
25. The speed of light in media M_1 and M_2 is 1.5×10^8 m/s and 2.0×10^8 m/s respectively: A ray of light enters from medium M_1 to M_2 at an incidence angle i. If the ray suffers total internal reflection, the value of i is: *(2010 Mains)*

a. Equal to $\sin^{-1}\left(\frac{2}{3}\right)$
b. Equal to or less than $\sin^{-1}\left(\frac{3}{5}\right)$
c. Equal to or greater than $\sin^{-1}\left(\frac{3}{4}\right)$
d. Less than $\sin^{-1}\left(\frac{2}{3}\right)$

26. A ray of light travelling in a transparent medium of refractive index μ, falls on a surface separating the medium from air at an angle of incidence of 45°. For which of the following value of μ the ray can undergo total internal reflection? *(2010 Pre)*

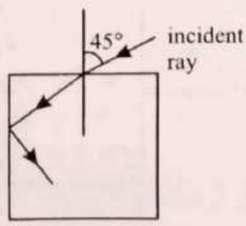
a. $\mu = 1.25$ b. $\mu = 1.33$
c. $\mu = 1.40$ d. $\mu = 1.50$

27. A small coin is resting on the bottom of a beaker filled with liquid. A ray of light from the coin travels up to the surface of the liquid and moves along its surface. How fast is the light travelling in the liquid? *(2007)*



a. 2.4×10^8 m/s b. 3.0×10^8 m/s
c. 1.2×10^8 m/s d. 1.8×10^8 m/s

28. For the given incident ray as shown in figure, the condition of total internal reflection of this ray the minimum refractive index of prism will be: *(2002)*



a. $\frac{\sqrt{3}+1}{2}$ b. $\frac{\sqrt{2}+1}{2}$
c. $\sqrt{\frac{3}{2}}$ d. $\sqrt{\frac{7}{6}}$

29. A disc is placed on a surface of pond which has refractive index $\frac{5}{3}$ A source of light is placed 4 m below the surface of liquid. The minimum radius of disc will be so light is not coming out: *(2001)*

a. ∞ b. 3 m
c. 6 m d. 4 m

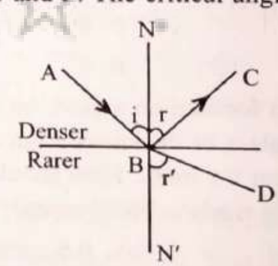
30. Optical fibre are based on: *(2001)*

a. Total internal reflection b. Less scattering
c. Refraction d. Less absorption coefficient

31. Light enters at an angle of incidence in a transparent rod of refractive index n. For what value of the refractive index of the material of the rod, the light once entered into it will not leave it through its lateral face whatsoever be the value of angle of incidence: *(1998)*

a. $n > \sqrt{2}$ b. 1.0
c. 1.3 d. 1.4

32. A ray of light from a denser medium strikes a rare medium as shown in figure. The reflected and refracted rays make an angle of 90° with each other. The angles of reflection and refraction are r and r'. The critical angle would be *(1996)*



a. $\sin^{-1}$ (tan r) b. $\sin^{-1}$ (sin r)
c. $\cos^{-1}$ (tan r) d. $\tan^{-1}$ (sin r)

33. A small source of light is 4 m below the surface of water of refractive index 5/3. In order to cut off all the light, coming out of water surface, minimum diameter of the disc placed on the surface of water is *(1994)*

a. 6 m b. ∞
c. 3 m v d. 4 m

Refraction at Spherical Surfaces

34. A plano-convex lens of unknown material and unknown focal length is given. With the help of a spherometer we can measure the, *(2020-Covid)*

a. Radius of curvature of the curved surface
b. Aperture of the lens
c. Refractive index of the material
d. Focal length of the lens

35. If f_v and f_R are the focal lengths of a convex lens for violet and red light respectively and F_v and F_R are the focal lengths of a concave lens for violet and red light respectively, then we must have (1996)

a. $f_v > f_R$ and $F_v > F_R$
b. $f_v < f_R$ and $F_v > F_R$
c. $f_v > f_R$ and $F_v < F_R$
d. $f_v < f_R$ and $F_v < F_R$

36. A lens is placed between a source of light and a wall. It forms images of area A_1 and A_2 on the wall, for its two different positions. The area of the source of light is (1995)

a. $\frac{A_1 - A_2}{2}$
b. $\frac{1}{A_1} + \frac{1}{A_2}$
c. $\sqrt{A_1 A_2}$
d. $\frac{A_1 + A_2}{2}$.

Lens Formula and Lens Maker's Formula

37. A plano-convex lens fits exactly into a planoconcave lens. Their plane surfaces are parallel to each other. If lenses are made of different materials of refractive indices μ_1 and μ_2 and R is the radius of curvature of the curved surface of the lenses, then the focal length of the combination is: (2013)

a. $\frac{2R}{(\mu_2 - \mu_1)}$
b. $\frac{R}{2(\mu_1 + \mu_2)}$
c. $\frac{R}{2(\mu_1 - \mu_2)}$
d. $\frac{R}{(\mu_1 - \mu_2)}$

38. When a biconvex lens of glass having refractive index 1.47 is dipped in a liquid, it acts as a plane sheet of glass. This implies that the liquid must have refractive index: (2012 Pre)

a. Equal to that of glass
b. Less then one
c. Greater than that of glass
d. Less then that of glass

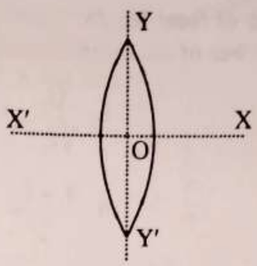
39. A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens on the opposite side. If the lens is removed the point where the rays meet will move 5 cm closer to the lens. The focal length of the lens is: (2011 Mains)

a. –10 cm
b. 20 cm
c. –30 cm
d. 5 cm

40. A convex lens is dipped in a liquid whose refractive index is equal to the refractive index of the lens. Then its focal length will: (2003)

a. Become zero
b. Become infinite
c. Become small, but non-zero
d. Remain unchanged

41. An equiconvex lens is cut into two halves along (i) XOX′ and (ii) YOY′ as shown in the figure. Let f, f′, f″ be the focal lengths of the complete lens, of each half in case (i), and of each half in case (ii), respectively. Choose the correct statement from the following: (2003)



a. f′ = f, f″ = 2f
b. f′ = 2f, f″ = f
c. f′ = f, f″ = f
d. f′ = 2f, f″ = 2f

42. The focal length of converging lens is measured for violet, green and red colours. It is respectively f_v, f_g, f_r We will get (1997)

a. $f_v < f_r$
b. $f_g > f_r$
c. $f_v = f_g$
d. $f_g > f_r$

43. Focal length of a convex lens of refractive index 1.5 is 2 cm. Focal length of lens when immersed in a liquid of refractive index of 1.25 will be (1988)

a. 10 cm
b. 2.5 cm
c. 5 cm
d. 7.5 cm

Combination of Lenses

44. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be: (2021)

a. 15
b. 50
c. 30
d. 25

45. Two similar thin equi-convex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F_1. When the space between the two lenses is filled with glycerine (which has the same refractive index ($\mu = 1.5$) as that of glass) then the equivalent focal length is F_2. The ratio $F_1 : F_2$ will be : (2019)

a. 2 : 1
b. 1 : 2
c. 2 : 3
d. 3 : 4

46. Two identical glass ($\mu_g = 3/2$) equiconvex lenses of focal length f each are kept in contact. The space between the two lenses is filled with water ($\mu_w = 4/3$). The focal length of the combination is: (2016 - II)

a. 4f/3
b. 3f/4
c. f/3
d. f

47. Two identical thin plano-convex glass lenses (refractive index 1.5) each having radius of curvature of 20 cm are placed with their convex surfaces in contact at the center. The intervening space is filled with oil of refractive index 1.7. The focal length of the combination is: (2015)

a. –25 cm
b. –50 cm
c. 50 cm
d. –20 cm

48. Two thin lenses of focal lengths f_1 and f_2 are in contact and coaxial. The power of the combinations is: *(2008)*

a. $\frac{f_1+f_2}{f_1f_2}$
b. $\sqrt{\frac{f_1}{f_2}}$
c. $\sqrt{\frac{f_2}{f_1}}$
d. $\frac{f_1+f_2}{2}$

49. A convex lens and a concave lens, each having same focal length of 25 cm, are put in contact to form a combination of lenses. The power in diopters of the combination is: *(2006)*

a. 25
b. 50
c. Infinite
d. Zero

50. For a plano convex lens ($\mu = 1.5$) has radius of curvature 10 cm. It is silvered on its plane surface. Find focal length after silvering: *(2000)*

a. 10 cm
b. 20 cm
c. 15 cm
d. 25 cm

51. If a convex lens of focal length 80 cm and a concave lens of focal length 50 cm are combined together, what will be their resulting power? *(1996)*

a. +7.5 D
b. – 0.75 D
c. +6.5 D
d. – 6.5 D

Refraction Through Prism and Dispersion Through Prism

52. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is m, then the angle of incidence is nearly equal to : *(2020)*

a. $\frac{2A}{\mu}$
b. μA
c. $\frac{2A}{2}$
d. $\frac{A}{2\mu}$

53. The refractive index of the material of a prism is 2 and the angle of the prism is 30°. One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is: *(2018)*

a. 30°
b. 45°
c. 60°
d. Zero

54. The angle of incidence for a ray of light at a refracting surface of a prism is 45°. The angle of prism is 60°. If the ray suffers minimum deviation through the prism, the angle of minimum deviation and refractive index of the material of the prism respectively, are: *(2016-II)*

a. $45°; \frac{1}{\sqrt{2}}$
b. $30°; \sqrt{2}$
c. $45°; \sqrt{2}$
d. $30°; \frac{1}{\sqrt{2}}$

55. The refracting angle of a prism is A, and refractive index of the material of the prism is cot(A/2). The angle of minimum deviation is: *(2015)*

a. $180° - 2A$
b. $90° - A$
c. $180° + 2A$
d. $180° - 3A$

56. The angle of a prism is A. One of its refracting surfaces is silvered. Light rays falling at an angle of incidence 2A on the first surface returns back through the same path after suffering reflection at the silvered surface. The refractive index μ, of the prism is: *(2014)*

a. 2sin A
b. 2 cos A
c. 1/2 cos A
d. tan A

57. For the angle of minimum deviation of a prism to be equal to its refracting angle, the prism must be made of a material whose refractive index: *(2012 Mains)*

a. Lies between $\sqrt{2}$ and 1
b. Lies between 2 and $\sqrt{2}$
c. Is less than 1
d. Is greater than 2

58. A ray of light is incident at an angle of incidence, i, on one face of prism of angle A (assumed to be small) and emerges normally from the opposite face. If the refractive index of the prism is μ, the angle of incidence i, is nearly equal to: *(2012 Pre)*

a. μA
b. $2/\mu A$
c. A/μ
d. $A/2\mu$

59. A thin prism of angle 15° made of glass of refractive index $\mu_1 = 1.5$ is combined with another prism of glass of refractive index ($\mu_2 = 1.75$). The combination of the prisms produces dispersion without deviation. The angle of the second prism should be: *(2011 Mains)*

a. 7°
b. 10°
c. 12°
d. 5°

60. A ray of light is incident on a 60° prism at the minimum deviation position. The angle of refraction at the first face (incident face) of the prism is: *(2010 Mains)*

a. Zero
b. 30°
c. 45°
d. 60°

61. The refractive index of the material of a prism is $\sqrt{2}$ and its refracting angle is 30°. One of the refracting surfaces of the prism is made a mirror inwards. A beam of monochromatic light entering the prism from the other face will retrace its path after reflection from the mirrored surface if its angle of incidence on the prism is: *(2004)*

a. 60°
b. 0°
c. 30°
d. 45°

62. For a prism its refractive index is cot A/2 then minimum angle of deviation is: *(1999)*

a. 180 – A
b. 180 – 2A
c. 90 – A
d. A/2

63. There is a prism with refractive index equal to $\sqrt{2}$ and the refractive angle equal to 30°. One of the refractive surface of the prism is polished. A beam of monochromatic light will retrace its path if its angle of incidence over the refracting surface of the prism is *(1992)*

a. 0°
b. 30°
c. 45°
d. 60°

64. A ray is incident at an angle of incidence i on one surface of a prism of small angle A and emerge normally from opposite surface. If the refractive index of the material of prism is μ, the angle of incidence *i* is nearly equal to: *(1989)*

a. $\frac{A}{\mu}$
b. $\frac{A}{2\mu}$
c. μA
d. $\frac{\mu A}{2}$

Natural Phenomenon

65. Pick the wrong answer in the context with rainbow. *(2019)*

a. When the light rays undergo two internal reflections in a water drop, a secondary rainbow is formed
b. The order of colours is reversed in the secondary rainbow
c. An observer can see a rainbow when his front is towards the sun
d. Rainbow is a combined effect of dispersion refraction and reflection of sunlight

66. Rainbow is formed due to: *(2000)*

a. Scattering and refraction
b. Total internal reflection and dispersion
c. Reflection only
d. Diffraction and dispersion

67. The blue colour of the sky is due to the phenomenon of *(1994)*

a. Scattering
b. Dispersion
c. Reflection
d. Refraction

Defects of Vision and Power of Lens

68. A biconvex lens has radii of curvature, 20 cm each. If the refractive index of the material of the lens is 1.5, the power of the lens is : *(2022)*

a. Infinity
b. +2D
c. +20D
d. +5D

69. The power of a biconvex lens is 10 dioptre and the radius of curvature of each surface is 10 cm. Then the refractive index of the material of the lens is, *(2020-Covid)*

a. $\frac{9}{8}$
b. $\frac{5}{3}$
c. $\frac{3}{2}$
d. $\frac{4}{3}$

70. A person can see clearly objects only when they lie between 50 cm and 400 cm from his eyes. In order to increase the maximum distance of distinct vision to infinity, the type and power of the correcting lens, the person has to use, will be: *(2016 - II)*

a. Concave, – 0.2 diopter
b. Convex, + 0.15 diopter
c. Convex, + 2.25 diopter
d. Concave, – 0.25 diopter

71. For a normal eye, the cornea of eye provides a converging power of 40 D and the least converging power of the eye lens behind the cornea is 20 D. Using this information, the distance between the retina and the cornea of eye lens can be estimated to be: *(2013)*

a. 1.5 cm
b. 5 cm
c. 2.5 cm
d. 1.67 cm

Simple and Compound Microscope

72. If the focal length of objective lens is increased then magnifying power of: *(2014)*

a. Microscope will increase but that of telescope decrease
b. Microscope and telescope both will increase
c. Microscope and telescope both will decrease
d. Microscope will decrease but that of telescope will increase

73. In compound microscope the magnification is 95, and the distance of object from objective lens 1/3.8 cm and focal length of objective is 1/4 cm. What is the magnification of eye pieces when final image is formed at least distance of distinct vision: *(1999)*

a. 5
b. 10
c. 100
d. None

Telescope

74. A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since: *(2021)*

a. a large aperture contributes to the quality and visibility of the images.
b. a large area of the objective ensures better light gathering power.
c. a large aperture provides a better resolution.
d. all of the above.

75. An astronomical telescope has objective and eyepiece of focal length 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, the lenses must be separated by a distance: *(2016 - I)*

a. 37.3 cm
b. 46.0 cm
c. 50.0 cm
d. 54.0 cm

76. In an astronomical telescope in normal adjustment a straight black line of length L is drawn on inside part of objective lens. The eye-piece forms a real image of this line. The length of this image is I. The magnification of the telescope is: *(2015 Pre)*

a. $\frac{L}{I}$ b. $\frac{L}{I}+1$

c. $\frac{L}{I}-1$ d. $\frac{L+1}{I-1}$

77. The magnifying power of a telescope is 9. When it is adjusted for parallel rays the distance between the objective and eyepiece is 20 cm. The focal length of lenses are: *(2012 Pre)*

a. 10 cm, 10 cm

b. 15 cm, 5 cm

c. 18 cm, 2 cm

d. 11 cm, 9 cm

78. A boy is trying to start a fire by focusing sunlight on a piece of paper using an equiconvex lens of focal length 10 cm. The diameter of the Sun is 1.39×10^9 m and its mean distance from the earth is 1.5×10^{11} m. What is the diameter of the Sun's image on the paper? *(2008)*

a. 12.4×10^{-4} m

b. 9.2×10^{-4} m

c. 6.5×10^{-4} m

d. 6.5×10^{-5} m

79. An astronomical telescope of tenfold angular magnification has a length of 44 cm. The focal length of the objective is *(1997)*

a. 44 cm b. 440 cm

c. 4 cm d. 40 cm

80. Four lenses of focal length +15 cm and ± 150 cm are available for making a telescope. To produce the largest magnification, the focal length of the eyepiece should be *(1994)*

a. +15 cm b. +150 cm

c. –150 cm d. –15 cm

Resolving Power of Lens

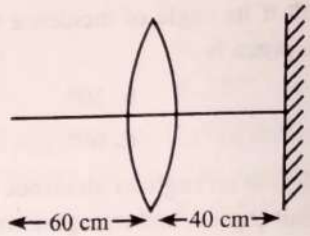
81. A telescope has an objective lens of 10 cm diameter and is situated at a distance of one kilometre from two objects. The minimum distance between these two objects, which can be resolved by the telescope, when the mean wavelength of light is 5000 Å, is of the order of: *(2004)*

a. 5 m b. 5 mm

c. 5 cm d. 0.5 m

Miscellaneous

82. A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of: *(2021)*



a. 30 cm from the lens, it would be a real image.

b. 30 cm from the plane mirror, it would be a virtual image.

c. 20 cm from the plane mirror, it would be a virtual image.

d. 20 cm from the lens, it would be a real image.

83. A beam of light from a source L is incident normally on a plane mirror fixed at a certain distance x from the source. The beam is reflected back as a spot on a scale placed just above the source L. When the mirror is rotated through a small angle θ, the spot of the light is found to move through a distance y on the scale. The angle θ is given by: *(2017-Delhi)*

a. $\frac{y}{x}$ b. $\frac{x}{2y}$

c. $\frac{x}{y}$ d. $\frac{y}{2x}$

84. A biconvex lens has a radius of curvature of magnitude 20 cm. Which one of the following options describe best the image formed of an object of height 2 cm placed 30 cm from the lens? *(2011 Pre)*

a. Virtual, upright, height = 1 cm

b. Virtual, upright, height = 0.5 cm

c. Real, inverted, height = 4 cm

d. Real, inverted, height = 1 cm

85. Exposure time of camera lens at $f/2.8$ setting is 1/200 second. The correct time of exposure at $f/5.6$ is *(1995)*

a. 0.20 second b. 0.40 second

c. 0.02 second d. 0.04 second

86. Ray optics is valid, when characteristic dimensions are *(1994, 89)*

a. Much smaller than the wavelength of light

b. Of the same order as the wavelength of light

c. Of the order of one millimetre

d. Much larger than the wavelength of light

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	d	b	a	d	d	b	b	b	d	d	d	a	a	a	b	c
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
a	c	c	a	d	a	b	c	d	d	c	b	a	a	a	a	a
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
b	c	d	a	c	b	a	a	c	a	b	b	b	a	d	a	b
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
b	b	b	a	b	b	a	b	b	d	b	c	c	c	b	a	d
69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
c	d	d	d	a	d	d	a	c	b	d	a	b	c	d	c	c
86																
d																

Explanations

1. (d) From lens maker formula

$$\frac{1}{f}=\left(\frac{\mu_2}{\mu_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

Since, lens material and surrounding media is same, therefore

$R_1 = R_2$

So, $F = F' = f$

Intensity is α to area, therefore

$$\frac{I_2}{I_1}=\frac{A_2}{A_1}$$

$$A_2=\frac{\pi 3d^2}{16}$$

$$A_1=\frac{\pi d^2}{4}$$

$$\frac{I_2}{I_1}=\frac{3}{4}$$

$$I_2=\frac{3}{4}I$$

2. (d) $u = -1.5f$

from mirror formula

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

$$\frac{1}{-1.5f}+\frac{1}{v}=\frac{-1}{f}$$

$$\Rightarrow \frac{1}{v}=\frac{-1}{f}+\frac{1}{1.5f}$$

$$\Rightarrow \frac{1}{v}=\frac{-1.5+1}{1.5f}=\frac{-0.5}{1.5f}\Rightarrow v=-3f$$

3. (b) Applying $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$$u=-40 \quad \Rightarrow \quad \frac{1}{v}-\frac{1}{40}=\frac{1}{-15}$$

$$\Rightarrow \quad v=-24 \text{ cm} \quad \text{...(i)}$$

Now $u = -20$

$$\Rightarrow \frac{1}{v}-\frac{1}{20}=-\frac{1}{15}\Rightarrow v=-60 \quad \text{...(ii)}$$

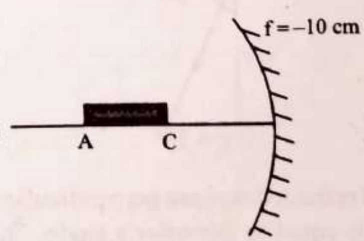
From Eq. (1) and (2)

Image moves 36 cm away from mirror

4. (a) A → 2 and 3; B → 2 and 3;

C → 2 and 4; D → 1 and 4.

5. (d)



From mirror formula $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

Image distance of A

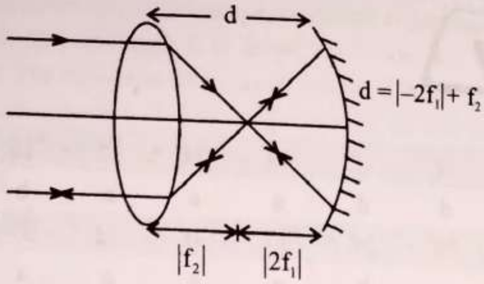
$$\frac{1}{v_A}-\frac{1}{(-30)}=\frac{1}{-10}\Rightarrow v_A=-15\text{cm}$$

Also image distance of C, $v_C = -20$ cm

The length of image

$$=|v_A-v_c|=|-15-(-20)|=5\text{cm}$$

6. (d)



$\therefore d = f_2 + 2f_1$

7. (b) The minimum height of mirror should be half of the object. Therefore, $h' = \frac{h}{2} = \frac{6}{2} = 3\,\text{feet}$

8. (b) Frequency (ν) = 2×10^{14} Hz

$\lambda = 5000\,\mathring{A} = 5 \times 10^{-7}\,m$

Velocity of light wave in material

$= \lambda \times \nu$

$= 5 \times 10^{-7} \times 2 \times 10^{14} = 10^8$ m/s

Refractive index of material

$= \frac{\text{Velocity of light vacuum}}{\text{Velocity of light material}} = \frac{3 \times 10^8}{10^8} = 3$

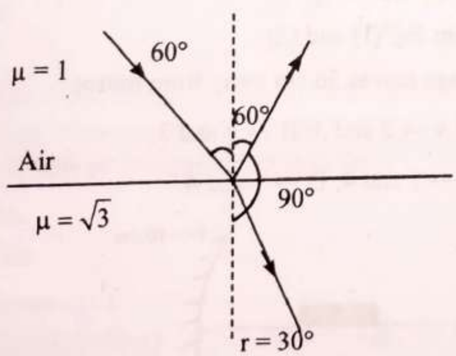
9. (b) Frequency does not change with medium, therefore remains same.

10. (d) Refractive index of water (μ_2) = 1.33.

$\frac{v_2}{v_1} = \frac{\mu_1}{\mu_2} = \frac{1}{1.33}$

Therefore $v_2 = \frac{v_1}{1.33} = \frac{3 \times 10^8}{1.33} = 2.25 \times 10^8$ m/s

11. (d)

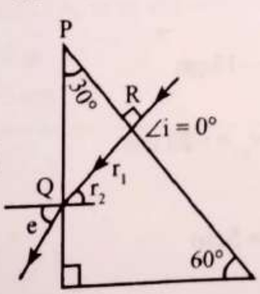


As the reflected and refracted rays are perpendicular therefore angle of incidence is equal to Brewster's angle.

$\Rightarrow \tan i_P = \mu = \sqrt{3}$

$i_p = 60° = i$

12. (d)



In ΔPQR

$\Rightarrow \angle PRQ + \angle RQP + \angle QPR = 180°$

$\Rightarrow 90° + \angle RQP + 30° = 180° \Rightarrow \angle RQP + 120° = 180°$

$\Rightarrow \angle RQP = 180° - 120° = 60°$

$\because \angle RQR + r_2 = 90° \Rightarrow r_2 = 90° - 60° = 30°$

$\Rightarrow r_2 = 30°$

Using snell's law,

$\Rightarrow \frac{\sin i}{\sin r} = \frac{\mu_{air}}{\mu_{prism}}$

$\Rightarrow \frac{\sin r_2}{\sin e} = \frac{1}{\sqrt{3}} \Rightarrow \frac{\sin 30}{\sin e} = \frac{1}{\sqrt{3}}$

$\Rightarrow \frac{\sqrt{3}}{2} = \sin e \Rightarrow e = \sin^{-}\left(\frac{\sqrt{3}}{2}\right) \Rightarrow e = \sin^{-1}(\sin 60°)$

$\Rightarrow e = 60°$

13. (a) For dispersion without deviation

$\delta_1 - \delta_2 = 0$

$\delta_1 = \delta_2$

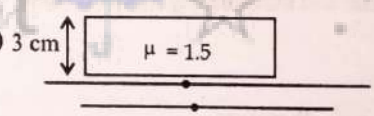
$A_1[\mu_1 - 1] = A_2[\mu_2 - 1]$

$10[1.42 - 1] = A_2[1.7 - 1]$

$10 \times 0.42 = A_2 \times 0.7$

$A_2 = 6°$

14. (a)



Glass slab of thickness = 3 cm

$\mu = 1.5$

As we know that,

In such case the shift caused $= t\left(1 - \frac{1}{\mu}\right)$

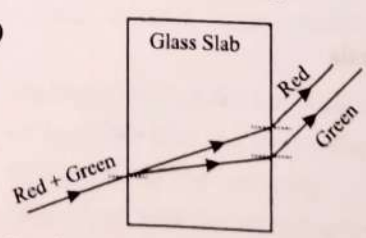
where t = thickness of slab.

μ =R.I. of slab.

$\text{Shift} = 3\left(1 - \frac{1}{1.5}\right) = 1\text{ cm}$

We have to move microscope by same distance (1 cm) in order to get mark in focus again.

15. (a)



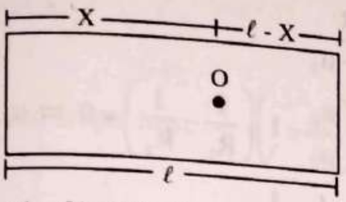
Red and green rays emerges from two points propagating in two different parallel directions.

16. (b) $\text{Time} = \frac{\text{distance}}{\text{velocity}} = \frac{t}{v} = \frac{t}{c/\mu} = \frac{\mu t}{c}$

17. (c) $v_g = \frac{c}{\mu} = \frac{3\times10^8}{\frac{3}{2}} = 2\times10^8\ \text{m/s}$

$t = \frac{x}{v_g} = \frac{4\times10^{-3}}{2\times10^8} = 2\times10^{-11}\ \text{s}$

18. (a)



Apparent depth of bubble from one surface,

$\frac{x}{\mu} = 5\ \text{cm}$ ...(i)

Real depth of the bubble from other surface,

$\frac{\ell - x}{\mu} = 3\ \text{cm}$ ...(ii)

From Eqs. (i) and (ii)

$\ell = (5+3)\mu = 12\ \text{cm}$

19. (c) From one side, $\frac{t-x}{5} = 1.5$

From other side, $\frac{x}{2} = 1.5 \Rightarrow x = 3$

$\therefore \frac{t-3}{5} = 1.5 \Rightarrow t = 7.5 + 3 = 10.5\ \text{cm}$

20. (c) $\mu = \frac{c}{u} \Rightarrow u \propto \frac{1}{\mu}$

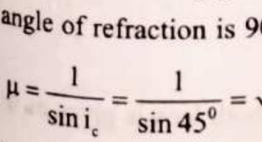
$\because \sin i_c = \frac{\mu_R}{\mu_D} = \frac{v_d}{v_r} = \frac{1.5}{2} = \frac{3}{4} \Rightarrow i_c = \sin^{-1}\left(\frac{3}{4}\right)$

21. (a) $\sin\theta = \frac{1}{\mu}$

$\mu = \frac{1}{\sin\theta} = \frac{1}{\sin 45^\circ} = \frac{1}{\left(1/\sqrt{2}\right)} = \sqrt{2}$

$\mu = \frac{c}{v} \Rightarrow v = \frac{c}{m} = \frac{3\times10}{\sqrt{2}}\ \text{m/sec}$

22. (d)



At $i = i_c$, refracted ray grazes with the surface.

So angle of refraction is 90°.

23. (a) $\mu = \frac{1}{\sin i_c} = \frac{1}{\sin 45^0} = \sqrt{2} = 1.414$

$\because (\mu_{red} = 1.39) < \mu, \mu_v > \mu; \mu_g > \mu$

Only red colour will not suffer total internal reflection

24. (b) Difference between real and apparent depth of pond is due to refraction at the air-water interface.

25. (c) $i_c = \sin^{-1}\left(\frac{\mu_F}{\mu_D}\right) = \sin^{-1}\left(\frac{C_{M_1}}{C_{M_2}}\right)$

For Total Internal Reflection $i \geq i_c$

26. (d) For Total Internal Reflection

$45^\circ \geq \theta_c \Rightarrow \sin 45^\circ \geq \sin\theta_c \Rightarrow \frac{1}{\sqrt{2}} \geq \frac{1}{\mu} \Rightarrow \mu \geq \sqrt{2}$

$\mu \geq 1.414$

$\therefore \mu = 1.50$

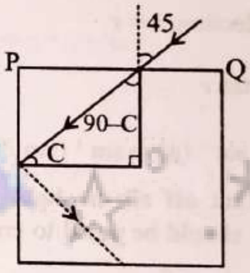
27. (d) As we know that for critical angle of incidence, the refracted ray will be come parallel to the surface.

$\Rightarrow \frac{1}{\mu} = \sin\theta = \frac{3}{5} \Rightarrow \mu = \frac{5}{3}$

$\mu = \frac{\text{Velocity of light in vacuum}}{\text{Velocity of light in fluid}}$

$\Rightarrow \frac{5}{3} = \frac{3\times10^8}{V} \Rightarrow V = 1.8\times10^8\ \text{m/s}$

28. (c) Snell's law at PQ $\mu_1 \sin 45 = \mu \sin(90 - C)$



$1\times\frac{1}{\sqrt{2}} = \mu\cos C$

Squaring both

$\frac{1}{2} = \mu^2\cos^2 c$

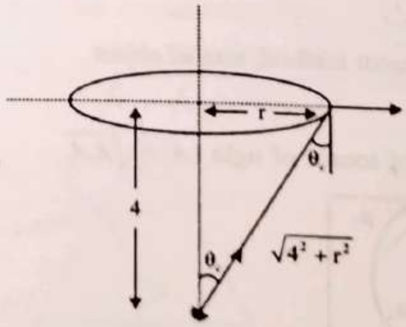
$\frac{1}{2} = \mu^2[1-\sin^2 c] \quad \left[\because \sin c = \frac{1}{\mu}\right]$

$\frac{1}{2} = \mu^2\left[1-\frac{1}{\mu^2}\right] \Rightarrow \frac{1}{2} = \mu^2 - \mu^2\times\frac{1}{\mu^2}$

$\frac{1}{2} = \mu^2 - 1 \Rightarrow \mu^2 = \frac{3}{2}$

$\mu = \sqrt{\frac{3}{2}}$

29. (b)

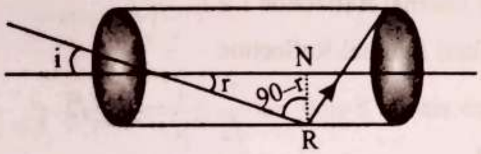


$\sin\theta_c = \frac{1}{\mu} = \frac{1}{5/3} \Rightarrow \frac{r}{\sqrt{4^2+r^2}} = \frac{3}{5}$

$\Rightarrow r = 3$

30. (a) Optical fibre are based on total internal reflection.

31. (a)



$\therefore 90° - r > i_c$ or $r < 90° - i_c$

According to Snell's law

$\sin i = n \sin r < n \sin(90° - i_c)$

$\Rightarrow \frac{\sin i}{\cos i_c} < n \Rightarrow \frac{\sin i}{\sqrt{1-\sin^2 i_c}} < n$

$\Rightarrow \frac{\sin i}{\sqrt{1-1/n^2}} < n \Rightarrow n^2 - 1 > 1 \Rightarrow n > \sqrt{2}$

32. (a) According to Snell's law,

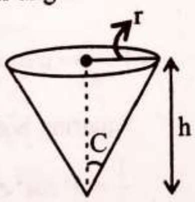
$\mu = \frac{\sin i}{\sin r'} = \frac{\sin i}{\sin(90° - r)} = \frac{\sin i}{\cos r}$

From law of reflection, i = r

$\therefore \mu = \frac{\sin r}{\cos r} = \tan r$

Critical angle = $\sin^{-1}(\mu) = \sin^{-1}(\tan r)$.

33. (a) In order to cut off all the light coming out of water surface, angle C should be equal to critical angle.



i.e. $\sin C = \frac{1}{\mu} = \frac{1}{5/3} = \frac{3}{5}$.

$\therefore \tan C = 3/4$

Now, $\tan C = \frac{r}{h}$;

$r = h\tan C = 4 \times \frac{3}{4} = 3$ m

Diameter of disc = 2r = 6 m.

34. (a) Spherometer is a device used to measure the radius of curvature of the curved surface.

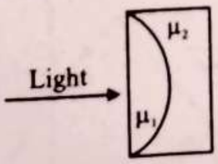
35. (b) For a convex lens, $f_R > f_V$ or $f_V < f_R$. For a concave lens, focal length is negative.

$\therefore |F_V| < |F_R|$ or $f_V > f_R$ as the smaller negative value is bigger.

36. (c) By displacement method, size of object $(O) = \sqrt{I_1 \times I_2}$.

Therefore area of source of light $(A) = \sqrt{A_1 A_2}$

37. (d)



Equivalent focal length is given by $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$

$\frac{1}{f_{eq}} = (\mu_1 - 1)\left(\frac{1}{\infty} - \frac{1}{-R}\right) + (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{\infty}\right)$

$\Rightarrow f_{eq} = \frac{R}{\mu_1 - \mu_2}$

38. (a) $P = \frac{1}{f} = \left(\frac{\mu_g}{\mu_\ell} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = 0 \Rightarrow \mu_\ell = \mu_g$

39. (c) Here, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

and v = +15 cm, u = +(15 – 5) = 10 cm

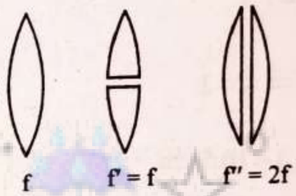
According to thin lens formula

$\Rightarrow \frac{1}{15} - \frac{1}{10} = \frac{1}{f} \Rightarrow f = -30$ cm

40. (b) Use $\frac{1}{f} = (\mu - 1)(1/R_1 - 1/R_2)$.

Here $\mu = \frac{\mu_{convexlens}}{\mu_{liquid}} = 1 \therefore f = \infty$

41. (a)



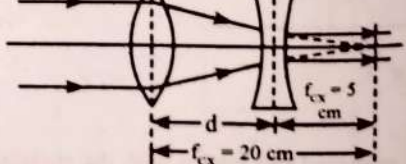
42. (a) $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

As the refractive index of violet colour (μ_v) is greater than the refractive index of red colour (μ_r), therefore focal length of violet colour is less than the focal length of red colour or in other words, $f_v > f_r$.

43. (c) $\frac{f_a}{f_e} = \frac{\left(\frac{\mu_g}{\mu_l} - 1\right)}{(\mu_g - 1)} = \frac{\left(\frac{1.5}{1.25} - 1\right)}{1.5 - 1} = \frac{\frac{1}{5}}{\frac{1}{2}} = \frac{2}{5}$

$f_e = \frac{5}{2} f_a = \frac{5}{2} \times 2 = 5$ cm

44. (a) Required ray diagram is given below:



Given,

$f_{cx} = 20$ cm and $f_{cv} = 5$ cm

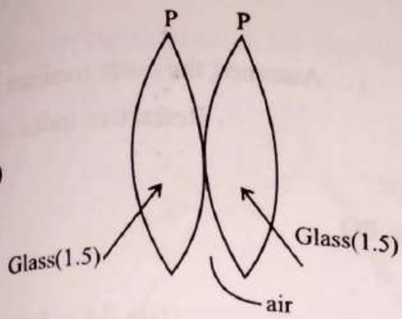
Required distance between both lens is given as below:

$d = f_{cx} - f_{cv}$

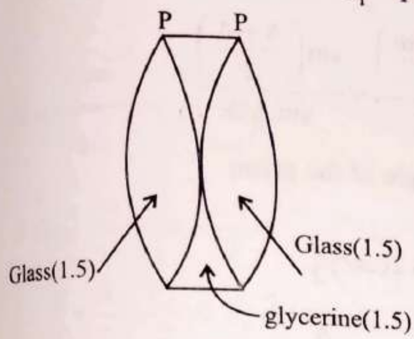
d = 20 – 5

d = 15 cm

45. (b)



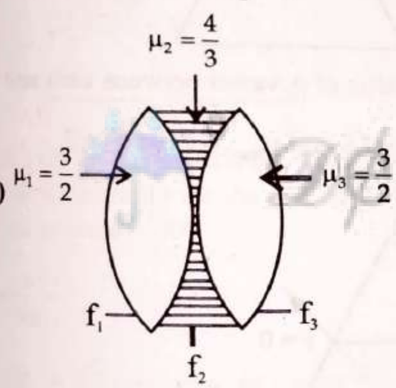
Equivalent focal length in air $\frac{1}{F_1}=\frac{1}{f}+\frac{1}{f}=\frac{2}{f}$



When glycerin is filled inside, glycerin lens behaves like a diverging lens of focal length (–f)

$$\frac{1}{F_2}=\frac{1}{f}+\frac{1}{f}-\frac{1}{f}=\frac{1}{f} \Rightarrow \frac{F_1}{F_2}=\frac{1}{2}$$

46. (b)

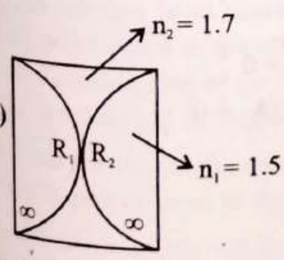


$$f_1=f_3=\frac{R}{2\left(\frac{3}{2}-1\right)}=R=f\,(\text{given})$$

$$f_2=\frac{-R}{2\left(\frac{4}{3}-1\right)}=\frac{-3}{2}R=\frac{-3}{2}f$$

$$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}=\frac{1}{f}+\left(-\frac{2}{3f}\right)+\frac{1}{f} \quad \Rightarrow \frac{1}{f_{eq}}=\frac{4}{3f} \Rightarrow f_{eq}=\frac{3f}{4}$$

47. (b)



From Lens Maker's formula

$$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_0}\right)$$

$R_1 = 20$

$R_0 = \infty$

We have $\frac{1}{f_1}=(1.5-1)\left(\frac{1}{20}\right)=\frac{1}{40}$

$$\frac{1}{f_2}=(1.5-1)\left(\frac{1}{20}\right)=\frac{1}{40}$$

& $\frac{1}{f_3}=(1.7-1)\left(\frac{2}{-20}\right)=\frac{-7}{100}$

Now $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3} \Rightarrow \frac{1}{f}=\frac{1}{40}+\frac{1}{40}-\frac{7}{100}$

$\Rightarrow f = -50$ cm

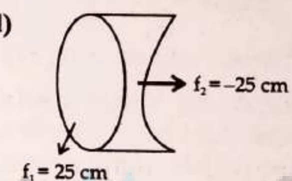
48. (a) Power for combination of lenses

$P = P_1 + P_2 - dP_1P_2$

$=\frac{1}{f_1}+\frac{1}{f_2}-0$ (d = distance between lenses)

$=\frac{f_1+f_2}{f_1f_2}$ (P_1 and P_2 are powers of lenses)

49. (d)



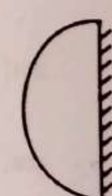
For combination of lenses,

$P = P_1 + P_2$

$=\frac{1}{F_1}+\frac{1}{F_2}$

$=\frac{1}{25}+\frac{1}{-25}=0$ [As f is positive for convex lens and negative for concave lens]

50. (a) Equivalent power of combination



$P_{eq} = 2P_L + P_M$

$=2(\mu-1)\left(\frac{1}{R}\right)+0$

The required focal length

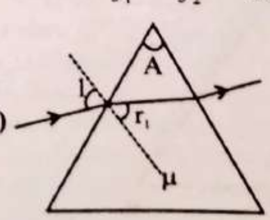
$$f=\frac{1}{P_{eq}}=\frac{R}{2(\mu-1)}=\frac{10}{2(1.5-1)}=10 \text{ cm}$$

51. (b) Focal length (f_1) = 80 cm and (f_2) = –50 cm (Minus sign due to concave lens)

Power of the combination (P)

$$=P_1+P_2=\frac{100}{f_1}+\frac{100}{f_2}=\frac{100}{80}-\frac{100}{50}=-0.75 \text{ D}$$

52. (b)



Formula of deviation,

$\sigma = i + e - A$

and $r_1 + r_2 = A$

$r_2 = O$

$r_1 = A$

Apply snell's law

$\sin 1 = \mu \sin r_1$

for small angle ($r_1 = A$)

$i = \mu A$

53. (b) $r_2 = 0$ (for retracing)

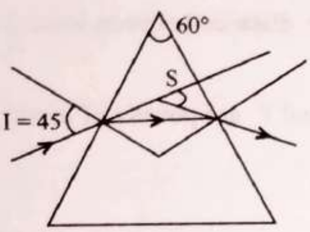
$A = r_1 + r_2 \Rightarrow r_1 = 30^\circ$

Applying Snell's law

$1 \times \sin i = \sqrt{2} \sin 30^\circ$

$\Rightarrow \sin i = \frac{1}{\sqrt{2}} \Rightarrow i = 45^\circ$

54. (b)



$\delta_{in} = 2i - A$

$= 2(45^\circ) - 60^\circ = 30^\circ$

$$\mu = \frac{\sin\left[\frac{A+\delta_{in}}{2}\right]}{\sin A/2} = \frac{\sin 45}{\sin 30} = \frac{1/\sqrt{2}}{1/2} = \sqrt{2}$$

55. (a) $\mu = \dfrac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

$\because \mu = \cot(A/2)$

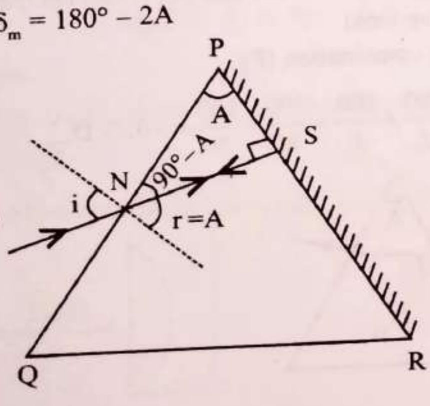
$$\cot(A/2) = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin(A/2)}$$

$$\cos(A/2) = \sin\left(\frac{\delta_m + A}{2}\right)$$

$$\sin\left[90 - \frac{A}{2}\right] = \sin\left[\frac{\delta_m + A}{2}\right] \Rightarrow 90^\circ - A/2 = \frac{\delta_m + A}{2}$$

$\Rightarrow \delta_m = 180^\circ - 2A$

56. (b)



$i = 2A$

$\mu_1 = 1$

$\mu_1 = \mu$

r = refractive angle

Apply Snell's law, PQ

$\mu_1 \sin i = \mu_2 \sin r$

$1 \times \sin 2A = \mu \sin A$

$\mu = 2 \cos A$

[$\because$ Assumed the outer medium is = air

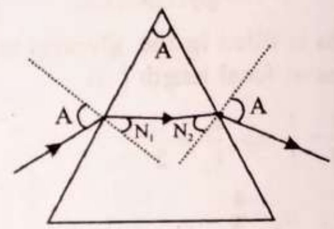
$\therefore$ Refractive index of prism]

[$\sin 2A = 2\sin A \cos A$]

57. (b) $\mu = \dfrac{\sin\left(\frac{A + \delta m}{2}\right)}{\sin A/2} = \dfrac{\sin\left(\frac{A+A}{2}\right)}{\sin A/2}$

Where A is the angle of the prism

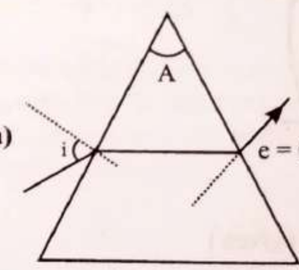
$$= \frac{2\sin A/2 \cos \frac{A}{2}}{\sin A/2} \Rightarrow 2\cos A/2$$



in this situation value of A varies between zero and 90° so

$\mu_{min} = 2\cos\frac{90^\circ}{2} = \sqrt{2}$ and $\mu_{max} = 2\cos 0^\circ = 2$

58. (a)



For a thin prism.

Equation $\delta = (\mu - 1)A$ holds good.

$i + e = \delta + A$

$i + 0 = (\mu - 1)A + A$

$i = \mu A$

59. (b) For condition dispersion without deviation in prism

$\delta_1 = \delta_2$

$\Rightarrow (\mu_1 - 1)A_1 - (\mu_2 - 1)A_2 = 0$

$\Rightarrow (1.5 - 1)15^\circ - (1.75 - 1)A_2 = 0$

$\Rightarrow 7.5^\circ - 0.75A_2 = 0$

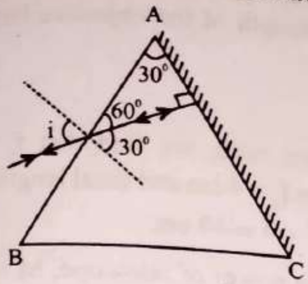
$\Rightarrow A_2 = 10^\circ$

60. (b) $r_1 + r_2 = A$

at minimum deviation $r_1 = r_2 = r$

$r = \frac{A}{2} = 30^\circ$

61. (d) According to question for Snell's law



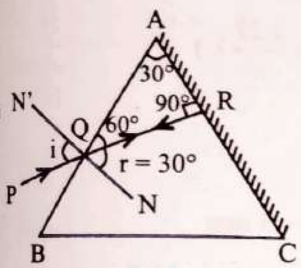
$1.\sin i = \mu \sin 30° \Rightarrow \sin i = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}} \Rightarrow i = 45°$

62. (b) $\mu = \dfrac{\cos\frac{A}{2}}{\sin\frac{A}{2}} = \dfrac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}$

$\Rightarrow \sin\frac{\pi}{2} - \frac{A}{2} = \sin\frac{A+\delta_m}{2}$

$\frac{\pi}{2} - \frac{A}{2} = \frac{A}{2} + \frac{\delta_m}{2} \Rightarrow \delta_m = 180 - 2A$

63. (c)



The path will be retraced when the refracted ray QR is incident normally on the polished surface AC. Thus angle of refraction r = 30°

$\mu = \dfrac{\sin i}{\sin r}$

$\therefore \sin i = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}}$ or $i = \sin^{-1}\frac{1}{\sqrt{2}} = 45°$

64. (c) It is given that refracted ray emerges normally from opposite surface, therefore $r_2 = 0$

As $A = r_1 + r_2$ $\quad \therefore r_1 = A$

Now, $\mu = \dfrac{\sin i_1}{\sin r_1} = \dfrac{i_1}{r_1} = \dfrac{i}{A}; i = \mu A$

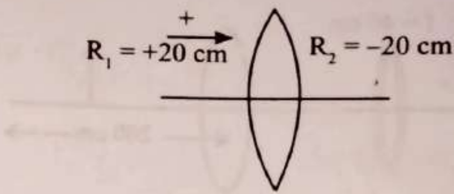
65. (c) Rainbow can't be observed when observer faces towards sun as it forms in the direction opposite of sun.

66. (b) The rainbow is an example of the dispersion of sunlight by the water drops in the atmosphere. This is a phenomenon due to a combination of the refraction of sunlight by spherical water droplets and of internal (total) reflection.

67. (a) From the Rayleigh principle, the amount of scattering is inversely proportional to the fourth power of the wavelength.

68. (d) $R_1 = R_2 = 20$ cm $= 0.2$ m and $\mu = 1.5$

As, $P = \frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$



$P = \left(\frac{3}{2} - 1\right)\left(\frac{1}{0.2} + \frac{1}{0.2}\right)$

69. (c) Power $(P) = \frac{100}{f} \Rightarrow f = \frac{100}{100} = 10$ cm

$f = \frac{R}{2(\mu - 1)} \Rightarrow 10 = \frac{10}{2(\mu - 1)} \Rightarrow (\mu - 1) = \frac{1}{2} \Rightarrow \mu = \frac{3}{2}$

70. (d) Lens formula and sign convention $= \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

As we want to correct myopia. So, far point must go to infinity.

$v = -4$m, $u = -\infty$, $P = ?$

$P = \frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{-4} - \frac{1}{\infty}$

$P = \frac{-1}{4} \times \frac{25}{25} = -0.25$ dioptre

(–) Implies concave mirror

71. (d) For a normal eye, rays coming from infinity should go the retina without effort when we look at infinity, lens offers minimum power and hence combination gives 40 D + 20 D = 60 D.

Distance between the retina and the cornea eye must be equal to focal length.

$f = \frac{1}{60}\text{m} = 1.67$ cm

72. (d) Magnifying power of Microscope $= \frac{LD}{f_0 f_e} \propto \frac{1}{f_0}$

Magnifying power of Telescope $= \frac{f_0}{f_e} \propto f_0$

73. (a) Compound microscope $M = m_0 \times m_e$

$M = \frac{F_0}{u + F_0} \times m_e \Rightarrow 95 = \frac{1/4}{-1/3.8 + 1/4} m_e$

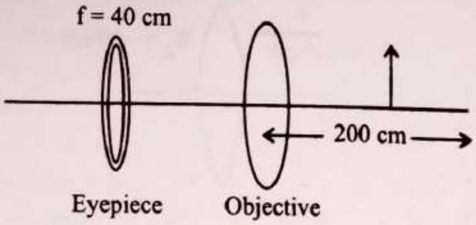
$\Rightarrow 95 = 19\, m_e$

$\Rightarrow m_e = \frac{95}{19} = 5$

74. (d) The magnifying power is given by, M.P $= \frac{f_o}{f_e}$ and resolving power is given by, R.P. $= \frac{d}{1.22\lambda}$.

Here, both magnifying power and resolving power is directly proportional to focal length of objective lens and also on diameter of the lens. And both focal length and diameter are fundamental characteristics for construction of objective lens. And large focal length and large diameter gives large aperture which in turn allows more light to enter or in other words ensures better light gathering power and also provides better resolution and good quality and visibility of the images are formed.

75. (d)



Using lens formula for objective lens

$$\frac{1}{v_0}-\frac{1}{u_0}=\frac{1}{f_0}$$

$$\frac{1}{v_0}=\frac{1}{u_0}+\frac{1}{f_0}$$

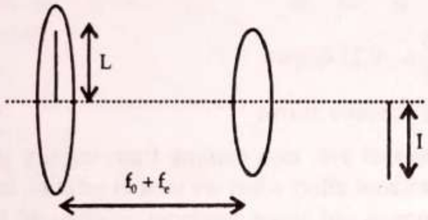
$$\frac{1}{V_0}=\frac{1}{40}-\frac{1}{200}$$

$$=\frac{5-1}{200}\Rightarrow V_0=50\,\text{cm}$$

Tube length $\ell=|V_0|+f_e$

$= 50 + 4 = 54$ cm

76. (a)



Magnification of telescope,

$$M=\frac{f_0}{f_e}$$

Here $\frac{f_0}{f_e+u}=-\frac{I}{L}$

$$\Rightarrow\frac{f_e}{f_e-(f_0+f_e)}=-\frac{I}{L}\Rightarrow\frac{f_e}{f_0}=\frac{I}{L}$$

Therefore $M=\frac{L}{I}$

77. (c) $M=\frac{f_0}{f_e}$ $\quad L=f_0+f_e$

$M=9 \quad L=20$

$9=\frac{f_0}{f_e}\Rightarrow f_0=9f_e$

$L=20 \Rightarrow 20=f_0+f_e$

$20=9f_e+f_e \Rightarrow f_e=2$ cm

$f_0=18$ cm

78. (b) Focal length of convex lens = 10 cm = 10×10^{-2} m

Diameter of sun = 1.39×10^9 m

Diameter of sun's image on the paper = $f\theta$

where $\theta=\frac{\text{Diameter of sun}}{\text{Distance of sun from earth}}$

$\Rightarrow$ Diameter of Sun's image

$$\boxed{=10\times10^{-2}\times\frac{1.39\times10^9}{1.5\times10^{11}}}=9.2\times10^{-4}\,\text{m}$$

79. (d) Length of astronomical telescope $(f_0+f_e)=44$ cm and ratio of focal length of the objective lens to that of the eye piece $\frac{f_0}{f_e}=10$.

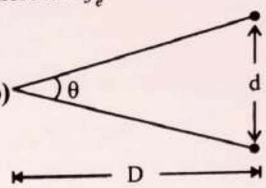
From the given ratio, we find that $f_0=10\,f_e$. Therefore $10f_e+f_e=44$ or $f_e=4$ cm and focal length of the objective (f_0) $=44-f_e=44-4=40$ cm.

80. (a) Magnifying power of telescope, $M=f_0/f_e$.

To produce largest magnifications $f_0>f_e$ and f_0 and f_e both should be positive (convex lens)

Therefore $f_e=+15$ cm.

81. (b)



$\theta=\frac{d}{D}$ but $\theta\approx\frac{\lambda}{a}$

(Note: exact relation $\theta=\frac{1.22}{a}\lambda$)

$$\frac{d}{D}=\frac{\lambda}{a}\Rightarrow d=\frac{\lambda D}{a}$$

$$\Rightarrow d=\frac{5000\times10^{-10}\times10^3}{10\times10^{-2}}=5\text{ mm}$$

82. (c) Applying lens formula for the formation of first image, after refracting from convex lens.

$$\frac{1}{v_2}-\frac{1}{u_1}=\frac{1}{f}$$

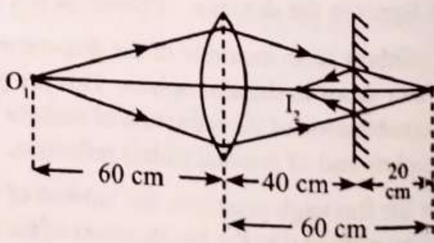
$u_1=-60$ cm [Given object is at distance of 60 cm in front of convex lens]

$f=30$ cm

$$\Rightarrow\frac{1}{v_1}-\frac{1}{(-60)}=\frac{1}{30}\quad\Rightarrow\frac{1}{v_1}=\frac{1}{30}-\frac{1}{60}=\frac{1}{60}$$

$\Rightarrow v_1=60$ cm

i.e., first image will formed 60 cm behind the convex lens. This real image formed by lens acts as virtual object for mirror.



The real image (I_2) formed in front of plane mirror. It is 20 cm from the plane mirror. That means, it will also be 20 cm from lens.

Again, applying lens law for second refraction from lens

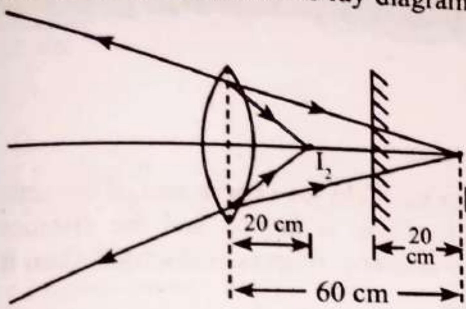
$u_2=-20$ cm, $f=30$ cm, $v_2=?$

$$\frac{1}{v_2}-\frac{1}{u_2}=\frac{1}{f}\Rightarrow\frac{1}{v_2}-\frac{1}{(-20)}=\frac{1}{30}$$

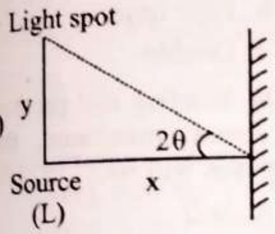
$$\Rightarrow\frac{1}{v_2}=\frac{1}{30}-\frac{1}{20}=-\frac{1}{60}\Rightarrow\frac{1}{v_2}=-\frac{1}{60}$$

$$\Rightarrow v_2=-60\,\text{cm}$$

i.e., final virtual image is at 60 cm from lens and 20 cm from the mirror as shown is ray diagram:



83. (d)



$$\tan 2\theta=\frac{y}{x}$$

$$2\theta=\frac{y}{x}$$

$$\theta=\frac{y}{2x}$$

84. (c) Let $\mu=\frac{3}{2}$ [Generally take this, if not given in question]

According to Lens maker's formula

$$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$\frac{1}{f}=\left(\frac{3}{2}-1\right)\left(\frac{1}{20}-\frac{1}{(-20)}\right)\qquad\text{[For biconvex lens]}$$

$R_1=+20$ cm

$R_2=-20$ cm

$$\Rightarrow\frac{1}{f}=\frac{1}{20}\Rightarrow f=20\text{ cm}$$

According to thin lens formula

$$\Rightarrow\frac{1}{f}=\frac{1}{v}-\frac{1}{u}\Rightarrow\frac{1}{20}=\frac{1}{v}-\frac{1}{-30}\Rightarrow v=60\text{ cm}$$

Magnification, $m=\frac{v}{u}=\frac{h_I}{h_u}$

$$\Rightarrow\frac{60}{-30}=\frac{h_I}{2}\Rightarrow h_I=-4\text{ cm}$$

So, the image formed will be real, inverted height 4 cm and at a distance of 60 cm from lens.

85. **(c)** Time of exposure $t\propto(\text{f – number})^2$

$$\therefore\ \frac{t}{\left(\frac{1}{200}\right)}=\left(\frac{5.6}{2.8}\right)^2$$

$t=0.02$ s

86. **(d)** Ray optics is valid, when characteristics dimensions are much larger than the wavelength of light.

CHAPTER 10

Wave Optics

Huygen's Wave Theory

1. Which one of the following phenomena is not explained by Huygen's construction of wavefront? *(1988)*
   a. Refraction b. Reflection
   c. Diffraction d. Origin of spectra

Intensities of Maxima and Minima

2. The interference pattern is obtained with two coherent light sources of intensity ratio n. In the interference pattern, the ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be: *(2016 - II)*
   a. $\frac{\sqrt{n}}{(n+1)^2}$ b. $\frac{2\sqrt{n}}{(n+1)^2}$
   c. $\frac{\sqrt{n}}{n+1}$ d. $\frac{2\sqrt{n}}{n+1}$
3. Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the same direction. The sum of the maximum and minimum intensities is: *(2008)*
   a. $2(I_1+I_2)$ b. I_1+I_2
   c. $\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$ d. $\left(\sqrt{I_1}-\sqrt{I_2}\right)^2$
4. Ratio of intensities of two waves are given by 4 : 1 Then ratio of the amplitudes of the two waves is: *(1991)*
   a. 2 : 1 b. 1 : 2
   c. 4 : 1 d. 1 : 4

Interference and Young's Double Slit Experiment

5. In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nn wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is *(2022)*
   a. 12 b. 6
   c. 8 d. 9
6. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes: *(2020)*
   a. Half b. Four times
   c. One-fourth d. Double
7. Two coherent sources of light interfere and produce fringe pattern on a screen. For central maximum, the phase difference between the two waves will be. *(2020-Covid)*
   a. π b. $3\pi/2$
   c. $\pi/2$ d. Zero
8. In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1 m away, was found to be 0.2°. What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water? ($\mu_{water} = 4/3$) *(2019)*
   a. 0.266° b. 0.15°
   c. 0.05° d. 0.1°
9. In Young's double slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is 5896 Å and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20°. To increase the fringe angular width to 0.21° (with same λ and D) the separation between the slits needs to be changed to *(2018)*
   a. 2.1 mm b. 1.9 mm
   c. 1.8 mm d. 1.7 mm
10. Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8^{th} bright fringe in the medium lies where 5^{th} dark fringe lies in air. The refractive index of the medium is nearly: *(2017-Delhi)*
    a. 1.59 b. 1.69
    c. 1.78 d. 1.25
11. The intensity at the maximum in a Young's double slit experiment is I_0. Distance between two slits is $d = 5\lambda$, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance $D = 10d$? *(2016 - I)*
    a. I_0 b. $\frac{I_0}{4}$
    c. $\frac{3}{4}I_0$ d. $\frac{I_0}{2}$

12. Two slits in Young's experiment have widths in the ratio 1 : 25. The ratio of intensity at the maxima and minima in the interference pattern, $\frac{I_{max}}{I_{min}}$ is: *(2015 Re)*

a. 4/9
b. 9/4
c. 12/149
d. 49/121

13. In the Young's double-slit experiment, the intensity of light at a point on the screen where the path difference is λ is K, (λ being the wavelength of light used). The intensity at a point where the path difference is $\lambda/4$,will be: *(2014)*

a. K
b. 4K
c. K/2
d. Zero

14. In Young's double slit experiment, the slits are 2 mm apart and are illuminated by photons of two wavelengths $\lambda_1 = 12000$ Å and $\lambda_2 = 10000$ Å. At what minimum distance from the common central bright fringe on the screen 2 m from the slit will a bright fringe from one interference pattern coincide with a bright fringe from the other? *(2013)*

a. 4 mm
b. 3 m
c. 8 mm
d. 6 mm

15. Interference was observed in interference chamber where air was present, now the chamber is evacuated, and if the same light is used, a careful observer will see *(1993)*

a. No interference
b. Interference with brighter bands
c. Interference with dark bands
d. Interference with larger width

16. If yellow light emitted by sodium lamp in Young's double slit experiment is replaced by monochromatic blue light of the same intensity *(1992)*

a. Fringe width will decrease
b. Fringe width will increase
c. Fringe width will remain unchanged
d. Fringes will becomes less intense

17. In Young's double slit experiment carried out with light of wavelength (λ) = 5000 Å, the distance between the slits is 0.2 mm and the screen is at 200 cm from the slits. The central maximum is at $x = 0$. The third maximum (taking the central maximum as zeroth maximum) will be at x equal to *(1992)*

a. 1.67 cm
b. 1.5 cm
c. 0.5 cm
d. 5.0 cm

18. In Young's experiment, two coherent sources are placed 0.90 mm apart and fringe are observed one metre away. If it produces second dark fringe at a distance of 1 mm from central fringe, the wavelength of monochromatic light is used would be: *(1991)*

a. 60×10^{-4} cm
b. 10×10^{-4} cm
c. 10×10^{-5} cm
d. 6×10^{-5} cm

19. In Young's double slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index – without disturbing the geometrical arrangement, the new fringe width will be: *(1990)*

a. 0.30 mm
b. 0.40 mm
c. 0.53 mm
d. 450 microns

20. The Young's double slit experiment is performed with blue and with green light of wavelengths 4360 Å and 5460 Å respectively. If x is the distance of 4th maxima from the central one, then: *(1990)*

a. x(blue) = x(green)
b. x(blue) > x(green)
c. x(blue) < x(green)
d. $\frac{x(\text{blue})}{x(\text{green})} = \frac{5460}{4360}$

21. Interference is possible in: *(1989)*

a. Light waves only
b. Sound waves only
c. Both light and sound waves
d. Neither light nor sound waves

Diffraction of Light From a Narrow Slit

22. In a diffraction pattern due to a single slit of width a, the first minima is observed at an angle 30° when light of wavelength 5000 Å is incident on the slit. The first secondary maximum is observed at an angle of: *(2016 - I)*

a. $\sin^{-1}\left(\frac{1}{4}\right)$
b. $\sin^{-1}\left(\frac{2}{3}\right)$
c. $\sin^{-1}\left(\frac{1}{2}\right)$
d. $\sin^{-1}\left(\frac{3}{4}\right)$

23. A linear aperture whose width is 0.02 cm is placed immediately in front of a lens of focal length 60 cm. The aperture is illuminated normally by a parallel beam of wavelength 5×10^{-5} cm. The distance of the first dark band of the diffraction pattern from the center of the screen is: *(2016 - II)*

a. 0.20 cm
b. 0.15 cm
c. 0.10 cm
d. 0.25 cm

24. For a parallel beam of monochromatic light of wavelength 'λ', diffraction is produced by a single slit whose width 'a' is of the order of the wavelength of the light. If 'D' is the distance of the screen from the slit, the width of the central maxima will be: *(2015)*

a. $\frac{D\lambda}{a}$
b. Da/λ
c. l2D/a
d. $\frac{2D\lambda}{a}$

25. In a double slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of double slit within the central maxima of single slit pattern? *(2015)*

a. 0.1 mm
b. 0.5 mm
c. 0.02 mm
d. 0.2 mm

26. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern the phase difference between the Huygens's wavelength from the edge of the slit and the wavelet from the mid point of the slit is: *(2015 Re)*

a. $\pi/8$
b. $\pi/4$
c. $\pi/2$
d. π

27. A beam of light of $\lambda = 600$ nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between first dark fringes on either side of the central bright fringe is: *(2014)*

a. 1.2 cm b. 1.2 mm
c. 2.4 cm d. 2.4 mm

28. A parallel beam of fast moving electrons is incident normally on a narrow slit. A fluorescent screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statements is correct? *(2013)*

a. The angular width of the central maximum will be unaffected
b. Diffraction pattern is not observed on the screen in the case of electrons
c. The angular width of the central maximum of the diffraction pattern will increase
d. The angular width of the central maximum will decrease

29. A parallel beam of monochromatic light of wavelength 5000 Å is incident normally on a single narrow slit of width 0.001 mm. The light is focused by a convex lens on a screen placed in focal plane. The first minimum will be formed for the angle of diffraction equal to *(1993)*

a. 0° b. 15°
c. 30° d. 50°

Resolving Power of Optical Instrument

30. Assume that light of wavelength 600 nm is coming from a star. The limit of resolution of telescope whose objective has a diameter of 2 m is: *(2020)*

a. 1.83×10^{-7} rad
b. 7.32×10^{-7} rad
c. 6.00×10^{-7} rad
d. 3.66×10^{-7} rad

31. An astronomical refracting telescope will have large angular magnification and high angular resolution, when it has an objective lens of *(2018)*

a. Large focal length and large diameter
b. Large focal length and small diameter
c. Small focal length and large diameter
d. Small focal length and small diameter

32. The ratio of resolving powers of an optical microscope for two wavelengths $\lambda_1 = 4000$ Å and $\lambda_2 = 6000$ Å is *(2017-Delhi)*

a. 9 : 4 b. 3 : 2
c. 16 : 81 d. 8 : 27

33. The angular resolution of a 10 cm diameter telescope at a wavelength of 5000 Å is of the order of: *(2005)*

a. 10^{6} rad b. 10^{-4} rad
c. 10^{4} rad d. 10^{-6} rad

34. Diameter of human eye lens is 2 mm. What will be the minimum distance between two points to resolve them, which are situated at a distance of 50 metre from eye. The wavelength of light is 5000 Å: *(2002)*

a. 2.32 m b. 4.28 mm
c. 1.25 cm d. 12.48 cm

Polarization Phenomenon, Law of Malus and Brewster's Law

35. The Brewsters angle i_b for an interface should be : *(2020)*

a. $30° < i_b < 45°$ b. $45° < i_b < 90°$
c. $i_b = 90°$ d. $0° < i_b < 30°$

36. Unpolarised light is incident from air on a plane surface of a material of refractive index 'μ'. At a particular angle of incidence 'i', it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is **correct** for this situation? *(2018)*

a. $i = \sin^{-1}\left(\frac{1}{\mu}\right)$
b. Reflected light is polarised with its electric vector perpendicular to the plane of incidence
c. Reflected light is polarised with its electric vector parallel to the plane incidence
d. $i = \tan^{-1}\left(\frac{1}{\mu}\right)$

37. Two Polaroids P_1 and P_2 are placed with their axis perpendicular to each other. Unpolarised light I_0 is incident on P_1. A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle 45° with that of P_1. The intensity of transmitted light through P_2 is: *(2017-Delhi)*

a. $\frac{I_0}{4}$ b. $\frac{I_0}{8}$
c. $\frac{I_0}{16}$ d. $\frac{I_0}{2}$

38. Which of the phenomenon is not common to sound and light waves? *(1988)*

a. Interference b. Diffraction
c. Coherence d. Polarisation

Various Parameters of EM Waves Due to Refraction

39. An electromagnetic radiation of frequency n, wavelength λ, travelling with velocity v in air, enters a glass slab of refractive index μ. The frequency, wavelength and velocity of light in the glass slab will be respectively *(1997)*

a. $n, 2\lambda$ and $\frac{v}{\mu}$ b. $\frac{2n}{\mu}, \frac{\lambda}{\mu}$ *and* v
c. $\frac{n}{\mu}, \frac{\lambda}{\mu}$ and $\frac{v}{\mu}$ d. $n, \frac{\lambda}{\mu}$ and $\frac{v}{\mu}$

40. A beam of monochromatic light is refracted from vacuum into a medium of refractive index 1.5. The wavelength of refracted light will be *(1992, 91)*

a. Depend on intensity of refracted light
b. Same
c. Smaller
d. Larger

41. Green light wavelength 5460 Å is incident on an air–glass interface. If the refractive index of glass is 1.5, the wavelength of light in glass would be ($c = 3 \times 10^8\ ms^{-1}$) *(1991)*

a. 3640 Å
b. 5460 Å
c. 4861 Å
d. None of these

Miscellaneous

42. A star, which is emitting radiation at a wavelength of 5000 Å, is approaching the earth with a velocity of 1.5×10^4 m/s. The change in wavelength of the radiation as received on the earth is: *(1995)*

a. 25 Å
b. 100 Å
c. Zero
d. 2.5 Å

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	d	a	a	a	b	d	b	b	c	d	b	c	d	d	a	b
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
d	a	c	c	d	b	d	d	d	d	d	c	d	a	b	d	c
35	36	37	38	39	40	41	42									
b	b	b	d	d	c	a	a									

Explanations

1. (d) Christiaan Huygens was a 17th century Dutch mathematician, physicist, and astronomer who made significant contributions to the understanding of optics. He explained the phenomena of refraction, reflection, and diffraction as follows:

Refraction: Huygens explained refraction in terms of his wave theory of light. According to this theory, light is composed of waves that move through a medium. When a wave of light passes from one medium to another, such as from air to water, the speed of the wave changes, and this causes the wave to bend or refract. Huygens used this theory to explain why a straw in a glass of water appears to be bent.

Reflection: Huygens explained reflection as the bouncing back of a wave of light from a surface. He proposed that when a wave of light hits a surface, it creates a new wave that travels back in the opposite direction. Huygens also explained that the angle at which the wave hits the surface affects the angle at which it is reflected. This principle is known as the law of reflection.

Diffraction: Huygens proposed that when a wave of light encounters an obstacle or a small opening, it spreads out in all directions beyond the obstacle or opening. This phenomenon is known as diffraction. Huygens explained that this occurs because each point on the wavefront acts as a new source of waves, which then spread out in all directions. Huygens used this theory to explain why light can bend around corners and edges.

2. (d) As, we know

Resultant intensity, $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

ϕ = Phase difference between intensities

So, $I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$ [For I_{max}, $\phi = 0$]

$I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$ [For I_{min}, $\phi = \pi$]

Let $\frac{I_1}{I_2} = \frac{n}{1}$

$$\frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2 - \left(\sqrt{I_1}-\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2 + \left(\sqrt{I_1}-\sqrt{I_2}\right)^2} = \frac{4\sqrt{I_1 I_2}}{2(I_1 + I_2)}$$

Dividing numerator and denominator by I_2,

Required ratio will be $= \dfrac{2\sqrt{I_1/I_2}}{\left(\dfrac{I_1}{I_2} + 1\right)} = \dfrac{2\sqrt{n}}{n+1}$

3. (a) As, we know

Resultant intensity, $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

ϕ = Phase difference between intensities

So, $I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$ [For I_{max}, $\phi = 0$]

$I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$ [For I_{min}, $\phi = \pi$]

$\Rightarrow I_{max} + I_{min} = 2(I_1 + I_2)$

4. (a) $\because$ Intensity, $I \propto (\text{Amplitude})^2$

$$\frac{I_1}{I_2} = \frac{a^2}{b^2} = \frac{4}{1}$$

$$\therefore \frac{a}{b} = \frac{2}{1}$$

5. (a) From Young's double slit experiment, we know that

$$\beta = (n\lambda)\left(\frac{D}{d}\right)$$

$$n_1\lambda_1\frac{D}{d} = n_2\lambda_2\frac{D}{d} \Rightarrow n_1\lambda_1 = n_2\lambda_2$$

It is given that, $n_1 = 8$, $\lambda_1 = 600$ nm, $\lambda_2 = 400$ nm

(8) (600 nm) = n_2(400 nm)

$\therefore$ $n_2 = 12$

6. (b) In Young's double slit experiment

Fringe width, $\beta = \frac{\lambda D}{d}$

Now, $d' = \frac{d}{2}$ and $D' = 2D$

$$\text{So}, \beta' = \frac{\lambda(2D)}{d/2} = \frac{4\lambda D}{d} \Rightarrow \beta' = 4\beta$$

7. (d) $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$ [$\because \phi = 0$]

Phase difference between the two waves will be zero for central maximum.

8. (b) Angular fringe width in air, $\theta_0 = \frac{\beta}{D}$

Angular fringe width in water, $\theta_w = \frac{\beta}{\mu D} = \frac{\theta_0}{\mu} = \frac{0.2^\circ}{\left(\frac{4}{3}\right)} = 0.15^\circ$

9. (b) Here, initial angular width, $\theta_1 = 0.20^\circ$

Final angular width, $\theta_2 = 0.21^\circ$

We know that,

Angular width of fringe, $\theta = \frac{\lambda}{d}$

$$\therefore \frac{\theta_1}{\theta_2} = \frac{\lambda_1}{\lambda_2} \times \frac{d_2}{d_1}$$

$$\Rightarrow \frac{0.20}{0.21} = \frac{d_2}{2} \quad [\because \lambda_2 = \lambda_1 = 400\text{nm}, d_1 = 2\text{mm}]$$

$$\therefore d_2 = \frac{0.20}{0.21} \times 2 = 1.9\text{mm}$$

10. (c) According to the question,

8^{th} bright fringe in the medium = 5^{th} dark fringe in air

$y_8 = y_5$

$$\frac{n_1\lambda_1 D}{d} = \frac{(2n_2 - 1)}{2}\frac{\lambda D}{d}$$

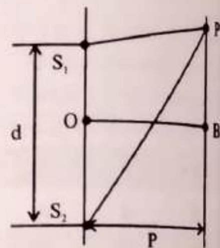
$$\Rightarrow \frac{8D}{d}\lambda_1 = \frac{(2\times5-1)}{2}\frac{\lambda D}{d}$$

$$\Rightarrow 8\lambda_1 = \frac{9}{2}\lambda \Rightarrow \lambda_1 = \frac{9}{16}\lambda$$

$$\Rightarrow \frac{\lambda}{\mu} = \frac{9}{16}\lambda \quad \left[\lambda_1 = \frac{\lambda_{air}}{\mu}\right]$$

$$\because \mu = \frac{16}{9} = 1.78$$

11. (d) Path difference $= S_2P - S_1P$



$= \sqrt{D^2 + d^2} - D$

$$= D\left\{1 + \frac{1}{2}\frac{d^2}{D^2}\right\} - D = \frac{d^2}{2D}$$

$$\Delta x = \frac{d^2}{2\times10d} \quad [\text{Given}, D = 10d]$$

$$= \frac{d}{20} = \frac{5\lambda}{20} = \frac{\lambda}{4}$$

$$\Delta\phi = \frac{2\pi}{\lambda}\frac{\lambda}{4} = \frac{\pi}{2}$$

So intensity at desired point is,

$$I = I_0\cos^2\frac{\phi}{2} = I_0\cos^2\frac{\pi}{4} = \frac{I_0}{2}$$

12. (b) Intensity at width of the slit

$$\frac{I_1}{I_2} = \frac{W_1}{W_2} = \frac{1}{25} \Rightarrow \frac{I_2}{I_1} = \frac{25}{1}$$

$$\frac{I_{max}}{I_{max}} = \frac{\left(\sqrt{I_2} + \sqrt{I_1}\right)^2}{\left(\sqrt{I_2} - \sqrt{I_1}\right)^2} = \left(\frac{\sqrt{\frac{I_2}{I_1}} + 1}{\sqrt{\frac{I_2}{I_1}} - 1}\right)^2$$

$$= \left(\frac{5+1}{5-1}\right)^2 = \left(\frac{6}{4}\right)^2 = \frac{9}{4}$$

13. (c) Phase difference, $\phi = \frac{2\pi}{\lambda} \times$ Path difference

When, path difference $= \lambda$

Phase difference, $\phi_1 = \frac{2\pi}{\lambda} \times \lambda = 2\pi$

Intensity,

$$K = 4I_0^2\cos^2\frac{\phi_1}{2} = 4I_0^2\cos 2\pi = 4I_0^2 \quad \text{...(i)}$$

When, Path difference $= \frac{\lambda}{4}$

Phase difference, $\phi_2 = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$

Intensity,

$I_2 = 4I_0^2 \cos^2\frac{\phi_2}{2} = 4I_0^2 \times \cos^2\frac{\pi}{4} = 4I_0^2 \times \frac{1}{2}$

$= 2I_0^2 = \frac{K}{2}$

14. (d) Let n_1 bright fringe of λ_1 coincides with n_2 bright fringe of λ_2. Then

$\frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d} \Rightarrow n_1\lambda_1 = n_2\lambda_2$

$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{10000}{12000} = \frac{5}{6}$

Let x be given distance.

$\therefore\ x = \frac{n_1\lambda_1 D}{d}$

Here, $n_1 = 5$, D = 2 m, d = 2 mm = 2×10^{-3} m

$\lambda_1 = 12000\ \text{Å} = 12000 \times 10^{-10}\ \text{m} = 12 \times 10^{-7}$ m

$x = \frac{5\times12\times10^{-7}\,\text{m}\times 2\ \text{m}}{2\times10^{-3}\,\text{m}} = 6\times10^{-3}\,\text{m} = 6\ \text{mm}$

15. (d) In vacuum, λ increases very slightly compared to that in air. As $\beta \propto \lambda$, therefore, width of interference fringe increase slightly.

16. (a) As $\beta = \frac{\lambda D}{d}$, $\beta \propto \lambda$ and $\lambda_b < \lambda_y$,

$\therefore$ Fringe width β will decrease.

17. (b) For n^{th} maximum

$x_n = (n)\lambda\frac{D}{d} = 3\times5000\times10^{-10}\times\frac{2}{0.2\times10^{-3}}$

$= 1.5 \times 10^{-2}$ m = 1.5 cm

18. (d) For dark fringe $x_n = (2n-1)\frac{\lambda D}{2d}$

$\therefore\ \lambda = \frac{2x_n d}{(2n-1)D} = \frac{2\times10^{-3}\times0.9\times10^{-3}}{(2\times2-1)\times1}$

$\lambda = 0.6 \times 10^{-6}$ m = 6×10^{-5} cm

19. (a) Fringe width when whole apparatus is immersed in a medium of refractive index μ is,

$\beta' = \frac{\beta}{\mu} = \frac{0.4}{\frac{4}{3}} = 0.3$ mm

20. (c) Distance of n^{th} maxima, $x = n\lambda\frac{D}{d} \propto \lambda$

As $\lambda_b < \lambda_g$

$\therefore$ x(blue) < x(green)

21. (c) Interference is a wave phenomenon shown by both the light waves and sound waves.

22. (d) For n^{th} minima,

a sinθ = nλ

For first minima, $\sin 30^\circ = \frac{\lambda}{a} = \frac{1}{2}$

For n^{th} maxima,

$a\sin\theta = (2n+1)\frac{\lambda}{2} \Rightarrow \sin\theta = \frac{(2n+1)\lambda}{2a}$

First secondary maxima will be at

$\sin\theta = \frac{3\lambda}{2a} = \frac{3}{2}\left(\frac{1}{2}\right) \Rightarrow \theta = \sin^{-1}\left(\frac{3}{4}\right)$

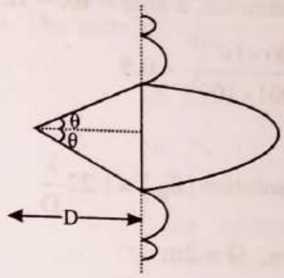
23. (b) $\beta = \frac{\lambda f}{a}$

f = D = 60 cm

For first minima,

$y = \frac{\lambda D}{a} = \frac{5\times10^{-5}\times60}{2\times10^{-2}} = \frac{5\times10^{-3}\times60}{2} = 0.15$ cm

24. (d) Linear width of central maxima = D2θ = 2Dθ $= 2D\frac{\lambda}{a}$



25. (d) In a double slit experiment, the two slits are 1 mm apart.

d = 1 mm = 10^{-3}m.

The screen is placed at a distance D = 1 m away. Monochromatic light of wave length, λ = 500 nm = 5×10^{-7}m is used.

The distance between two successive maxima or two successive minima is

$\frac{\lambda D}{d} = \frac{5\times10^{-7}}{10^{-3}} = 5\times10^{-4}\text{m} = 0.5\text{mm}$

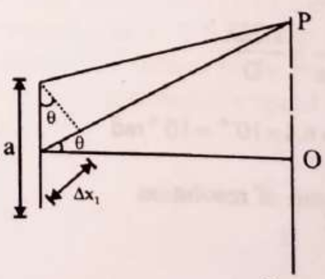
Ten maxima are contained within a distance, 10 × 0.5 mm = 5 mm

For a single slit pattern we have $\sin\theta = \frac{\lambda}{a}$

The width of the central maxima is $2D\sin\theta = \frac{2D\lambda}{a} = 5\text{mm}$

$\therefore\ a = \frac{2D\lambda}{5\times10^{-3}} = \frac{2\times5\times10^{-7}}{5\times10^{-3}} = 2\times10^{-4}\text{m} = 0.2\text{mm}$

26. (d)



Screen

For first minima at P, a sin θ = λ

So, phase difference = $\frac{2\pi}{\lambda}\times$ Phase difference

$\Delta\phi_1 = \frac{\Delta x_1}{\lambda}\times 2\pi$

$= \frac{(a/2)\sin\theta}{\lambda} \times 2\pi$

$\Delta\phi_1 = \frac{\lambda}{2\lambda} \times 2\pi = \pi \text{radian}$

27. (d) Width of central bright fringe

$= \frac{2\lambda D}{d} = \frac{2 \times 600 \times 10^{-9} \times 2}{1 \times 10^{-3}} \text{m}$

$= 2.4 \times 10^{-3}$ m

= 2.4 mm

28. (d) When speed of electrons is increased the wavelength of electrons will decreases. Therefore the angular width of the central maximum of diffraction pattern will decrease.

29. (c) For first minimum, a $\sin\theta = n\lambda = 1\lambda$

$\sin\theta = \frac{\lambda}{a} = \frac{5000 \times 10^{-10}}{0.001 \times 10^{-3}} = 0.5$

$\theta = 30°$

30. (d) Limit of resolution $(\theta_R) = 1.22\frac{\lambda}{D}$

$\lambda = 600 \times 10^{-9}\text{m},\ D = 2\text{m}$

$\therefore \theta_R = \frac{1.22 \times 600 \times 10^{-9}}{2} = 3.66 \times 10^{-7}\text{ rad}$

31. (a) $\uparrow m = \frac{f_0 \uparrow}{f_e}$ and $\uparrow \theta = \frac{1.22\lambda}{d\uparrow}$

32. (b) R.P. (microscope) $= \frac{D}{1.22\lambda}$

R.P. (microscope) $\propto \frac{}{\lambda}$

$\lambda_1 = 4000\text{ Å},\ \lambda_2 = 6000\text{ Å}$

$\frac{RP_1}{RP_2} = \frac{6000\text{Å}}{4000\text{Å}} = \frac{3}{2}$

33. (d) Resolving power $= \frac{D}{1.22\lambda}$

The angular resolution

$\Delta\phi = \frac{1}{\text{Resolving Power}} = \frac{1.22\lambda}{D}$

$= \frac{1.22 \times 5000 \times 10^{-10}}{0.1} = 6.1 \times 10^{-6} \approx 10^{-6}\text{rad}$

34. (c) For the aperture, limit of resolution

$y \geq \frac{\lambda D}{d}$

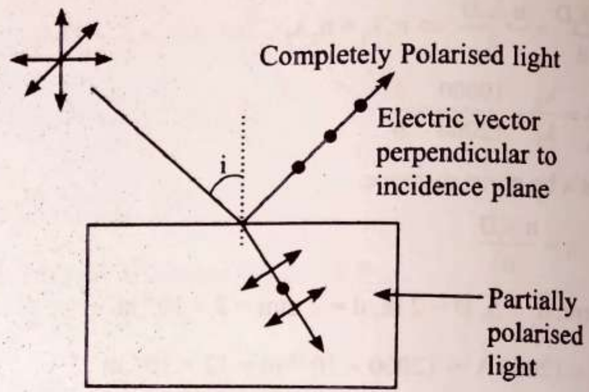
$y \geq \frac{5 \times 10^{-7} \times 50}{2 \times 10^{-3}} \geq 1.25\text{cm}$

35. (b) $\tan i_b = \frac{\mu_2}{\mu_1} = \frac{\mu_2}{1}$

$\therefore \tan i_b > 1$ $[\because \mu_2 > 1]$

$\therefore 90° > i_b > 45°$

36. (b) Reflected light is polarised with its electric vector perpendicular to the plane of incidence.



37. (b) $I = I_0\cos^2(\phi) \Rightarrow \phi = 45°$

$I_1 = \frac{I_0}{2}$

then

$I_2 = I_1\cos^2(\phi)$

$I_2 = \frac{I_0}{2}\cos^2 45° \Rightarrow I_2 = \frac{I_0}{4}$

$I_3 = I_2\cos^2 45°$

$I_3 = \frac{I_0}{8}$

38. (d) Light waves can be polarised as they are transverse. Sound waves can not be polarized as they are longitudinal.

39. (d) Frequency = n; Wavelength = λ; Velocity of air = v and refractive index of glass slab = μ.

Frequency remains the same, when light changes the medium. Refractive index is the ratio of wavelengths in vacuum and in the given medium. Similarly refractive index is also the ratio of velocities in vacuum and in the given medium.

40. (c) λ' of refracted light is smaller, because $\lambda' = \frac{\lambda}{\mu}$

41. (a) $\lambda_g = \frac{\lambda_a}{\mu} = \frac{5460}{1.5} = 3460\text{ Å}$

42. (a) Wavelength $(\lambda) = 5000\text{ Å}$ and velocity (v) = 1.5×10^6 m/s. Wavelength of the approaching star,

$(\lambda') = \lambda\frac{c - v_s}{c}$ or $\frac{\lambda'}{\lambda} = 1 - \frac{v_s}{c}$

or $\frac{v_s}{c} = 1 - \frac{\lambda'}{\lambda} = \frac{\lambda - \lambda'}{\lambda} = \frac{\Delta\lambda}{\lambda}$

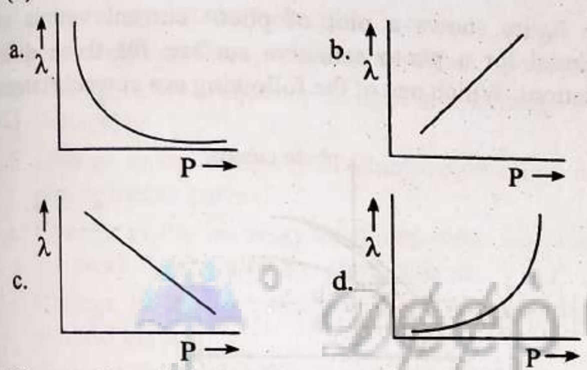
Therefore $\Delta\lambda = \lambda \times \frac{v}{c} = 5000\text{Å} \times \frac{1.5 \times 10^6}{3 \times 10^8} = 25\text{Å}$.

CHAPTER 11

Dual Nature of Radiation and Matter

Photoelectric Effect

1. The graph which shows the variation of the de-Broglie wavelength (λ) of a particle and its associated momentum (P) is: *(2022)*



2. The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of 3.3×10^{-3} watt will be: ($h = 6.6 \times 10^{-34}$ Js) *(2021)*
   a. 10^{17} b. 10^{16}
   c. 10^{15} d. 10^{18}

3. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled? *(2020)*
   a. Four times b. One-fourth
   c. Zero d. Doubled

4. When the light of frequency $2v_0$ (where v_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5v_0$, the maximum velocity of electrons emitted from the same plate is v_2. The ratio of v_1 to v_2 is: *(2018)*
   a. 4 : 1 b. 1 : 4
   c. 1 : 2 d. 2 : 1

5. The photoelectric threshold wavelength of silver is 3250×10^{-10} m. The velocity of the electron ejected from a silver surface by ultraviolet light of wavelength 2536×10^{-10} m is: *(2017-Delhi)*
   (Given $h = 4.14 \times 10^{-15}$ eV and $c = 3 \times 10^8$ ms^{-1})
   a. $\approx 0.6 \times 10^6$ ms^{-1} b. $\approx 61 \times 10^3$ ms^{-1}
   c. $\approx 0.3 \times 10^6$ ms^{-1} d. $\approx 6 \times 10^5$ ms^{-1}

6. When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is V/4. The threshold wavelength for the metallic surface is: *(2016-I)*
   a. 4λ b. 5λ
   c. $\frac{5}{2}\lambda$ d. 3λ

7. Photons with energy 5 eV are incident on a cathode C in a photoelectric cell. The maximum energy of emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A, if the stopping potential of A relative to C is: *(2016-II)*
   a. -1 V b. -3 V
   c. $+3$ V d. $+4$ V

8. A certain metallic surface is illuminated with monochromatic light of wavelength λ. The stopping potential for photo-electric current for this light is $3V_0$. If the same surface is illuminated with light of wavelength 2λ, the stopping potential is V_0. The threshold wavelength for this surface for photoelectric effect is: *(2015)*
   a. 4λ b. $\frac{\lambda}{4}$
   c. $\frac{\lambda}{6}$ d. 6λ

9. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\lambda/2$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is:
   (h = Plank's constant, c = speed of light) *(2015 Re)*
   a. $\frac{hc}{3\lambda}$ b. $\frac{hc}{2\lambda}$
   c. $\frac{hc}{\lambda}$ d. $\frac{2hc}{\lambda}$

10. When the energy of the incident radiation is increased by 20%, the kinetic energy of the photoelectrons emitted from a metal surface increased from 0.5 eV to 0.8 eV. The work function of the metal is: *(2014)*
    a. 0.65 eV b. 1.0 eV
    c. 1.3 eV d. 1.5 eV

11. For photoelectric emission from certain metal the cut-off frequency is ν. If radiation of frequency 2ν impinges on the metal plate, the maximum possible velocity of the emitted electron will be: (m is the electron mass) *(2013)*

a. $2\sqrt{\frac{h\nu}{m}}$ b. $\sqrt{\frac{h\nu}{(2m)}}$
c. $\sqrt{\frac{h\nu}{m}}$ d. $\sqrt{\frac{2h\nu}{m}}$

12. Two radiations of photons energies 1 eV and 2.5 eV, successively illuminate a photosensitive metallic surface of work function 0.5 eV. The ratio of the maximum speeds of the emitted electrons is: *(2012 Mains)*

a. 1 : 4 b. 1 : 2
c. 1 : 1 d. 1 : 5

13. A 200 W sodium street lamp emits yellow light of wavelength 0.6 μm. Assuming it to be 25% efficient in converting electrical energy to light, the number of photons of yellow light it emits per second is. *(2012 Pre)*

a. 1.5×10^{20} b. 6×10^{18}
c. 62×10^{29} d. 3×10^{19}

14. Monochromatic radiation emitted when electron on hydrogen atom jumps from first excited to the ground state irradiates a photosensitive material. The stopping potential is measured to be 3.57 V. The threshold frequency of the materials is: *(2012 Pre)*

a. 4×10^{15} Hz b. 5×10^{15} Hz
c. 1.6×10^{15} Hz d. 2.5×10^{15} Hz

15. The threshold frequency for a photosensitive metal is 3.3×10^{14} Hz. If light of frequency 8.2×10^{14} Hz is incident on this metal, the cut-off voltage for the photoelectric emission is nearly: *(2011 Mains)*

a. 2 V b. 3 V
c. 5 V d. 1 V

16. In photoelectric emission process from a metal of work function 1.8 eV, the kinetic energy of most energetic electrons is 0.5 eV. The corresponding stopping potential is: *(2011 Pre)*

a. 1.8 V b. 1.3 V
c. 0.5 V d. 2.3 V

17. Light of two different frequencies whose photons have energies 1 eV and 2.5 eV respectively illuminate a metallic surface whose work function is 0.5 eV successively. Ratio of maximum speeds of emitted electrons will be: *(2011 Pre)*

a. 1 : 4 b. 1 : 2
c. 1 : 1 d. 1 : 5

18. Photoelectric emission occurs only when the incident light has more than a certain minimum: *(2011 Pre)*

a. Power b. Wavelength
c. Intensity d. Frequency

19. When monochromatic radiation of intensity I falls on a metal surface, the number of photoelectron and their maximum kinetic energy are N and T respectively. If the intensity of radiation is 2I, the number of emitted electrons and their maximum kinetic energy are respectively: *(2010 Mains)*

a. N and 2T b. 2N and T
c. 2N and 2T d. N and T

20. The potential difference that must be applied to stop the fastest photo electrons emitted by a nickel surface, having work function 5.01 eV, when ultraviolet light of 200 nm falls on it, must be: *(2010 Pre)*

a. 1.2 V b. 2.4 V
c. –1.2 V d. –2.4 V

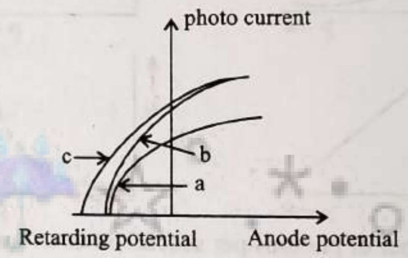
21. A source S_1 is producing 10^{15} photons per second of wavelength 5000 Å. Another source S_2 is producing 1.02×10^{15} photons per second of wavelength 5100 Å Then (power of S_2)/(power of S_1) is equal to: *(2010 Pre)*

a. 0.98 b. 1.00
c. 1.02 d. 1.04

22. Monochromatic light of wavelength 667 nm is produced by a helium neon laser. The power emitted is 9 mW. The number of photons arriving per sec. on the average at a target irradiated by this beam is: *(2009)*

a. 3×10^{16} b. 9×10^{15}
c. 3×10^{19} d. 9×10^{17}

23. The figure shows a plot of photo current versus anode potential for a photo sensitive surface for three different radiations. Which one of the following is a correct statement? *(2009)*



a. Curves (a) and (b) represent incident radiations of same frequency but of different intensities.
b. Curves (b) and (c) represent incident radiations of different frequencies and different intensities.
c. Curves (b) and (c) represent incident radiations of same frequency having same intensity.
d. Curves (a) and (b) represent incident radiations of different frequencies and different intensities.

24. The number of photo electrons emitted for light of a frequency ν (higher than the threshold frequency ν_0) is proportional to: *(2009)*

a. Threshold frequency (ν_0) b. Intensity of light
c. Frequency of light (ν) d. $\nu - \nu_0$

25. The work function of a surface of a photosensitive material is 6.2 eV. The wavelength of the incident radiation for which the stopping potential is 5 V lies in the: *(2008)*

a. X-ray region b. Ultraviolet region
c. Visible region d. Infrared region

26. Monochromatic light of frequency 6.0×10^{14} Hz is produced by a laser. The power emitted is 2×10^{-3} W. The number of photons emitted, on the average, by the sources per second is: *(2007)*

a. 5×10^{16} b. 5×10^{17}
c. 5×10^{14} d. 5×10^{15}

27. A 5 watt source emits monochromatic light of wavelength 5000. When placed 0.5 m away, it liberates photoelectrons from a photosensitive metallic surface. When the source is moved to a distance of 1.0 m, the number of photoelectrons liberated will be reduced by a factor of (2007)

a. 8 b. 16
c. 2 d. 4

28. When photons of energy $h\nu$ fall on an aluminum plate (of work function E_0), photoelectrons of maximum kinetic energy K are ejected. If the frequency of the radiation is doubled, the maximum kinetic energy of the ejected photoelectrons will be: (2006)

a. $K + E_0$ b. 2K
c. K d. $K + h\nu$

29. In a discharge tube ionization of enclosed gas is produced due to collisions between: (2006)

a. Positive ions and neutral atoms/molecules
b. Negative electrons and neutral atoms molecules
c. Photons and neutral atoms/molecules
d. Neutral gas atoms/molecules

30. A photo-cell employs photoelectric effect to convert: (2006)

a. Change in the frequency of light into a change in electric voltage
b. Change in the intensity of illumination into a change in photoelectric current
c. Change in the intensity of illumination into a change in the work function of the photocathode
d. Change in the frequency of light into a change in the electric current

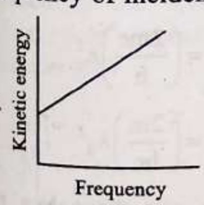
31. A photosensitive metallic surface has work function, $h\nu_0$. If photons of energy $2h\nu_0$ fall on this surface, the electrons come out with a maximum velocity of 4×10^6 m/s. When the photon energy is increased to $5h\nu_0$, then maximum velocity of photoelectrons will be: (2005)

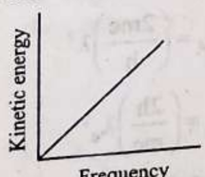
a. 16×10^6 m/s b. 8×10^7 m/s
c. 4×10^5 m/s d. 8×10^6 m/s

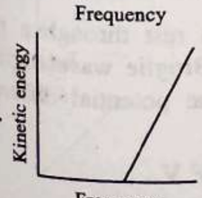
32. The work functions for metals A, B and C are respectively 1.92 eV, 2.0 eV and 5 eV. According to Einstein's equation, the metals which will emit photoelectrons for a radiation of wavelength 4100 Å is/are: (2005)

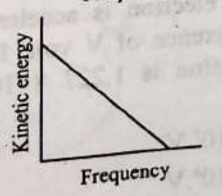
a. None b. A only
c. A and B only d. All the three metals

33. According to Einstein's photoelectric equation, the graph between the kinetic energy of photoelectrons ejected and the frequency of incident radiation is: (2004)

a.


b.


c.


d.


34. A photoelectric cell is illuminated by a point source of light 1 m away. When the source is shifted to 2 m then: (2003)

a. Each emitted electron carries one quarter of the Initial energy
b. Number of electrons emitted is half the initial number
c. Each emitted electron carries half the initial energy
d. Number of electrons emitted is a quarter of the initial number

35. When ultraviolet rays incident on metal plate then photoelectric effect does not occur, it occurs by incidence of: (2002)

a. Infrared rays b. X-rays
c. Radio wave d. Light wave

36. Which of the following is not the property of cathode rays: (2002)

a. It produces heating effect
b. It does not deflect in electric field
c. It casts shadow
d. It produces fluorescence

37. Which one among the following shows particle nature of light: (2001)

a. Photo electric effect b. Interference
c. Refraction d. Polarization

38. A photo-cell is illuminated by a source of light, which is placed at a distance d from the cell. If the distance become d/2, then number of electrons emitted per second will be: (2001)

a. Remain same b. Four times
c. Two times d. One-fourth

39. By photoelectric effect, Einstein proved: (2000)

a. $E = h\nu$ b. $KE = \frac{1}{2}mv^2$
c. $E = mc^2$ d. $E = \frac{-Rhc^2}{n^2}$

40. Who evaluated the mass of electron indirectly with the help of charge: (2000)

a. Thomson b. Millikan
c. Rutherford d. Newton

41. The current conduction in a discharge tube is due to: (1999)

a. Electrons only b. + ve ions and – ve ions
c. (– ve) ions and electrons d. (+ ve) ions, and electrons

42. Light of wavelength 3000 Å in Photoelectric effect gives electron of max. K.E. 0.5 eV. If wavelength change to 2000 Å then max. K.E. of emitted electrons will be: (1999)

a. Less than 0.5 eV b. 0.5 eV
c. Greater than 0.5 eV d. PEE does not occurs

43. If the light of wavelength λ is incident on metal surface, the ejected fastest electron has speed v. If the wavelength is changed to $\frac{3\lambda}{4}$ the speed of the fastest emitted electron will be: (1998)

a. Smaller than $\sqrt{\frac{4}{3}}v$ b. Greater than $\sqrt{\frac{4}{3}}v$
c. 2v d. Zero

44. Work function of a metal surface is $\phi = 1.5$ eV. If a light of wavelength 5000 Å falls on it then the maximum K.E. of ejected electron will be: *(1998)*

a. 1.2 eV b. 0.98 eV
c. 0.45 eV d. 0 eV

45. Which of the following statement is correct? *(1997)*

a. The photocurrent increases with intensity or light
b. The stopping potential increases with increase of incident light
c. The current in photocell increases with increasing frequency
d. The photo current is proportional to the applied voltage.

46. The kinetic energy of an electron, which is accelerated in the potential difference of 100 volts, is *(1997)*

a. 416.6 cal b. 6.636 cal
c. 1.602×10^{-17} J d. 1.6×10^{4} J

47. An electron of mass m and charge e is accelerated from rest through a potential difference V in vacuum. Its final velocity will be *(1996)*

a. $\sqrt{\frac{2eV}{m}}$ b. $\sqrt{\frac{eV}{m}}$
c. $\frac{eV}{2m}$ d. $\frac{eV}{m}$

48. The velocity of photons is proportional to (where υ = frequency) *(1996)*

a. $1/\upsilon$ b. υ^2
c. υ d. $\sqrt{\upsilon}$

49. When light of wavelength 300 nm (nanometer) falls on a photoelectric emitter, photoelectrons are liberated. For another emitter, however, light of 600 nm wavelength is sufficient for creating photoemission. What is the ratio of the work functions of the two emitters? *(1993)*

a. 1 : 2 b. 2 : 1
c. 4 : 1 d. 1 : 4

50. Number of ejected photoelectrons increases with increase *(1993)*

a. In intensity of light b. In wavelength of light
c. In frequency of light d. Never

51. The cathode of a photoelectric cell is changed such that the work function changes from W_1 to W_2 ($W_2 > W_1$). If the current before and after changes are I_1 and I_2, all other conditions remaining unchanged, then (assuming $\nu > W_2$) *(1992)*

a. $I_1 = I_2$ b. $I_1 < I_2$
c. $I_1 > I_2$ d. $I_1 < I_2 < 2I_1$

52. Photoelectric work function of a metal is 1 eV. Light of wavelength $\lambda = 3000$ Å falls on it. The photo electrons come out with a maximum velocity *(1991)*

a. 10 metres/sec b. 10^2 metres/sec
c. 10^4 metres/sec d. 10^6 metres/sec

53. A radio transmitter operates at a frequency 880 kHz and a power of 10 kW. The number of photons emitted per second is *(1990)*

a. 1.72×10^{31} b. 1.327×10^{25}
c. 1.327×10^{37} d. 1.327×10^{45}

54. In which of the following, emission of electrons does not take place *(1990)*

a. Thermionic emission b. X-rays emission
c. Photoelectric emission d. Secondary emission

55. Ultraviolet radiations of 6.2 eV falls on an aluminium surface. Kinetic energy of fastest electron emitted is (work function = 4.2 eV) *(1989)*

a. 3.2×10^{-21} J b. 3.2×10^{-19} J
c. 7×10^{-25} J d. 9×10^{-32} J

56. Thermions are *(1988)*

a. Protons b. Electrons
c. Photons d. Positrons

57. The threshold frequency for photoelectric effect on sodium corresponds to a wavelength of 5000Å. Its work function is *(1988)*

a. 4×10^{-19} J b. 1 J
c. 2×10^{-19} J d. 3×10^{-19} J

Unit Conversion

58. The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly. (2020)

a. 0.6 b. 0.06
c. 0.006 d. 6

Wave Nature of Matter (De-Broglie Wavelength)

59. When two monochromatic light of frequency, ν and ν are incident on a photoelectric metal, their stopping potential becomes $V_s/2$ and V_s respectively. The threshold frequency for this metal is: *(2022)*

a. $\frac{3}{2}\nu$ b. 2ν
c. 3ν d. $\frac{2}{3}\nu$

60. An electromagnetic wave of wavelength 'λ' is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d, then: *(2021)*

a. $\lambda_d = \left(\frac{2mc}{h}\right)\lambda^2$ b. $\lambda = \left(\frac{2mc}{h}\right)\lambda_d^2$
c. $\lambda = \left(\frac{2h}{mc}\right)\lambda_d^2$ d. $\lambda = \left(\frac{2m}{hc}\right)\lambda_d^2$

61. An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is: *(2020)*

a. 10^2 V b. 10^3 V
c. 10^4 V d. 10 V

62. The de Broglie wavelength of an electron moving with kinetic energy of 144 eV is nearly, (*2020-Covid*)
a. 102×10^{-4} nm b. 102×10^{-5} nm
c. 102×10^{-2} nm d. 102×10^{-3} nm

63. An electron is accelerated through a potential difference of 10,000 V. Its de Broglie wavelength is, (nearly) : ($m_e = 9 \times 10^{-31}$ kg) (*2019*)
a. 12.2×10^{-13} m b. 12.2×10^{-12} m
c. 12.2×10^{-14} m d. 12.2 nm

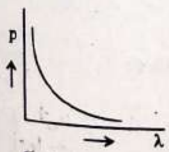
64. An electron of mass m with an initial velocity $\vec{V} = V_0\hat{i}(V_0 > 0)$ enters an electric field $\vec{E} = -E_0\hat{i}(E_0 = \text{constant} > 0)$ at t = 0. If λ_0 is its de-Broglie wavelength initially, then its de-Broglie wavelength at time t is (*2018*)
a. $\lambda_0 t$ b. $\lambda_0\left(1+\frac{eE_0}{mVf_0}t\right)$
c. $\frac{\lambda_0}{\left(1+\frac{eE_0}{mV_0}t\right)}$ d. λ_0

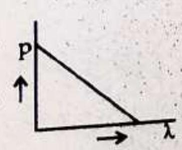
65. The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (Kelvin) and mass m, is: (*2017-Delhi*)
a. $\frac{h}{\sqrt{3mkT}}$ b. $\frac{2h}{\sqrt{3mkT}}$
c. $\frac{2h}{\sqrt{mkT}}$ d. $\frac{h}{\sqrt{mkT}}$

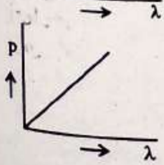
66. An electron of mass m and a photon have same energy E. The ratio of de-Broglie wavelengths associated with them is (c being velocity of light) (*2016-I*)
a. $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ b. $\left(\frac{E}{2m}\right)^{\frac{1}{2}}$
c. $c(2mE)^{\frac{1}{2}}$ d. $\frac{1}{c}\left(\frac{2m}{E}\right)^{\frac{1}{2}}$

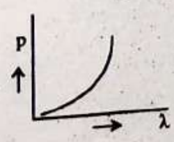
67. Electrons of mass m with de-Broglie wavelength λ fall on the target in an X-ray tube. The cutoff wavelength (λ_0) of the emitted X-ray is: (*2016-II*)
a. $\lambda_0 = \frac{2m^2c^2\lambda^2}{h^2}$ b. $\lambda_0 = \lambda$
c. $\lambda_0 = \frac{2mc\lambda^2}{h}$ d. $\lambda_0 = \frac{2h}{mc}$

68. Which of the following figures represent the variation of particle momentum and the associated de-Broglie wavelength? (*2015*)

a. 

b. 

c. 

d. 

69. If the kinetic energy of the particle is increased to 16 times its previous value, the percentage change in the de-Broglie wavelength of the particle is (*2014*)
a. 25 b. 75
c. 60 d. 50

70. The wavelength λ_e of an electron and λ_p of a photon of same energy E are related by: (*2013*)
a. $\lambda_p \propto \frac{1}{\sqrt{\lambda_e}}$ b. $\lambda_p \propto \lambda_e^2$
c. $\lambda_p \propto \lambda_e$ d. $\lambda_p \propto \sqrt{\lambda_e}$

71. An α-particle moves in a circular path of radius 0.83 cm in the presence of a magnetic field of 0.25 Wb/m². The de Broglie wavelength associated with the particle will be: (*2012 Pre*)
a. 1 Å b. 0.1 Å
c. 10 Å d. 0.01 Å

72. Electrons used in a electron microscope are accelerated by a voltage of 25 kV. If the voltage is increased to 100 kV then the de Broglie wavelength associated with the electrons would: (*2011 Pre*)
a. Increases by 2 times b. Decrease by 2 times
c. Decrease by 4 times d. Increases by 4 times

73. A radioactive nucleus of mass M emits a photon of frequency n and the nucleus recoils. The recoil energy will be: (*2011 Pre*)
a. $Mc^2 - h\nu$ b. $h^2\nu^2/2Mc^2$
c. Zero d. $h\nu$

74. A particle of mass 1 mg has the same wavelength as an electron moving with a velocity of 3×10^6 ms⁻¹. The velocity of the particle is (mass of electron = 9.1×10^{-31} kg): (*2008*)
a. 2.7×10^{-21} ms^{-1} b. 2.7×10^{-18} ms^{-1}
c. 9×10^{-2} ms^{-1} d. 3×10^{-31} ms^{-1}

75. If particles are moving with same velocity, then de-Broglie wavelength is maximum for: (*2002*)
a. Proton b. α-particle
c. Neutron d. β-particle

76. The K.E. of electron and photon is same then relation between their De-Broglie wavelength: (*1999*)
a. $\lambda_p < \lambda_e$ b. $\lambda_p = \lambda_e$
c. $\lambda_p > \lambda_e$ d. $\lambda_p = 2\lambda_e$

77. An electron beam has a kinetic energy equal to 100 eV. Find its wavelength associated with a beam, if mass of electron = 9.1×10^{-31} kg and 1 eV = 1.6×10^{-19} J/eV. (Planck's constant = 6.6×10^{-34} Js). (*1996*)
a. 24.6 Å b. 0.12 Å
c. 1.2 Å d. 6.3 Å

78. An electron of mass m, when accelerated through a potential difference V, has de-Broglie wavelength λ. The de-Broglie wavelength associated with a proton of mass M accelerated through the same potential difference, will be (*1995*)
a. $\frac{M}{m}$ b. $\lambda\frac{m}{M}$
c. $\lambda\sqrt{\frac{M}{m}}$ d. $\lambda\sqrt{\frac{m}{M}}$

79. The de Broglie wave corresponding to a particle of mass m and velocity v has a wavelength associated with it *(1989)*

a. $\frac{h}{mv}$ b. hmv

c. $\frac{mh}{v}$ d. $\frac{m}{hv}$

Parameters of Photon (Momentum, Pressure and Energy)

80. Light of wavelength 5000 nm is incident on a metal with work function 2.28 eV. The de-Broglie wavelength of the emitted electron is: *(2015 Re)*

a. $\leq 2.8 \times 10^{-12}$ m

b. $< 2.8 \times 10^{-10}$ m

c. $< 2.8 \times 10^{-9}$ m

d. $\geq 2.8 \times 10^{-9}$ m

81. If the momentum of an electron is changed by *P*, then the de-Broglie wavelength associated with it changes by 0.5%. The initial momentum of electron will be: *(2012 Mains)*

a. 200 P b. 400 P

c. P/200 d. 100 P

82. The momentum of a photon of energy 1 MeV in kg m/s will be *(2006)*

a. 5×10^{-22} b. 0.33×10^{6}

c. 7×10^{-24} d. 10^{-22}

83. The value of Planck's constant is: *(2002)*

a. 6.63×10^{-34} J/s b. 6.63×10^{-34} kg-m²/s

c. 6.63×10^{-34} kg-m² d. 6.63×10^{-34} J–s⁻¹

84. In a discharge tube at 0.02 mm, there is formation of*(1996)*

a. Crooke's dark space

b. Faraday's dark space

c. Both space partly

d. None of these.

85. If a photon has velocity c and frequency υ, then which of the following represents its wavelength? *(1996)*

a. $\frac{h\upsilon}{c^2}$ b. hυ

c. $\frac{hc}{E}$ d. $\frac{h\upsilon}{c}$

86. If we consider electrons and photons of same wavelength, then they will have same *(1995)*

a. Momentum b. Angular momentum

c. Energy d. Velocity

87. Momentum of photon wavelength λ is *(1993)*

a. $\frac{h\upsilon}{c}$ b. Zero

c. $\frac{hy}{c^2}$ d. $\frac{hy}{c}$

88. The wavelength of a 1 keV photon is 1.24×10^{-9} m. What is the frequency of 1 MeV photon? *(1991)*

a. 1.24×10^{15} b. 2.4×10^{20}

c. 1.24×10^{18} d. 2.4×10^{23}

89. The momentum of a photon of an electromagnetic radiation is 3.3×10^{-29} kg ms⁻¹. What is the frequency of the associated waves? *(1990)*

[$h = 6.6 \times 10^{-34}$ Js ; $c = 3 \times 10^{8}$ ms⁻¹]

a. 1.5×10^{13} Hz b. 7.5×10^{12} Hz

c. 6×10^{3} Hz d. 3×10^{3} Hz

90. The energy of a photon of wavelength λ is *(1988)*

a. hcλ b. $\frac{hc}{\lambda}$

c. $\frac{hc}{\lambda^2}$ d. $\frac{\lambda}{hc}$

Davisson and Germer Experiment

91. The wave nature of electrons was experimentally verified by. *(2020-Covid)*

a. Hertz b. Einstein

c. Davisson and Germer d. de-Broglie

92. In the Davisson and Germer experiment, the velocity of electrons emitted from the electron gun can be increased by *(2011 Pre)*

a. Increasing the potential difference between the anode and filament

b. Increasing the filament current

c. Decreasing the filament current

d. Decreasing the potential difference between the anode and filament

93. The interplanar distance in a crystal is 2.8×10^{-8} m. The value of maximum wavelength which can be diffracted: *(2001)*

a. 2.8×10^{-8} m b. 5.6×10^{-8} m

c. 1.4×10^{-8} m d. 7.6×10^{-8} m

Answer Key

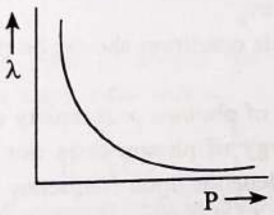
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	c	c	a, d	d	b	a	b	b	d	b	a	c	a	c	b
18	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**
d	b	c	b	a	a	b	b	d	d	d	b	b	d	c	c	d
35	**36**	**37**	**39**	**39**	**40**	**41**	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**	**51**
b	b	a	b	a	b	d	c	b	b	a	c	a	None	b	a	a
52	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**	**61**	**62**	**63**	**64**	**65**	**66**	**67**	**68**
d	a	b	b	b	a	b	None	b	c	d	b	c	a	a	c	a
69	**70**	**71**	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**	**81**	**82**	**83**	**84**	**85**
b	b	d	b	b	b	d	c	c	d	a	d	a	a	b	a	c
86	**87**	**88**	**89**	**90**	**91**	**91**	**92**	**93**								
a	a	b	a	b	c	a	b	a								

Explanations

1. (a) According to de-Broglie's hypothesis

$$\lambda = \frac{h}{p} \Rightarrow \lambda \propto \frac{1}{P}$$

The above equation shows that the de-Broglie wavelength (λ) of a particle is inversely proportional to its momentum. Therefore, the graph of wavelength (λ) and momentum (P) represents a hyperbolic characteristics.



2. (b) Power is defined as energy per unit time i.e.,

$$P = \frac{E}{t}$$

$$\because E = h\nu = \frac{hc}{\lambda}$$

$$\Rightarrow P = \frac{n}{t}\left[\frac{hc}{\lambda}\right]$$ [Say 'n' is no. of photons emitted]

$$\Rightarrow \frac{n}{t} = \frac{P}{\left(\frac{hc}{\lambda}\right)} = \frac{P\lambda}{hc}$$

Here, $\left(\frac{n}{t}\right)$ is number of photons per second.

$$\Rightarrow \frac{n}{t} = \frac{3.3\times10^{-3}\times600\times10^{-9}}{6.6\times10^{-34}\times3\times10^{8}}$$

$$\Rightarrow \frac{n}{t} = \frac{3.3\times6\times10^{-10}}{6.6\times3\times10^{-26}} = 10^{-10+26} = 10^{16}$$

$\Rightarrow n = 10^{16}$ photons per second.

3. (c) According to the questions, $\nu = \frac{3}{2}\nu_0$

When frequency is halved

$$\therefore \nu' = \frac{\nu}{2} = \frac{3}{4}\nu_0$$

Hence, $\nu' < \nu_0$, where ν_0 is threshold frequency

So, no photoelectric emission will take place.

Hence, photoelectric current = zero.

4. (c) From Einstein equation

$$2\,h\nu_0 = h\nu_0 + \frac{1}{2}mv_1^2$$

$$2h\nu_0 = m{v_1}^2 \quad \text{...(i)}$$

If frequency is increased to 5 ν_0 then

$$5h\nu_0 = h\nu_0 + \frac{1}{2}mv_2^2$$

$$4h\nu_0 = \frac{1}{2}mv_2^2 \Rightarrow 8h\nu_0 = mv_2^2 \quad \text{...(ii)}$$

From Eqs. (*i*) and (*ii*)

$$\frac{2h\nu_0}{8h\nu_0} = \frac{mv_1^2}{mv_2^2} \Rightarrow \frac{v_1}{v_2} = \frac{1}{2}$$

5. (a,d) $\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + \frac{1}{2}mv^2$

$$v = \sqrt{\frac{2hc}{m}\left[\frac{1}{\lambda} - \frac{1}{\lambda_0}\right]}$$

$$v = \sqrt{\frac{2\times12400\times1.6\times10^{-19}}{9.1\times10^{-31}}\left[\frac{714}{2536\times3250}\right]}$$

[$\because$ hc = 12400 ev]

$v = 6 \times 10^5$ m/s or 0.6×10^6 m/s

6. **(d)** $eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$ ...(i)

$\frac{eV}{4} = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0}$ ...(ii)

From equation (i) and (ii)

on solving $\lambda_0 = 3\lambda$

7. **(b)** $eV_3 = \frac{1}{2}mv_{max}^2 = h\nu - \phi_0$

$2 = 5 - \phi_0 \Rightarrow \phi_0 = 3$ eV

In second case

$eV_s = 6 - 3 = 3$eV

$\Rightarrow V_s = 3$ V

$\therefore V_{AC} = -3$V

8. **(a)** Using formula, $eV_0 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$

We have, $3eV_0 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$ ...(i)

$eV_0 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$ ...(ii)

On solving (i) and (ii), we get, $\lambda_0 = 4\lambda$

9. **(b)** $KE_1 = \frac{hc}{\lambda} - \phi$

$KE_2 = \frac{hc}{\lambda/2} - \phi = \frac{2hc}{\lambda} - \phi$

$KE_2 = 3KE_1$

$\Rightarrow \frac{2hc}{\lambda} - \phi = 3\left(\frac{hc}{\lambda} - \phi\right)$

$\Rightarrow 2\phi = \frac{hc}{\lambda} \Rightarrow \phi = \frac{hc}{2\lambda}$

10. **(b)** By using, $h\nu = \phi_0 + K_{max}$

We have,

$h\nu = \phi_0 + 0.5$ ...(i)

And, $1.2h\nu = \phi_0 + 0.8$ ...(ii)

On solving (i) and (ii), $\phi_0 = 1.0$ eV

11. **(d)** Using formula, $KE = h\nu - \phi_0$

We have, $\frac{1}{2}mv_{max}^2 = h(2\nu) - h\nu$ [$\because$ Given, $\nu_0 = \nu$]

$2h(2\nu) = h\nu + \frac{1}{2}mv_{max}^2 \Rightarrow v_{max} = \sqrt{\frac{2h\nu}{m}}$

12. **(b)** $\frac{1}{2}mv_{max}^2 = E - \phi$

$\frac{1}{2}mv_{1max}^2 = (1 - 0.5)\text{eV} = 0.5$ ev ...(i)

$\frac{1}{2}mv_{2max}^2 = (2.5 - 0.5)\text{eV} = 2$ ev ...(ii)

On solving (i) and (ii)

$\Rightarrow \frac{v_{1max}^2}{v_{2max}^2} = \frac{1}{4} \Rightarrow \frac{v_{1max}}{v_{2max}} = \frac{1}{2}$

13. **(a)** Number of photons per second $= \frac{\text{Power}}{\text{Energy of photon}}$

$\therefore \frac{n}{t} = \frac{P}{\frac{hc}{\lambda}} = \frac{200 \times \frac{25}{100}}{\frac{6.6 \times 10^{-34} \times 3 \times 10^8}{0.6 \times 10^{-6}}} = 1.5 \times 10^{20}$

14. **(c)** Work function, $\phi = h\nu - eV_0 = 10.2 - 3.57 = 6.63$ eV

Threshold frequency

$= \frac{6.63 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} = 1.6 \times 10^{15}$ Hz

15. **(a)** According to Einstein photoelectric equation,

$eV_0 = h\nu - h\nu_0 \Rightarrow V_0 = \frac{h\nu - h\nu_0}{e} = \frac{h}{e}(\nu - \nu_0)$

$= \frac{6.63 \times 10^{-34}}{1.6 \times 10^{-19}}(8.2 \times 10^{14} - 3.3 \times 10^{14}) \approx 2$V

16. **(c)** Stopping potential $= K_{max}$

$eV_0 = K.E \Rightarrow 0.5 \text{ eV} = eV_0 \Rightarrow V_0 = 0.5$ V

17. **(b)** Einstein's equation

$K_{max} = h\nu - \phi_0 \Rightarrow K_{max_1} = 1 - 0.5 = 0.5$ ev

& $K_{max_2} = 2.5 - 0.5 = 2$ ev

Now, $\frac{K_{max_1}}{K_{max_2}} = \frac{0.5}{2} \Rightarrow \frac{\frac{1}{2}Mv_1^2}{\frac{1}{2}Mv_2^2} = \frac{1}{4} \Rightarrow \left(\frac{v_1}{v_2}\right)^2 = \frac{1}{4} \Rightarrow \frac{v_1}{v_2} = \frac{1}{2}$

18. **(d)** Einstein's photoelectric equation

$K_{max} = h\nu - h\nu_0$

$\nu > \nu_0 \rightarrow$ This condition should be present for photoelectric emission

19. **(b)** Number of photons $\propto$ Intensity of light

Kinetic energy of photon does not depend on intensity of light but it depends upon frequency of incident light.

20. **(c)** $h\nu = \phi_0 + eV_0$ where $h\nu = \frac{12400}{2000} = 6.2$ eV

$\Rightarrow V_0 = 6.2 - 5.01 = 1.19 \approx 1.20$ V

Thus the potential difference that must be applied to stop photoelectrons $= -V_0 = -1.2$ V

21. **(b)** $\frac{\text{Power of } S_2}{\text{Power of } S_1} = \frac{n_2\left(\frac{hc}{\lambda_2}\right)}{n_1\left(\frac{hc}{\lambda_1}\right)}$ (Put values and solve)

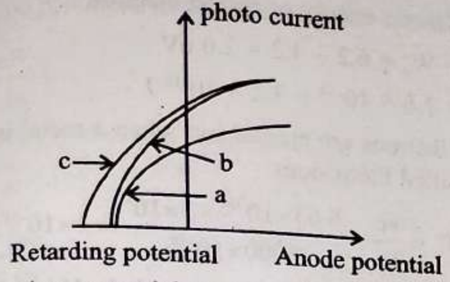
$= \frac{n_2\lambda_1}{n_1\lambda_2} = 1$

22. **(a)** No. of photons per second

$= \frac{\text{Power}}{\text{Energy of each photon}} = \frac{P}{\left(\frac{hc}{\lambda}\right)}$

$\Rightarrow \frac{n}{t} = \frac{9 \times 10^{-3} \times 667 \times 10^{-9}}{6.63 \times 10^{-34} \times 3 \times 10^8} = 3 \times 10^{16}$

23. (a)



Here, stopping potential is same for both but saturation current is different which depends on intensity therefore, (a) and (b) have different intensities.

24. (b) No of photoelectrons emitted of intensity of light.

25. (b) Stopping potential = Incident energy-work function

$\Rightarrow$ Incident energy = 6.2 + 5 = 11.2 eV

$\Rightarrow \frac{12400}{\lambda} = 11.2$ [Here, hc = 12400, λ in Å]

$\Rightarrow \lambda = 1107\ \text{Å} = 110.7$ nm

This wavelength lies in ultraviolet region.

26. (d) Let the number of photons emitted, on the average, by the source per second be N

Power emitted $= \frac{\text{Energy}}{\text{Time}}$

$\Rightarrow P = Nh\nu \Rightarrow 2 \times 10^{-3} = N \times 6.62 \times 10^{-34} \times 6 \times 10^{14}$

$\Rightarrow N = 5 \times 10^{15}$

27. (d) For a light source of power P watt, the intensity at a distance d is given by

$$I = \frac{P}{4\pi d^2}$$

Where we assume light to spread out uniformly in all directions, i.e., it is a spherical source.

$\therefore I \propto \frac{1}{d^2}$ or $\frac{I_1}{I_2} = \frac{d_2^2}{d_1^2} \Rightarrow \frac{I_1}{I_2} = \left(\frac{1}{0.5}\right)^2 \Rightarrow \frac{I_1}{I_2} = 4$ or, $I_2 = \frac{I_1}{4}$

When a intensity of source is reduced by a factor of four, the number of photoelectrons is also reduced by a factor of 4.

28. (d) As we know that

K = Incident Energy –Work function

$K = h\nu - E_0$ ...(*i*)

Now, the frequency is doubled,

$K' = h(2\nu) - E_0$

$= 2h\nu - E_0$

$= h\nu + (h\nu - E_0)$

$= h\nu + K$

Hence, New kinetic energy will be K + hν.

29. (b) In the discharge tube, electrons travelling towards anode collide with the neutral atoms/ molecules present in the tube and produce ionized gas molecules.

30. (b) Photo-cell employs photoelectric effect to change the intensity of illumination into a change in photoelectric current.

31. (d) When photons of energy $2h\nu_0$ fall on the surface,

$K.E = h\nu - W \Rightarrow \frac{1}{2}mv_{1\,max}^2 = 2h\nu_0 - h\nu_0$ ...(*i*)

$\Rightarrow \frac{1}{2}m(4\times10^6)^2 = h\nu_0$

Now, incident energy changed from $2h\nu_0$ to $5h\nu_0$.

$\Rightarrow \frac{1}{2}mv_{1\,max}^2 = 4h\nu_0$

Putting value of $h\nu_0$

$\Rightarrow \frac{1}{2}mv_{1\,max}^2 = 4 \times \frac{1}{2}m\left(4\times10^6\right)^2$

$\Rightarrow v_{1\,max} = 2\times4\times10^6 = 8\times10^6$ m/s

32. (c) Work function, $\phi_A = 1.92$ eV, $\phi_B = 2.0$ eV, $\phi_C = 5$ eV

Incident energy $= \frac{hc}{\lambda} = \frac{12400}{4100}\text{Å} = 3.03$ eV

As energy of incident radiation is nearly equal to 3.0 eV, thus electrons are emitted only from metal A and B which have work function lower than incident energy.

33. (c) Einstein's equation

$hf = hf_0 + E \Rightarrow hf = \phi + E$

$E = hf - \phi \Rightarrow y = mx + c$

y = E, slope (m) = h, c = – ϕ

34. (d) For a point source $I \propto \frac{1}{r^2}$

35. (b) When ultraviolet rays fail to emit photo-electrons, rays like X-rays with higher frequency should be used.

36. (b) Cathode rays are basically negatively charged particles (electrons). If the cathode rays are allowed to pass between two plates kept at a difference of potential, the rays are found to be deflected from the rectilinear path. The direction of deflection shows that the rays carry negative charges.

37. (a) Photoelectric effect shows particle nature of light

38. (b) Intensity becomes 4 times. So number increases.

39. (a) With the help of photoelectric effect Einstein showed that E = hν.

40. (b) Millikan evaluated the mass of electron indirectly with the help of charge.

41. (d) Current conduction in a discharge tube is due to electrons and positive ions.

42. (c) $K.E._{max.} = \frac{hc}{\lambda} - \phi$

Then K.E. will be greater than 0.5 eV

43. (b) From Einstein's photoelectric effect equation

$$\frac{hc}{\lambda} = \phi_0 + \frac{1}{2}mv^2$$

When wavelength is changed to $\frac{3\lambda}{4}$

$$\frac{4hc}{3\lambda} = \phi_0 + \frac{1}{2}mv_1^2 \Rightarrow \frac{4}{3}\left(\phi_0 + \frac{1}{2}mv^2\right) = \phi_0 + \frac{1}{2}mv_1^2$$

$$\Rightarrow \frac{1}{2}mv_1^2 = \frac{\phi_0}{3} + \frac{1}{2}m\left(\sqrt{\frac{4}{3}}v\right)^2 \Rightarrow \boxed{v_1 > \sqrt{\frac{4}{3}}v}$$

44. (b) $K.E_{max} = \frac{hc}{\lambda} - \phi = \frac{12400\ eV}{5000\ \mathring{A}} - 1.5\ eV = (2.48 - 1.5)\ eV = 0.98\ eV$

45. (a) Since the emission of photoelectrons is directly proportional to the intensity of the incident light, therefore photo current increases with the intensity of light.

46. (c) Potential difference (V) = 100 volts.

Kinetic energy of an electron (K.E.) = eV = $(1.6 \times 10^{-19}) \times 100 = 1.6 \times 10^{-17}$ joule.

47. (a) The kinetic energy of an electron

$$\frac{1}{2} \times mv^2 = eV$$

Or final velocity of electron, $(v) = \sqrt{\frac{2eV}{m}}$.

48. (None) The velocity of a photon in vacuum is a constant, $c = \upsilon\lambda$. But c = constant and one cannot say that it is proportional to υ or λ but only $c = \upsilon\lambda$.

In media, for a particular medium, υ remain the same, velocity changes. Therefore λ changes.

49. (b) $W_0 = \frac{hc}{\lambda_0}$ or $W_0 \propto \frac{1}{\lambda_0}$; $\Rightarrow \frac{W_1}{W_2} = \frac{\lambda_2}{\lambda_1} = \frac{600}{300} = 2$

50. (a) Photoelectric current is directly proportional to the intensity of incident light.

51. (a) Photoelectric current is not affected by the work function as long as $h\nu > W_0$. The photoelectric current is proportional to the intensity of incident light. Since there is no change in the intensity of light, hence $I_1 = I_2$.

52. (d) $h\nu = W + \frac{1}{2}mv^2$ or $\frac{hc}{\lambda} = W + \frac{1}{2}mv^2$

Here $\lambda = 3000\ \mathring{A} = 3000 \times 10^{-10}$ m

And W = 1eV = 1.6×10^{-19} joule

$$\frac{hc}{\lambda} = \frac{(6.6\times10^{-34})(3\times10^{8})}{3000\times10^{-10}}$$

$$\frac{hc}{\lambda} = (1.6\times10^{-19}) + \frac{1}{2}\times(9.1\times10^{-31})v^2$$

Solving we get $v \cong 10^6$ m/s

53. (a) No. of photons emitted per sec,

$$\frac{n}{t} = \frac{\text{Power}}{\text{Energy of photon}}$$

$$= \frac{P}{h\nu} = \frac{10000}{6.6\times10^{-34}\times880\times10^{3}} = 1.72\times10^{31}$$

54. (b) X-rays emission: They are due to transitions in the inner energy levels of the atom.

Photoelectric emission: Emission of electrons from the metal surface on irradiation with radiation of suitable frequency.

Secondary emission: When an electron strikes the surface of a metallic plate, it emits other electrons from the surface.

Thermionic emission: When a metal is heated to a high temperature, the free electron gain kinetic energy and escape from the surface of the metal

55. (b) Kinetic energy of fastest electron

$= E - W_0 = 6.2 - 4.2 = 2.0$ eV

$= 2 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-19}$ J

56. (b) Electrons are ejected out when a metal is heated, which are called therm-ions.

57. (a) $W_0 = \frac{hc}{\lambda_0} = \frac{6.63\times10^{-34}\times3\times10^{8}}{5000\times10^{-10}} = 4\times10^{-19}$ J

58. (b) As we know one electron volt 1 eV is 1.6×10^{-19}J

$$1J = \frac{1}{1.6\times10^{-19}} eV$$

$$10^{-20} J = \frac{10^{-20}}{1.6\times10^{-19}} eV = 0.06\,eV$$

59. (None) Using the equation

$$\frac{eV_s}{2} = h\nu - \phi \quad ...(i)$$

$$eV_s = \frac{h\nu}{2} - \phi \quad ...(ii)$$

This equation cannot be solved using given data.

60. (b) Using Einstein's photoelectric equation;

$$\frac{hc}{\lambda} = K_{max} + \phi$$

Given, $\phi = 0$ then,

$$\frac{hc}{\lambda} = K_{max} \Rightarrow \frac{hc}{\lambda} = \frac{P^2}{2m} \Rightarrow P = \left(\frac{2mhc}{\lambda}\right)^{\frac{1}{2}}$$

Using De-broglie wavelength,

$$\lambda_d = \frac{h}{P} = \frac{h}{\left(\frac{2mhc}{\lambda}\right)^{\frac{1}{2}}} \Rightarrow \lambda_d = \left(\frac{h\lambda}{2mc}\right)^{\frac{1}{2}}$$

$$\Rightarrow \lambda_d^2 = \frac{h\lambda}{2mc} \Rightarrow \lambda = \left(\frac{2mc}{h}\right)\lambda_d^2$$

61. (c) De-broglie wavelength of the electron

$\lambda = \frac{12.27}{\sqrt{V}}\mathring{A}$ where, V is potential difference

$$\sqrt{V} = \frac{12.27\times10^{-10}}{1.227\times10^{-11}} = 10^2$$

$\therefore V = 10^4$ volts

62. (d) $\lambda = \frac{12.27}{\sqrt{V}}\mathring{A} = \frac{12.27}{\sqrt{144}}\mathring{A} = 1.02\times10^{-10} m = 102\times10^{-3} nm$

63. (b) For an electron accelerated through a potential V

De Broglie wavelength $(\lambda) = \frac{12.27}{\sqrt{V}}\mathring{A} = \frac{12.27\times10^{-10}}{\sqrt{10000}}$

$= 12.27\times10^{-12}$ m

64. (c) Here $\vec{E} = -E_0\hat{i}$, initial velocity $\vec{v} = v_0\hat{i}$

Force action on electron due to electric field.

$\vec{F} = (-e)(-E_0\hat{i}) = eE_0\hat{i}$

Acceleration produced in the electron,

$$\vec{a}=\frac{\vec{F}}{m}=\frac{eE_0}{m}\hat{i}$$

Now, velocity of electron after time t,

$$\vec{v}=\vec{v}_0+\vec{a}t$$

$$\Rightarrow v=v_0+\frac{eE_0}{m}t$$

$$\Rightarrow \lambda=\frac{h}{m\left(v_0+\frac{eE_0}{m}t\right)}=\frac{h}{(mv_0+eE_0t)}=\frac{\lambda_0}{\left(1+\frac{eE_0}{mv_0}t\right)}$$

65. (a) At equilibrium thermal energy possessed by neutron $=\frac{3}{2}kT$

$$\lambda=\frac{h}{p}=\frac{h}{\sqrt{2mE}}\Rightarrow\frac{h}{\sqrt{2m\frac{3}{2}kT}}$$

$$\lambda=\frac{h}{\sqrt{3mkT}}$$

66. (a) De-Broglie wavelength $=\frac{h}{mv}$

m = Mass of particle

v = Velocity of particle

h = Planks constant

$E=\frac{1}{2}mv^2$; (Mass of electron = m)

$$\sqrt{\frac{2E}{m}}=v$$

$$\lambda_E=\frac{h}{m\sqrt{\frac{2E}{m}}}=\frac{h}{\sqrt{2mE}}$$

Now for photon, $E=Pc\Rightarrow P=\frac{E}{c}$

$$\lambda_p=\frac{h}{mv}=\frac{h}{P}=\frac{hc}{E}$$

$$\lambda_E/\lambda_P=\frac{h}{\sqrt{2mE}}\frac{E}{hc}=\frac{1}{c}\sqrt{\frac{E}{2m}}$$

67. (c) $\lambda=\frac{h}{P}\quad\Rightarrow\quad P=\frac{h}{\lambda}$

K.E. of electrons $=E=\frac{P^2}{2m}=\frac{h^2}{2m\lambda^2}$

Also in X-ray, $\lambda_0=\frac{hc}{E}\Rightarrow\lambda_0=\frac{2mc\lambda^2}{h}$

68. (a) $P=\frac{h}{\lambda}\Rightarrow P\propto\frac{1}{\lambda}$ (Rectangular hyperbola)

69. (b) $\lambda=\frac{h}{\sqrt{2mK}}$

$$\frac{\lambda_1}{\lambda_2}=\sqrt{\frac{K_2}{K_1}}=\sqrt{\frac{16K}{K}}=\frac{4}{1}$$

Percentage change $=\frac{4-1}{4}\times100=75\%$

70. (b) $\lambda_p=\frac{h}{P}=\frac{hc}{E}$ and $\lambda_e=\frac{h}{P}=\frac{h}{\sqrt{2mE}}\Rightarrow\lambda_p\propto\lambda_e^2$

71. (d) $\lambda=\frac{h}{P}=\frac{h}{mv}=\frac{h}{qBR}$

$$=\frac{6.6\times10^{-34}}{2\times1.6\times10^{-19}\times0.25\times0.83\times10^{-2}}=1\times10^{-12}\,m=0.01\,Å$$

72. (b) As we know

$\lambda=\frac{1.227}{\sqrt{V}}$ nm, where V is potential difference

$$\lambda\propto\frac{1}{\sqrt{V}}$$

So, $\frac{\lambda_1}{\lambda_2}=\sqrt{\frac{V_2}{V_1}}\Rightarrow\frac{\lambda_1}{\lambda_2}=\sqrt{\frac{100}{25}}\Rightarrow\frac{\lambda_1}{\lambda_2}=2\Rightarrow\lambda_2=\frac{\lambda_1}{2}$

So, wavelength of electrons decrease by 2 times

73. (b) Using law of conservation of momentum

$$P_{nucleus}=P_{photon}$$

$$Mu=\frac{E}{c}=\frac{h\nu}{c}$$

Recoil energy

$$\frac{1}{2}Mu^2=\frac{1}{2}\frac{M^2u^2}{M}=\frac{1}{2M}\left(\frac{h\nu}{c}\right)^2$$

where u is recoil speed of nucleus

Recoil K.E. of nucleus $=\frac{h^2\nu^2}{2Mc^2}$

74. (b) $\because\lambda=\frac{h}{mv}=\frac{h}{p}$

Since the wavelength of particle and electron are equal, momentum of electron should be equal to momentum of particle

$\Rightarrow m_1v_1=m_2v_2\Rightarrow 1mg\times v_1=9.1\times10^{-31}\times3\times10^6$

$\Rightarrow 10^{-3}\times v_1=3\times10^6\times9.1\times10^{-31}$

$v_1=2.7\times10^{-18}$ m/s

75. (d) $\lambda_{max}=\frac{h}{p}$

$$\lambda_m=\frac{h}{mv}$$

$$\lambda_m\propto\frac{1}{m}$$

β-particle consists of electrons which is lightest of other particles. Hence, λ is maximum for β-particle.

76. (c) $(K.E.)_e=E_{photon}$

$$\frac{1}{2}mv^2=\frac{hc}{\lambda_{photon}}\Rightarrow\frac{1}{2}\left(\frac{h}{\lambda_e v}\right)v^2=\frac{hc}{\lambda_{photon}}$$

$$\frac{\lambda_e}{\lambda_{photon}}=\frac{v}{2c}$$

$c>v$, $\lambda_{photon}>\lambda_e$

77. **(c)** Kinetic energy (E) = 100 eV;

Mass of electron (m) = 9.1×10^{-31} kg;

1 eV = 1.6×10^{-19} J and

Planck's constant (h) = 6.6×10^{-34} J–s.

Energy of an electron (E) = $100 \times (1.6 \times 10^{-19})$ J

Or

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.6\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times100\times1.6\times10^{-19}}}$$

$= 1.2 \times 10^{-10}$ m = 1.2 Å.

78. **(d)** Momentum of electrons $(P_e) = \sqrt{2meV}$ and momentum for proton $(P_p) = \sqrt{2MeV}$

Therefore,

$$\frac{\lambda_p}{\lambda_e} = \frac{h/p_p}{h/p_e} = \frac{p_e}{p_p} = \frac{\sqrt{2meV}}{\sqrt{2MeV}} = \sqrt{\left(\frac{m}{M}\right)}.$$

Therefore, $\lambda_p = \lambda_e\sqrt{\left(\frac{m}{M}\right)} = \lambda\sqrt{\left(\frac{m}{M}\right)}$ $[\because \lambda_e = \lambda]$

79. **(a)** de Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{mv}$.

80. **(d)** Energy of photon $(E) = \frac{12400}{5000} = 2.48$ eV

Work function $(\phi_0) = 2.28$ eV

According to Einstein equation

$E = \phi_0 + (K.E.)_{max} \Rightarrow 2.48 = 2.28 + (K.E.)_{max}$

$\Rightarrow (K.E.)_{max} = 0.20$ eV

For electron $\lambda_{min} = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda_{min} \approx 28\,\text{Å}$

So $\lambda \geq 2.8 \times 10^{-9}$ m

81. **(a)** $p \propto \frac{1}{\lambda} \Rightarrow \frac{\Delta p}{p} = -\frac{\Delta\lambda}{\lambda} \Rightarrow \frac{P}{P_{inital}} = \frac{0.5}{100} \Rightarrow P_{initial} = 200$ P

82. **(a)** Energy of photon, E = 1 MeV

Momentum of photon, $P = \frac{E}{c}$

$\therefore\ P = \frac{E}{c} = \frac{1\times10^6\times1.6\times10^{-19}\,J}{3\times10^8\ ms^{-1}} = 0.53\times10^{-21}$

83. **(b)** From Einstein's equation, E = hν

h = 6.63×10^{-34} kg–m²/s

Value based on experiments.

84. **(a)** Crooke's dark space is formed in a discharge tube at 0.02 mm.

85. **(c)** Energy of the photon, $E = \frac{hc}{\lambda}$ or $\lambda = \frac{hc}{E}$, where λ is the wavelength.

86. **(a)** Wavelength, $\lambda = \frac{h}{mv} = \frac{h}{p}$. Therefore same wavelength of electrons and photons, the momentum should be same.

87. **(a)** Momentum of the photon $= \frac{h\upsilon}{c}$

88. **(b)** Here, $\frac{hc}{\lambda} = 10^3$ eV and $h\upsilon = 10^6$ eV

Dividing both the equations

Hence, $\upsilon = \frac{10^3 c}{\lambda} = \frac{10^3\times3\times10^8}{1.24\times10^{-9}}$

$= 2.4 \times 10^{20}$ Hz

89. **(a)** No. of photons emitted per sec,

$\Rightarrow \frac{c}{\upsilon} = \frac{h}{p} = \lambda$

$\upsilon = \frac{c}{\lambda} = \frac{cp}{h} = 3\times10^8 \times \frac{3.3\times10^{-29}}{6.6\times10^{-34}}$

$= 1.5 \times 10^{13}$ Hz

Where, υ = frequency of radiation

90. **(b)** Energy of a photon, $E = h\upsilon = \frac{hc}{\lambda}$.

91. **(c)** Wave nature of electrons was experimentally verified by Davisson and Germer.

92. **(a)** The velocity of electrons emitted from the electron gun can be increased by Increasing the potential difference between the anode and filament.

93. **(b)** $2d\sin\theta = n\lambda$

$\because\ -1 \leq \sin\theta \leq 1$

Therefore $\lambda_{max} = 2d$

$\Rightarrow \lambda_{max} = 2 \times 2.8 \times 10^{-8}$ m $\Rightarrow \lambda_{max} = 5.6 \times 10^{-8}$ m

Chapter 12

Atoms

Alpha-Particle Scattering and Rutherford's Nuclear Model of Atom (Distance of Closest Approach and Impact Parameter, Electron Orbits)

1. When an α particle of mass m moving with velocity v bombards on a heavy nucleus of charge 'Ze', its distance of closest approach from the nucleus depends on mass: *(2016 - I)*

a. $\frac{1}{m}$ b. $\frac{1}{\sqrt{m}}$

c. $\frac{1}{m^2}$ d. m

2. An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to: *(2010 Pre)*

a. $1/v^4$ b. $1/Ze$

c. v^2 d. $1/m$

3. In a Rutherford scattering experiment, when a projectile of charge z_1 and mass M_1 approaches a target nucleus of charge z_2 and mass M_2, the distance of closest approach is r_0. The energy of the projectile is *(2009)*

a. Directly proportional to z_1z_2

b. Inversely proportional to z_1

c. Directly proportional to mass M_1

d. Directly proportional to $M_1 \times M_2$

Bohr H-atom Model, Radius of Orbit, Velocity and Energy of Electrons, Wavelengths of Hydrogen Spectrum and Ionisation Potential

4. Let T_1 and T_2 be the energy of an electron in the first and second excited states of hydrogen atom, respectively. According to the Bohr's model of an atom, the ratio $T_1 : T_2$ is : *(2022)*

a. 9 : 4 b. 1 : 4

c. 4 : 1 d. 4 : 9

5. For which one of the following, Bohr's model is not valid? *(2020)*

a. Singly ionised helium atom (He^+)

b. Deuteron atom

c. Singly ionised neon atom (Ne^+)

d. Hydrogen atom

6. The total energy of an electron in the n^{th} stationary orbit of the hydrogen atom can be obtained by. *(2020-Covid)*

a. $E_n = -\frac{13.6}{n^2}eV$ b. $E_n = -\frac{1.36}{n^2}eV$

c. $E_n = -13.6 \times n^2 eV$ d. $E_n = \frac{13.6}{n^2}eV$

7. The total energy of an electron in n atom in an orbit is –3.4 eV. Its kinetic and potential energies are, respectively: *(2019)*

a. –3.4 eV, –3.4 eV b. –3.4 eV, –6.8 eV

c. 3.4 eV, –6.8 eV d. 3.4 eV, 3.4 eV

8. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom, is *(2018)*

a. 2 : –1 b. 1 : –1

c. 1 : 1 d. 1 : –2

9. The ratio of wavelengths of the last line of Balmer series and the last line of Lyman series is: *(2017-Delhi)*

a. 1 b. 4

c. 0.5 d. 2

10. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it emits a photon of wavelength λ. When it jumps from the 4th orbit to the 3rd orbit, the corresponding wavelength of the photon will be: *(2016 - II)*

a. $\frac{20}{7}\lambda$ b. $\frac{20}{13}\lambda$

c. $\frac{16}{25}\lambda$ d. $\frac{9}{16}\lambda$

11. Given the value of Rydberg constant is $10^7 m^{-1}$, the wave number of the last line of the Balmer series in hydrogen spectrum will be: *(2016 - I)*

a. $0.025 \times 10^4 m^{-1}$ b. $0.5 \times 10^7 m^{-1}$

c. $0.25 \times 10^7 m^{-1}$ d. $2.5 \times 10^7 m^{-1}$

12. Consider 3rd orbit of He^+ (Helium) using non relativistic approach the speed of electron in this orbit will be (given $K = 9 \times 10^9$ constant $Z = 2$ and h (Planck's constant) $= 6.6 \times 10^{-34}$ Js): *(2015)*

a. 1.46×10^6 m/s b. 0.73×10^6 m/s
c. 3.0×10^8 m/s d. 2.92×10^6 m/s

13. In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is: *(2015 Pre)*

a. 5/27 b. 4/9
c. 9/4 d. 27/5

14. Hydrogen atom in ground state is excited by a monochromatic radiation of $\lambda = 975$ Å. Number of spectral lines in the resulting spectrum emitted will be: *(2014)*

a. 3 b. 2
c. 6 d. 10

15. Ratio of longest wavelengths corresponding to Lyman and Balmer series in hydrogen spectrum is: *(2013)*

a. 9/31 b. 5/27
c. 3/ 23 d. 7/29

16. The transition from the state n = 3 to n = 1 in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition from: *(2012 Mains)*

a. $2 \to 1$ b. $3 \to 2$
c. $4 \to 3$ d. $4 \to 2$

17. The transition from the state n = 3 to n = 1 in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition from *(2012 Mains)*

a. $2 \to 1$ b. $3 \to 2$
c. $4 \to 2$ d. $4 \to 3$

18. An electron of a stationary hydrogen atom passes from the fifth energy level to the ground level. The velocity that the atom acquired as a result of photon emission will be *(2012 Pre)*

a. $\frac{24hR}{25m}$ b. $\frac{25hR}{24m}$
c. $\frac{25m}{24hR}$ d. $\frac{24m}{25hR}$

(m is the mass of electron, R, Rydberg constant and h Planck's constant)

19. Electron in hydrogen atom first jumps from third excited state to second excited state and then from second excited to the first excited state. The ratio of the wavelength $\lambda_1 : \lambda_2$ emitted in the two cases is: *(2012 Pre)*

a. 7/5 b. 27/20
c. 27/5 d. 20/7

20. Out of the following which one is not a possible energy for a photon to be emitted by hydrogen atom according to Bohr's atomic model? *(2011 Mains)*

a. 1.9 eV b. 11.1 eV
c. 13.6 eV d. 0.65 eV

21. The wavelength of the first line of Lyman series for hydrogen atom is equal to that of the second line of Balmer series for a hydrogen like ion. The atomic number Z of hydrogen like ion is: *(2011 Pre)*

a. 3 b. 4
c. 1 d. 2

22. The electron in the hydrogen atom jumps from excited state (n = 3) to its ground state (n = 1) and the photons thus emitted irradiate a photosensitive material. If the work function of the material is 5.1 eV, the stopping potential is estimated to be: (the energy of the electron in n^{th} state) *(2010 Mains)*

a. 5.1 V b. 12.1 V
c. 17.2 V d. 7 V

23. The energy of a hydrogen atom in the ground state is −13.6 eV. The energy of a He^+ ion in the first excited state will be: *(2010 Pre)*

a. −6.8 eV b. −13.6 eV
c. −27.2 eV d. −54.4 eV

24. The ionisation energy of the electron in the hydrogen atom in its ground state is 13.6 eV. The atoms are excited to higher energy levels to emit radiations of 6 wavelengths. Maximum wavelength of emitted radiation corresponds to the transition between: *(2009)*

a. n = 3 to n = 1 states b. n = 2 to n = 1 states
c. n = 4 to n = 3 states d. n = 3 to n = 2 states

25. The ground state energy of hydrogen atom is −13.6 eV. When its electron is in the first excited state, its excitation energy is: *(2008)*

a. 0 b. 3.4 eV
c. 6.8 eV d. 10.2 eV

26. The total energy of electron in the ground state of hydrogen atom is −13.6 eV. The kinetic energy of an electron in the first excited state is: *(2007)*

a. 6.8 eV b. 13.6 eV
c. 1.7 eV d. 3.4 eV

27. Ionisation potential of hydrogen atom is 13.6 eV. Hydrogen atoms in the ground state are excited by monochromatic radiation of photon energy 12.1 eV. According to Bohr's theory, the spectral lines emitted by hydrogen will be: *(2006)*

a. Two b. Three
c. Four d. One

28. The total energy of an electron in the first excited state of hydrogen is about −3.4 eV. Its kinetic energy in this state is: *(2005)*

a. −3.4 eV b. −1.7 eV
c. 1.7 eV d. 3.4 eV

29. Energy levels A, B and C of a certain atom correspond to increasing values of energy i.e., $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are wavelengths of radiations corresponding to transitions C to B, B to A and C to A respectively, which of the following relations is correct? *(2005)*

a. $\lambda_3 = \lambda_1 + \lambda_2$ b. $\lambda_3 = (\lambda_1\lambda_2)/(\lambda_1 + \lambda_2)$
c. $\lambda_1 + \lambda_2 + \lambda_3 = 0$ d. $\lambda^2_3 = \lambda^2_1 + \lambda^2_2$

30. Energy E of a hydrogen atom with principal quantum number n is given by $\frac{-13.6}{n^2}$eV. The energy of a photon ejected when the electron jumps from n = 3 state to n = 2 state of hydrogen is approximately: *(2004)*

a. 0.85 eV
b. 3.4 eV
c. 1.9 eV
d. 1.5 eV

31. The Bohr model of atoms: *(2004)*

a. Uses Einstein's photo electric equation
b. Predicts continuous emission spectra for atoms
c. Predicts the same emission spectra for all types of atoms
d. Assumes that the angular momentum of electrons is quantized

32. In which of the following systems will be radius of the first orbit (n = 1) be minimum: *(2003)*

a. Doubly ionised lithium
b. Singly ionised helium
c. Deuterium atom
d. Hydrogen atom

33. The energy of hydrogen atom in n^{th} orbit is E_n then the energy in n^{th} orbit of singly ionised helium atom will be: *(2001)*

a. $4E_n$
b. $E_n/4$
c. $2E_n$
d. $E_n/2$

34. Maximum frequency of emission is obtained for the transition: *(2000)*

a. n = 2 to n = 1
b. n = 6 to n = 2
c. n = 1 to n = 2
d. n = 2 to n = 6

35. When an electron do transition from n = 4 to n = 2, then emitted line in spectrum will be: *(2000)*

a. First line of Lyman series
b. Second line of Balmer series
c. First line of Paschen series
d. Second line of Paschen series

36. In the Bohr model of H-atom, an electron (e) is revolving around a proton (p) with velocity v, if r is the radius of orbit and m is mass and ε_0 is vacuum permittivity, the value of v is: *(1998)*

a. $\frac{e}{\sqrt{4\pi m \epsilon_0 r}}$
b. $\frac{2e}{\sqrt{\pi m \epsilon_0 r}}$
c. $\frac{e}{\sqrt{\pi m \epsilon_0 r}}$
d. $\frac{e}{\pi m \epsilon_0 r}$

37. The energy of the ground electronic state of hydrogen atom is – 13.6 eV. The energy of the first excited state will be *(1997)*

a. – 27.2 eV
b. –52.4 eV
c. – 3.4 eV
d. – 6.8 eV

38. When hydrogen atom is in its first excited level, its radius is _______ of the Bohr radius. *(1997)*

a. Twice
b. 4 times
c. Same
d. Half

39. The energy of a hydrogen atom in its ground state is –13.6 eV. The energy of the level corresponding to the quantum number n = 2 in the hydrogen atom is *(1996)*

a. –0.54 eV
b. –3.4 eV
c. –2.72 eV
d. –0.85 eV

40. According to Bohr's principle, the relation between principal quantum number (*n*) and radius of orbit (r) is *(1996)*

a. $r \propto \frac{1}{n}$
b. $r \propto \frac{1}{n^2}$
c. $r \propto n$
d. $r \propto n^2$

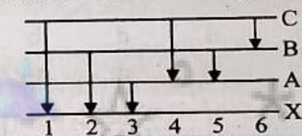
41. An electron makes a transition from orbit n = 4 to the orbit n = 2 of a hydrogen atom. What is the wavelength of the emitted radiations? (R – Rydberg's constant) *(1995)*

a. $\frac{16}{4R}$
b. $\frac{16}{5R}$
c. $\frac{16}{2R}$
d. $\frac{16}{3R}$

42. When a hydrogen atom is raised from the ground state to an excited state, *(1995)*

a. Both K.E. and P.E. increase
b. Both K.E. and P.E. decrease
c. The P.E. decreases and K.E. increases
d. The P.E. increases and K.E. decreases.

43. The figure indicates the energy level diagram of an atom and the origin of six spectral lines in emission (e.g. line no. 5 arises from the transition from level B to A). Which of the following spectral lines will occur in the absorption spectrum? *(1995)*



a. 4, 5, 6
b. 1, 2, 3, 4, 5, 6
c. 1, 2, 3
d. 1, 4, 6.

44. Hydrogen atoms are excited from ground state of the principle quantum number 4. Then the number of spectral lines observed will be: *(1993)*

a. 3
b. 6
c. 5
d. 2

45. Which source is associated with a line emission spectrum? *(1993)*

a. Electric fire
b. Neon street sign
c. Red traffic light
d. Sun

46. In terms of Bohr radius a_0, the radius of the second Bohr orbit of a hydrogen atom is given by: *(1992)*

a. $4a_0$
b. $8a_0$
c. $\sqrt{2a_0}$
d. $2a_0$

47. The ionisation energy of hydrogen atom is 13.6 eV. Following Bohr's theory, the energy corresponding to a transition between 3rd and 4th orbit is *(1992)*

a. 3.40 eV
b. 1.51 eV
c. 0.85 eV
d. 0.66 eV

48. The ground state energy of H-atom is 13.6 eV. The energy needed to ionize H-atom from its second excited state is: *(1991)*

a. 1.51 eV
b. 3.4 eV
c. 13.6 eV
d. None of these

De-Broglie's Explanation of Bohr's Second Postulate of Quantisation

49. Consider an electron in the n^{th} orbit of a hydrogen atom in the Bohr model. The circumference of the orbit can be expressed in terms of de Broglie wavelength λ of that electron as *(1990)*

a. $(0.529)n\lambda$

b. $\sqrt{\ }$

c. $(13.6)\lambda$

d. $n\lambda$

50. To explain his theory, Bohr used *(1989)*

a. Conservation of linear momentum

b. Quantisation of angular momentum

c. Conservation of quantum frequency

d. None of these

X-Ray

51. The minimum wavelength of the X-rays produced by electrons accelerated through a potential difference of V volts is directly proportional to *(1996)*

a. $\frac{1}{\sqrt{V}}$

b. $\frac{1}{V}$

c. $\sqrt{V}$

d. V^2

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	d	a	a	c	b	c	b	b	a	c	a	a	c	b	c	d
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
a	d	b	d	d	b	c	d	d	b	d	b	c	d	a	a	a
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
b	a	c	b	b	d	d	d	c	b	b	a	d	a	d	b	b

Explanations

1. (a) From the conservation of energy, K.E = P.E

$$\frac{1}{2}mu^2 = \frac{1}{4\pi\varepsilon_0}\frac{(ze)(ze)}{r}$$

$$\therefore r \propto \frac{1}{m}$$

When an α particle of mass m moving with velocity v bombards on a heavy nucleus of charge 'Ze', its distance of closest approach from the nucleus depends on mass $\frac{1}{m}$.

2. (d) $\frac{1}{2}mv^2 = \frac{(Ze)(2e)}{4\pi\varepsilon_0 d_{min}}$ then $d_{min.} \propto \frac{1}{m}$

3. (a) Energy of the projectile is the potential energy at closest approach, $\frac{1}{4\pi\varepsilon_0}\frac{z_1 z_2}{r_0}$

Therefore, energy $\propto z_1 z_2$.

4. (a) For the First excited state n = 2

$$T_1 = -13.6\frac{z^2}{n^2} = -\frac{13.6}{4}eV$$

For the second excited state

$$T_2 = -13.6\frac{z^2}{n^2} = -\frac{13.6}{9}eV$$

$$\Rightarrow T_1 : T_2 = 9 : 4$$

5. (c) Bohr model is not valid for singly ionised neon atom (net) since singly ionised neon atom has more than one electron in orbit.

6. (b) For hydrogen atom total energy of an electron in the n^{th} stationary orbit is

$$E = \frac{-13.6}{n^2}eV$$

7. (c) In Bohr's model of H atom

$$\therefore K.E. = |TE| = \frac{|U|}{2}$$

T.E. = K.E. + U

$\therefore$ K.E. = 3.4 eV

U = –6.8 eV

8. (b) In Hydrogen atom from Bohr's postulates

$$T.E. = -KE \Rightarrow \frac{K.E}{T.E} = 1 : -1$$

9. (b) Last line of Balmer

$n_1 = 2 \quad n_2 = \infty$

$$\frac{1}{\lambda_B} = Rz^2\left[\frac{1}{2^2} - \frac{1}{\infty}\right]$$

$$\frac{1}{\lambda_B} = \frac{Rz^2}{4}$$

Last line of Lyman

$n_1 = 1 \quad n_2 = \infty$

$$\frac{1}{\lambda_L} = Rz^2\left[\frac{1}{1^2} - \frac{1}{\infty}\right]$$

$$\frac{1}{\lambda_L} = Rz^2$$

$$\frac{\lambda_B}{\lambda_L} = 4$$

10. (a) $\frac{1}{\lambda} = \nu = RZ^2\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]$

Transition: $3 \rightarrow 2 \Rightarrow$ Wavelength λ

Transition: $4 \rightarrow 3 \Rightarrow$ Wavelength $\lambda' = ?$

$$\frac{\frac{1}{\lambda} = RZ^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}{\frac{1}{\lambda'} = RZ^2\left(\frac{1}{3^2} - \frac{1}{4^2}\right)} \Rightarrow \frac{\lambda'}{\lambda} = \frac{20}{7} \Rightarrow \lambda' = \frac{20\lambda}{7}$$

11. (c) $\frac{1}{\lambda} = RZ^2\left\{\frac{1}{n_2^2} - \frac{1}{n_1^2}\right\} = 10^7 \times 1^2\left\{\frac{1}{2^2} - \frac{1}{\infty^2}\right\}$

$= 0.25 \times 10^7\,m^{-1}$

12. (a) For H-like atoms

$$v = \frac{Z}{n} \times 2.188 \times 10^6\ m/s$$

here $Z = 2, n = 3$

$v = 1.46 \times 10^6\,m/s$

13. (a) For Lyman series

For longest wave length, $n_f = 1, n_i = 2$

$$\left(\frac{1}{\lambda_{max}}\right)_L = R(1)^2\left[\frac{1}{(1)^2} - \frac{1}{(2)^2}\right]$$

$$(\lambda_{max})_L = \frac{4}{3R}$$

For Balmer series

longest wave length, $n_f = 2, n_i = 3$

$$\left(\frac{1}{\lambda_{max}}\right)_L = R(1)^2\left[\frac{1}{(2)^2} - \frac{1}{(3)^2}\right]$$

$$(\lambda_{max})_B = \frac{36}{5R}$$

$$\frac{(\lambda_{max})_L}{(\lambda_{max})_B} = \frac{4}{3R} \times \frac{5R}{36} = \frac{5}{27}$$

14. (c) Energy incident $= \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{975 \times 10^{-10} \times 1.6 \times 10^{-19}}\,eV$

$= 12.75\ eV$

The hydrogen atom will be excited to $n = 4$

Number of spectral lines $= \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$

15. (b) $\left(\frac{\lambda_{Lyman}}{\lambda_{Balmer}}\right)_{max} = \frac{\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}{\left(\frac{1}{1^2} - \frac{1}{2^2}\right)} = \frac{5/36}{3/4} = \frac{5}{27}$

16. (c) Lyman series produces U.V. radiation. Balmer series produces visible radiation. Pachan series produces Infrared radiation so correct answer is $4 \rightarrow 3$

17. (d) Infrared radiation is found in Paschen, Brackett and Pfund series and it is obtain when electron transition occur from high energy level to minimum third level.

18. (a) According to Rydberg formula

$$\frac{1}{\lambda} = R\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$$

Here, $n_f = 1, n_i = 5$

$$\therefore \quad \frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{5^2}\right] = R\left[\frac{1}{1} - \frac{1}{25}\right] = \frac{24}{25}R$$

According to conservation of linear momentum, we get Momentum of photon = Momentum of atom

$$\frac{h}{\lambda} = mv \quad \text{or} \quad v = \frac{h}{m\lambda} = \frac{h}{m}\left(\frac{24R}{25}\right) = \frac{24hR}{25m}$$

19. (d) 1^{st} excited $n = 2$

2^{nd} excited $n = 3$

3^{rd} excited state $n = 4$

$$\frac{\lambda_1}{\lambda_2} = \frac{\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}{\left(\frac{1}{3^2} - \frac{1}{4^2}\right)} = \frac{\frac{1}{4} - \frac{1}{9}}{\frac{1}{9} - \frac{1}{16}} = \frac{\frac{5}{36}}{\frac{7}{144}} = \frac{20}{7}$$

as $\frac{1}{\lambda} \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

20. (b) The energy of n^{th} orbit of H-atom is given as

$$E_n = \frac{-13.6}{n^2}\,eV$$

$E_1 = -13.6\ eV \quad E_2 = -3.4\,eV \quad E_3 = -1.5\ eV$

$E_4 = -0.85\ eV$

$\therefore E_3 - E_2 = -1.5 - (-3.4) = 1.9\ eV$

$E_4 - E_3 = -0.85 - (1.5) = 0.65\ eV$

21. (d) Wavelength of first line of Lyman series of hydrogen atom is

$$\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{2^2}\right] \Rightarrow \frac{1}{\lambda} = \frac{3R}{4}$$

Wavelength of second line of Balmer series of hydrogen like ion

$$\Rightarrow \frac{1}{\lambda'} = R.Z^2\left(\frac{1}{2^2}-\frac{1}{4^2}\right) \Rightarrow \frac{1}{\lambda'} = \frac{3R.Z^2}{16}$$

Since, $\lambda = \lambda' \Rightarrow \frac{3R}{4} = \frac{3RZ^2}{16} \Rightarrow Z^2 = 4$

$Z = 2$

22. (d) Energy released when electron in the atom jumps from excited state (n = 3) to ground state (n = 1) is

$$E = h\nu = E_3 - E_1 = \frac{-13.6}{3^2} - \left(\frac{-13.6}{1^2}\right)$$

$$= \frac{-13.6}{9} + 13.6 = 12.1 \text{ eV}$$

Therefore, stopping potential

$eV_s = h\nu - \phi_0$

$= 12.1 - 5.1$ [$\because$ Work function $\phi_0 = 5.1$]

$V_s = 7$ V

23. (b) $E_n = -13.6\left(\frac{Z^2}{n^2}\right) = (-13.6)\left(\frac{4}{4}\right) = -13.6 \text{eV}$

24. (c) $n_i = 1$

Number of spectral lines $\Rightarrow 6 = \frac{n_f(n_f-1)}{2}$ $\quad n_f = 4$

$$\lambda = \frac{n_i^2 n_f^2}{R\left[n_f^2 - n_i^2\right]}$$

$3 \to 1 \quad \lambda = \frac{9}{8R}$

$2 \to 1 \quad \lambda = \frac{4}{R}$

$4 \to 3 \quad \lambda = \frac{144}{7R} \quad \lambda_{max}$ for $4 \to 3$ transition

$3 \to 2 \quad \lambda = \frac{36}{5R}$

25. (d) Total energy of e^- in ground state of H-atom

$= -13.6$ eV

Excitation energy:

$$= -13.6 \times z^2\left(\frac{1}{n_z^2} - \frac{1}{n_1^2}\right) = -13.6 \times (1)^2\left(\frac{1}{4}-1\right)$$

$$= -13.6 \times \left(\frac{-3}{4}\right) = +10.2 \text{ eV}$$

26. (d) Total energy of e^- in H-atom in ground state $= -13.6$ eV

For first excited state, n = 2

Total energy in first excited state

$$= -13.6\frac{z^2}{n^2}$$

$$= \frac{-13.6(1)^2}{4} = -3.4 \text{ eV}$$

K.E of an electron in first excited state

$= -(\text{T.E.}) = -(-3.4) = 3.4$ eV

27. (b) Let us consider the atoms excited to n^{th} shell by the monochromatic radiation of photon energy 12.1 eV. Using bohr equation for transition between states,

$$12.1 = 13.6\left[\frac{1}{1^2} - \frac{1}{n^2}\right] \Rightarrow 12.1 = 13.6 - \frac{13.6}{n^2}$$

$$\Rightarrow \frac{13.6}{n^2} = 1.5 \Rightarrow n^2 \approx 9 \Rightarrow n = 3$$

No. of spectral lines emitted

$$= \frac{n(n-1)}{2} = \frac{3\times 2}{2} = 3$$

28. (d) $\because \text{K.E.} = \frac{1}{2}|\text{P.E.}|$

But Potential energy is negative.

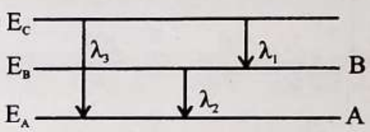
So, total energy = K.E + P.E

$$-3.4 = \frac{1}{2}\text{P.E.} - \text{P.E.} \Rightarrow -3.4 = \frac{-\text{P.E.}}{2}$$

$\Rightarrow$ K.E. = +3.4 eV

Basically, $\text{K.E.} = |\text{T.E.}| \Rightarrow \text{K.E.} = \frac{1}{2}|\text{P.E.}|$

29. (b)



$E_A < E_B < E_C \Rightarrow E_{CA} = E_{CB} + E_{BA}$

or, $\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} \Rightarrow \frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

$$\Rightarrow \lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$$

30. (c) Energy of photon $= E_3 - E_2$

$$= \frac{-13.6}{9} - \left(\frac{-13.6}{4}\right) = \frac{5}{36} \times 13.6 = 1.9 \text{ eV}$$

31. (d) Bohr model of atoms assumes that the angular momentum of electrons is quantised.

32. (a) $r_n = 0.529 \mathring{A}\left(\frac{n^2}{Z}\right)$

33. (a) $E_n = \frac{Z^2 E_0}{n^2}$

for Hydrogen atom Z = 1

$$E_n = \frac{(1)^2 E_0}{n^2} \quad \text{...(1)}$$

Similarly for ionised (singly) Helium atom

Z = 2

$$E_n = \frac{(2)^2 E_0}{n^2} \quad \text{...(2)}$$

From Eqs. (1) and (2)

$E_n^{'} = 4E_n$

34. (a) $\frac{1}{\lambda} = R\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]$

$n_1 = 2 \quad n_2 = 1$

$\frac{1}{\lambda} = R\left[\frac{1}{1} - \frac{1}{4}\right]$

$\frac{1}{\lambda} = \frac{3R}{4} \quad f = \frac{hc}{\lambda} = \frac{3hRc}{4}$

$n_1 = 6 \quad n_2 = 2$

$\frac{1}{\lambda} = R\left[\frac{1}{4} - \frac{1}{36}\right] = \frac{8R}{36}$

$f = \frac{8hRc}{36}$

$n_1 = 1 \quad n_2 = 2$

$\frac{1}{\lambda} = R\left[\frac{1}{4} - 1\right] = \frac{-3R}{4}$

$n_2 = 2$ to $n = 6$

$\frac{1}{\lambda} = R\left[\frac{1}{36} - \frac{1}{4}\right] = \frac{-8R}{36}$

$f \propto \frac{1}{\lambda}$

f is maximum for n = 2 to n = 1

35. (b) Jump to second orbit leads to Balmer series. The jump from 4th orbit shall give rise to second line of Balmer series.

36. (a) $\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \times \frac{e^2}{r^2} \Rightarrow v = \frac{e}{\sqrt{4\pi\epsilon_0 r m}}$

37. (c) Energy of the ground electronic state of hydrogen atom E = – 13.6 eV.

We know that energy of the first excited state for second orbit (where n = 2)

$E_n = -\frac{13.6}{(n)^2} = -\frac{13.6}{(2)^2} = -3.4\,\text{eV}.$

38. (b) When a hydrogen atom is in its excited level, then n = 2. Therefore radius of hydrogen atom in its first excited level (r) $\propto n^2 r_0 \Rightarrow r \propto (2)^2 r_0$

39. (b) Energy of hydrogen atom in ground state = – 13.6 eV and quantum number (n) = 2.

We know that energy of hydrogen atom

$(E_n) = -\frac{13.6}{n^2} = -\frac{13.6}{(2)^2} = -3.4$ eV.

40. (d) According to Bohr's principle, radius of orbit $(r) = 4\pi\varepsilon_0 \times \frac{n^2h^2}{4\pi^2 me^2} \propto n^2.$ where N = principal quantum number.

41. (d) Transition of hydrogen atom from orbit $n_1 = 4$ to $n_2 = 2$.

Wave number $= \frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$= R\left[\frac{1}{(2)^2} - \frac{1}{(4)^2}\right] = R\left[\frac{1}{4} - \frac{1}{16}\right]$

$= R\left[\frac{4-1}{16}\right] = \frac{3R}{16}.$

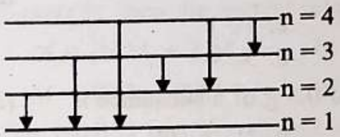
$\Rightarrow \lambda = 16/3R.$

42. (d) P.E. $= -\frac{k\,Ze^2}{R}$ and K.E. $= \frac{1}{2}mv^2 = \frac{k\,Ze^2}{2R}.$

Therefore when a hydrogen atom is raised from the ground, it increases the value of the radius R. As a result of this, K.E. decreases and potential energy increases.

43. (c) In Absorption spectrum excitation from ground level to higher level takes place. Therefore spectral lines 1, 2, 3 will occur in the absorption spectrum.

44. (b) No. of spectral lines $= \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$



45. (b) Neon street sign is a source of line emission spectrum.

46. (a) As $r \propto n^2$, therefore, radius of 2nd Bohr orbit $= 4a_0$

47. (d) $E = E_4 - E_3$

$= \frac{13.6}{4^2} - \left(-\frac{13.6}{3^2}\right) = -0.85 + 1.51 = 0.66$ eV

48. (a) Second excited state corresponds to n = 3

$\therefore\ E = \frac{13.6}{3^2}\,\text{eV} = 1.51\,\text{eV}$

But one has to ionise only from ground state. Even if one has to excite an atom from n = 3, one has to excite from n = 1.

49. (d) In terms of wavelength of wave associated with electron the circumference of an orbit in an atoms can be given by the relation,

Circumference = $n\lambda$

where n = 1, 2, 3, ...

50. (b) Bohr used quantisation of angular momentum.

For stationary orbits, Angular momentum $I\omega = \frac{nh}{2\pi}$

where n = 1, 2, 3,...etc.

51. (b) By the law of photoelectric effect

$\frac{hc}{\lambda} = eV$ or $\lambda = \frac{hc}{eV} \propto \frac{1}{V}.$

CHAPTER 13

Nuclei

Atomic Nucleus, Its Size, Density and Weight

1. A nucleus of mass number 189 splits into two nuclei having mass number 125 and 64. The ratio of radius of two daughter nuclei respectively is : *(2022)*

a. 25 : 16 b. 1 : 1
c. 4 : 5 d. 5 : 4

2. The energy equivalent of 0.5 g of a substance is : *(2020)*

a. 4.5×10^{13} J b. 1.5×10^{13} J
c. 0.5×10^{13} J d. 4.5×10^{16} J

3. If radius of the $^{27}_{12}Al$ nucleus is taken to be R_{Al}, then the radius of $^{125}_{53}Te$ nucleus is nearly: *(2015)*

a. $\frac{5}{3}R_{Al}$ b. $\frac{3}{5}R_{Al}$
c. $\left(\frac{13}{53}\right)^{1/3} R_{Al}$ d. $\left(\frac{53}{13}\right)^{1/3} R_{Al}$

4. If the nuclear radius of ^{27}Al is 3.6 Fermi, the approximate nuclear radius of ^{64}Cu in Fermi is: *(2012 Pre)*

a. 2.4 b. 1.2
c. 4.8 d. 3.6

5. Two nuclei have their mass numbers in the ratio of 1 : 3. The ratio of their nuclear densities would be: *(2008)*

a. 1 : 1 b. 1 : 3
c. 3 : 1 d. $3^{1/3} : 1$

6. If the nucleus $^{27}_{13}Al$ has nuclear radius of about 3.6 fm, then $^{125}_{32}Te$ would have its radius approximately as: *(2007)*

a. 9.6 fm b. 12.0 fm
c. 4.8 fm d. 6.0 fm

7. The radius of germanium (Ge) nuclide is measured to be twice the radius of $^{9}Be_{4}$. The number of nucleons in Ge are: *(2006)*

a. 73 b. 74
c. 76 d. 72

8. The nuclei of which one of the following pairs of nuclei are isotones? *(2005)*

a. $^{34}Se_{74}$, $^{31}Ga_{71}$ b. $^{42}Mo_{92}$, $^{40}Zr_{92}$
c. $^{38}Sr_{84}$, $^{38}Sr_{86}$ d. $^{20}Ca_{40}$, $^{16}S_{32}$

9. A nucleus represented by the symbol $^{A}_{Z}X$ has: *(2004)*

a. Z protons and A – Z neutrons
b. Z protons and A neutrons
c. A protons and Z – A neutrons
d. Z neutrons and A – Z protons

10. The mass number of a nucleus is: *(2003)*

a. Always less than its atomic number
b. Always more than its atomic number
c. Sometimes equal to its atomic number
d. Sometimes less than and sometimes more than its atomic number

11. The volume occupied by an atom is greater than the volume of the nucleus by a factor of about: *(2003)*

a. 10^{1} b. 10^{5}
c. 10^{10} d. 10^{15}

12. Boron has two isotopes $_5B^{10}$ and $_5B^{11}$. If atomic weight of Boron is 10.81 then ratio of $_5B^{10}$ to $_5B^{11}$ in nature will be: *(1998)*

a. 15 : 16 b. 19 : 81
c. 81 : 19 d. 20 : 53

13. A nucleus ruptures into two nuclear parts, which have their velocity ratio equal to 2 : 1. What will be the ratio of their nuclear size (nuclear radius)? *(1996)*

a. $3^{1/2} : 1$ b. $1 : 3^{1/2}$
c. $2^{1/3} : 1$ d. $1 : 2^{1/3}$

14. The mass number of He is 4 and that of sulphur is 32. The radius of sulphur nucleus is larger than that of helium by the factor of *(1995)*

a. 4 b. 2
c. 8 d. $\sqrt{8}$

15. The mass density of a nucleus varies with mass number A as *(1992)*

a. A^2 b. A
c. Constant d. 1/A

16. The constituents of atomic nuclei are believed to be *(1991)*

a. Neutrons and protons
b. Protons only
c. Electron and protons
d. Electrons, protons and neutrons

17. In the nucleus of $_{11}Na^{23}$, the number of protons, neutrons and electrons are (1991)

a. 11, 12, 0
b. 23, 12, 11
c. 12, 11, 0
d. 23, 11, 12

18. The nuclei $_6C^{13}$ and $_7N^{14}$ can be described as (1990)

a. Isotones
b. Isobars
c. Isotopes of carbon
d. Isotopes of nitrogen

19. The ratio of the radii of the nuclei $_{13}Al^{27}$ and $_{52}Te^{125}$ approximately (1990)

a. 6 : 10
b. 13 : 52
c. 40 : 177
d. 14 : 73

Mass Energy, Nuclear Binding Energy and Nuclear Force

20. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is: (2021)

a. 9.4 MeV
b. 804 MeV
c. 216 MeV
d. 0.9 MeV

21. The power obtained in a reactor using U^{235} disintegration is 1000 kW. The mass decay of U^{235} per hour is: (2011 Pre)

a. 10 microgram
b. 20 microgram
c. 40 microgram
d. 1 microgram

22. The binding energy per nucleon in deuterium and helium nuclei are 1.1 MeV and 7.0 MeV, respectively. When two deuterium nuclei fuse to form a helium nucleus the energy released in the fusion is: (2010 Mains)

a. 23.6 MeV
b. 2.2 MeV
c. 28.0 MeV
d. 30.2 MeV

23. The mass of a ^7_3Li nucleus is 0.042 u less than the sum of the masses of all its nucleons. The binding energy per nucleon of $^7_3$ Li nucleus is nearly: (2010 Pre)

a. 23 MeV
b. 46 MeV
c. 5.6 MeV
d. 3.9 MeV

24. If M(A; Z), M_p and M_n denote the masses of the nucleus ^A_ZX, proton and neutron respectively in units of u (1 u = 931.5 MeV/c^2) and BE represents its binding energy in MeV, then: (2008)

a. $M(A,Z)=Z_{M_p}+(A-Z)M_n+B.E/c^2$

b. $M(A,Z)=Z_{M_p}+(A-Z)M_n-B.E/c^2$

c. $M(A,Z)=Z_{M_p}+(A-Z)M_n+B.E$

d. $M(A,Z)=Z_{M_p}+(A-Z)M_n-B.E$

25. A nucleus ^A_ZX has mass represented by M(A, Z). If M_p and M_n denote the mass of proton and neutron respectively and B.E. The binding energy in MeV, then: (2007)

a. $B.E. = [Z_{Mp} + (A - Z)M_p - M(A, Z)]c^2$

b. $B.E. = [Z_{Mp} + Z_{Mn} - M(A, Z)]c^2$

c. $B.E. = M(A, Z) - Z_{Mp} - (A - Z)M_n$

d. $B.E. = [M(A, Z) - Z_{Mp} - (A - Z)M_n]c^2$

26. M_p denotes the mass of a proton and M_n that of a neutron. A given nucleus, of binding energy B, contains Z protons and N neutrons. The mass M (N, Z) of the nucleus is given by (c is velocity of light): (2004)

a. $M(N, Z) = NM_n + ZM_p + Bc^2$
b. $M(N, Z) = NM_n + ZM_p - B/c^2$
c. $M(N, Z) = NM_n + ZM_p + B/c^2$
d. $M(N, Z) = NM_n + ZM_p - Bc^2$

27. The mass of proton is 1.0073 u and that of neutron is 1.0087 u (u = atomic mass unit). The binding energy of ^4_2He is (Given: helium nucleus mass ≈ 4.0015 u) (2003)

a. 0.0305 J
b. 0.0305 erg
c. 28.4 MeV
d. 0.061 u

28. M_n and M_p represent the mass of neutron and proton respectively. An element having mass M has N neutron and Z-protons, then the correct relation will be: (2001)

a. $M < \{N.M_n + Z.M_p\}$
b. $M > \{N.M_n + Z.M_p\}$
c. $M = \{N.M_n + Z.M_p\}$
d. $M = N\{M_n + M_p\}$

29. The binding energy per nucleon is maximum in case of (1993)

a. ^4_2He
b. $^{56}_{26}Fe$
c. $^{141}_{56}Ba$
d. $^{235}_{92}U$

30. The energy equivalent of one atomic mass unit is (1992)

a. 1.6×10^{-19} J
b. 6.02×10^{23} J
c. 931 MeV
d. 9.31 MeV

31. The mass of α-particle is (1992)

a. Less than the sum of masses of two protons and two neutrons
b. Equal to mass of four protons
c. Equal to mass of four neutrons
d. Equal to sum of masses of two protons and two neutrons

32. If the nuclear force between two protons, two neutrons and between proton and neutron is denoted by F_{pp}, F_{nn} and F_{pn} respectively, then (1990)

a. $F_{pp} \approx F_{nn} \approx F_{pn}$
b. $F_{pp} \neq F_{nn}$ and $F_{pp} = F_{nn}$
c. $F_{pp} = F_{nn} = F_{pn}$
d. $F_{pp} \neq F_{nn} \neq F_{pn}$

33. Which of the following statements is true for nuclear forces? (1990)

a. They obey the inverse square law of distance.
b. They obey the inverse third power law of distance.
c. They are short range forces.
d. They are equal in strength to electromagnetic forces.

34. The average binding energy of a nucleon inside an atomic nucleus is about (1989)

a. 8 MeV
b. 8 eV
c. 8 J
d. 8 erg

Radioactive Decay Law, Half-Life and Average Life and Activity of a Radioactive Substance

35. The half life of a radioactive sample undergoing α-decay is 1.4×10^{17} s. If the number of nuclei in the sample is 2.0×10^{21}, the activity of the sample is nearly. *(2020-Covid)*

a. 10^5 Bq
b. 10^6 Bq
c. 10^3 Bq
d. 10^4 Bq

36. For a radioactive material, half-life is 10 minutes. If initially there are 600 number of nuclei, the time taken (in minutes) for the distintegration of 450 nuclei is . *(2018)*

a. 30
b. 10
c. 20
d. 15

37. Radioactive material 'A' has decay constant '8λ' and material 'B' has decay constant 'λ'. Initially they have same number of nuclei. After what time, the ratio of number of nuclei of material 'A' to that of 'B' will be 1/e? *(2017-Delhi)*

a. $\frac{1}{7\lambda}$
b. $\frac{1}{8\lambda}$
c. $\frac{1}{9\lambda}$
d. $\frac{1}{\lambda}$

38. The half-life of a radioactive substance is 30 minutes. The time (in minutes) taken between 40% decay and 85% decay of the same radioactive substance is: *(2016 - II)*

a. 45
b. 60
c. 15
d. 30

39. A radio isotope *X* with a half life of 1.4×10^9 years decays to Y which is stable. A sample of the rock from a cave was found to contain *X* and *Y* in the ratio 1 : 7. The age of the rock is: *(2014)*

a. 1.96×10^9 years
b. 3.92×10^9 years
c. 4.20×10^9 years
d. 8.40×10^9 years

40. The half life of a radioactive isotope '*X*' is 20 years. It decays to another element '*Y*' which is stable. The two elements '*X*' and '*Y*' were found to be in the ratio 1 : 7 in a sample of a given rock. The age of the rock is estimated to be: *(2013)*

a. 100 years
b. 40 years
c. 60 years
d. 80 years

41. The half life of a radioactive nucleus is 50 days. The time interval $(t_2 - t_1)$ between the time t_2 when 2 / 3 of it has decayed and the time t_1 when 1 / 3 of it had decayed is: *(2012 Mains)*

a. 30 days
b. 50 days
c. 60 days
d. 15 days

42. A mixture consists of two radioactive materials A_1 and A_2 with half lives of 20 s and 10 s respectively. Initially the mixture has 40 g of A_1 and 160 g of A_2. The amount of the two in the mixture will become equal after: *(2012 Pre)*

a. 40 s
b. 60 s
c. 80 s
d. 20 s

43. Two radioactive nuclei P and Q, in a given sample decay into a stable nucleus R. At time t = 0, number of P species are $4\,N_0$ and that of Q are N_0. Half-life of P (for conversion to R) is 1 minute where as that of Q is 2 minutes. Initially there are no nuclei of R present in the sample. When number of nuclei of P and Q are equal, the number of nuclei of R present in the sample would be: *(2011 Mains)*

a. $3N_0$
b. $\frac{9N_0}{2}$
c. $\frac{5N_0}{2}$
d. $2N_0$

44. The half life of a radioactive isotope 'X' is 50 years. It decay to another element 'Y' which is stable. The two elements 'X' and 'Y' were found to be in the ratio of 1 : 15 in a sample of a given rock. The age of the rock was estimated to be: *(2011 Pre)*

a. 150 years
b. 200 years
c. 250 years
d. 100 years

45. A radioactive nucleus of mass M emits a photon of frequency ν and the nucleus recoils. The recoil energy will be *(2011 Pre)*

a. $Mc^2 - h\nu$
b. $h^2\nu^2/2Mc^2$
c. Zero
d. $h\nu$

46. The decay constant of a radio isotope is λ. If A_1 and A_2 are its activities at times t_1 and t_2 respectively, the number of nuclei which have decayed during the time $(t_1 - t_2)$: *(2010 Mains)*

a. $A_1t_1 - A_2t_2$
b. $A_1 - A_2$
c. $(A_1 - A_2)/\lambda$
d. $\lambda(A_1 - A_2)$

47. The activity of a radioactive sample is measured as N_0 counts per minute at t = 0 and N_0/e counts per minute at t = 5 minutes. The time (in minutes) at which the activity reduces to half its value is: *(2010 Pre)*

a. $5\log_e 2$
b. $\log_e \frac{2}{5}$
c. $\frac{5}{\log_e 2}$
d. $5\log_{10} 5$

48. Two radioactive materials X_1 and X_2 have decay constants 5λ and λ respectively. Initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be $\frac{1}{e}$ after a time: *(2008)*

a. $\frac{e}{\lambda}$
b. λ
c. $\frac{1}{2}\lambda$
d. $\frac{1}{4\lambda}$

49. Two radioactive substances A and B have decay constants 5λ and λ respectively. At t = 0 they have the same number of nuclei. The ratio of number of nuclei of A to those of B will be $(1/e)^2$ after a time interval: *(2007)*

a. 4λ
b. 2λ
c. 1/2λ
d. 1/4λ

50. In a radioactive material the activity at time t_1 is R_1 and at a later time t_2, it is R_2. If the decay constant of the material is λ, then: (2006)

a. $R_1 = R_2 e^{-\lambda(t_1-t_2)}$ b. $R_1 = R_2^{e^{\lambda(t_1-t_2)}}$

c. $R_1 = R_2 e^{(t_2-t_1)}$ d. $R_1 = R_2$

51. The half life of radium is about 1600 years. Out of 100 g of radium existing now, 25 g will remain undecayed after: (2004)

a. 6400 years b. 2400 years

c. 3200 years d. 4800 years

52. A sample of radioactive element has a mass of 10 gm at an instant $t = 0$. The approximate mass of this element in the sample after two mean lives is: (2003)

a. 1.35 gm b. 2.50 gm

c. 3.70 gm d. 6.30 gm

53. A sample of radioactive element containing 4×10^{16} active nuclei. Half life of element is 10 days, then number of decayed nuclei after 30 days: (2002)

a. 0.5×10^{16} b. 2×10^{16}

c. 3.5×10^{16} d. 1×10^{16}

54. Half life of radioactive element is 12.5 hour and its quantity is 256 gm. After how much time its quantity will remain 1 gm: (2001)

a. 50 Hrs b. 100 Hrs

c. 150 Hrs d. 200 Hrs

55. The relation between λ and $T_{1/2}$ as ($T_{1/2} \rightarrow$ half life): (2000)

a. $T_{1/2} = \dfrac{\ln 2}{\lambda}$ b. $T_{1/2} \ln 2 = \lambda$

c. $T_{1/2} = \dfrac{1}{\lambda}$ d. $(\lambda + T_{1/2}) = \ln 2$

56. The life span of atomic hydrogen is: (2000)

a. Fraction of one sec b. One year

c. One hour d. One day

57. The half life of a radio nuclide is 77 days then its decay constant is: (1999)

a. 0.003/day b. 0.006/day

c. 0.009/day d. 0.012/day

58. Half life period of two elements are 40 minute and 20 minute respectively, then after 80 minute ratio of the remaining nuclei will be (Initially both have equal active nuclei): (1998)

a. 4 : 1 b. 1 : 2

c. 8 : 1 d. 16 : 1

59. The count rate of a Geiger Muller counter for the radiation of a radioactive material of half-life of 30 minutes decreases to 5 second^{-1} after 2 hours. The initial count rate was (1995)

a. 80 second^{-1} b. 625 second^{-1}

c. 20 second^{-1} d. 25 second^{-1}

60. The half life of radium is 1600 years. The fraction of a sample of radium that would remain after 6400 years (1991)

a. 1/4 b. 1/2

c. 1/8 d. 1/16

61. An element A decays into element C by a two step process (1989)

$A \rightarrow B + {}_2He^4$; $B \rightarrow C + 2e^-$ Then

a. A and C are isotopes b. A and C are isobars

c. A and B are isotopes d. A and B are isobars

62. Curie is a unit of (1989)

a. Energy of gamma rays b. Half-life

c. Radioactivity d. Intensity of gamma rays

63. A radioactive element has half life period 800 years. After 6400 years what amount will remain? (1989)

a. 1/2 b. 1/16

c. 1/8 d. 1/256

64. A radioactive sample with a half life of 1 month has the label: 'Activity = 2 micro curies on 1 – 8 – 1991'. What would be its activity two months earlier? (1988)

a. 1.0 micro curie b. 0.5 micro curie

c. 4 micro curie d. 8 micro curie

Alpha, Beta and Gamma Decay

65. What happens to the mass number and atomic number of an element when it emits γ-radiation? (2020-Covid)

a. Mass number and atomic number remain unchanged.

b. Mass number remains unchanged while atomic number decreases by one

c. Mass number increases by four and atomic number increases by two

d. Mass number decreases by four and atomic number decreases by two

66. α-particle consists of : (2019)

a. 2 protons and 2 neutrons only

b. 2 electrons, 2 protons and 2 neutrons

c. 2 electrons and 4 protons only

d. 2 protons only

67. A nucleus ${}^m_nX$ emits one α particle and two β - particles. The resulting nucleus is: (2011 Pre)

a. ${}^{m-4}_{n-2}Y$ b. ${}^{m-6}_{n-4}Z$

c. ${}^{m-6}_{n}Z$ d. ${}^{m-4}_{n}X$

68. The number of beta particles emitted by a radioactive substance is twice the number of alpha particles emitted by it. The resulting daughter is an: (2009)

a. Isomer of parent b. Isotone of parent

c. Isotope of parent d. Isobar of parent

69. In the nuclear decay given below: ${}^A_ZX \rightarrow {}^{A}_{Z+1}Y \rightarrow {}^{A-4}_{Z-1}B^* \rightarrow B$, the particles emitted in the sequence are: (2009)

a. γ, β, α b. β, γ, α

c. α, β, γ d. β, α, γ

70. In a radioactive decay process, the negatively charged emitted β-particles are: *(2007)*

a. The electrons produced as a result of the decay of neutrons inside the nucleus
b. The electrons produced as a result of collisions between atoms
c. The electrons orbiting around the nucleus
d. The electrons present inside the nucleus

71. A nuclear reaction given by $_zX^A \rightarrow _{z+1}Y^A + _{-1}e^0 + \bar{v}$ represents: *(2003)*

a. β-decay b. γ-decay
c. Fusion d. Fission

72. A deutron is bombarded on $_8O^{16}$ nucleus then α-particle is emitted then product nucleus is: *(2002)*

a. $_7N^{13}$ b. $_5B^{10}$
c. $_4Be^9$ d. $_7N^{14}$

73. Which rays contain (+ ve) charged particle: *(2001)*

a. α-rays b. β-rays
c. γ-rays d. X-rays

74. The most penetrating radiation out of the following are *(1997)*

a. β-rays b. γ-rays
c. X-rays d. α-rays

75. What is the respective number of α and β particles emitted in the following radioactive decay? *(1995)*

$^{200}X_{90} \rightarrow ^{168}Y_{80}$

a. 8 and 8 b. 8 and 6
c. 6 and 8 d. 6 and 6

76. The nucleus $_6C^{12}$ absorbs an energetic neutron and emits a beta–particle (β). The resulting nucleus is *(1990)*

a. $_7N^{14}$ b. $_7N^{13}$
b. $_5N^{13}$ d. $_6C^{13}$

77. The nucleus $^{115}_{48}Cd$, after two successive β-decay will give *(1988)*

a. $^{115}_{46}Pa$ b. $^{114}_{49}In$
c. $^{113}_{50}Sn$ d. $^{115}_{50}Sn$

Nuclear Reactions

78. In the given nuclear reaction, the element X is :
$^{22}_{11}Na \rightarrow X + e^+ + \nu$ *(2022)*

a. $^{22}_{12}Mg$ b. $^{22}_{11}Na$
c. $^{22}_{10}Ne$ d. $^{22}_{10}Ne$

79. A radioactive nucleus ^A_ZX undergoes spontaneous decay in the sequence

$^A_ZX \rightarrow _{Z-1}B \rightarrow _{Z-3}C \rightarrow _{Z-2}D$, where Z is the atomic number of element X. The possible decay particles in the sequence are: *(2021)*

a. α, $β^+$, $β^-$ b. $β^+$, α, $β^-$
c. $β^-$, α, $β^+$ d. α, $β^-$, $β^+$

80. When a uranium isotope $^{235}_{92}U$ is bombarded with a neutron, it generates $^{89}_{36}Kr$, three neutrons and : *(2020)*

a. $^{91}_{40}Zr$ b. $^{101}_{36}Kr$
c. $^{103}_{36}Kr$ d. $^{144}_{56}Br$

81. The binding energy per nucleon of ^7_3Li and ^4_2He nuclei are 5.60 MeV and 7.06 MeV, respectively. In the nuclear reaction $^7_3Li + ^1_1H \rightarrow ^4_2He + ^4_2He + Q$ the value of energy Q released is: *(2014)*

a. 19.6 MeV b. –2.4 MeV
c. 8.4 MeV d. 17.3 MeV

82. The binding energy of deuteron is 2.2 MeV and that of ^4_2He is 28 MeV. If two deuteron are fused to form one ^4_2He then the energy released is: *(2006)*

a. 21.6 MeV b. 23.6 MeV
c. 17.2 MeV d. 28.2 MeV

83. In the reaction $^2H_1 + ^3H_1 \rightarrow ^4He_2 + ^1n_0$, if the binding energies of $^2H_1$, $^3H_1$ and $^4He_2$ are respectively a, b and c (in MeV), then the energy (in MeV) released in this reaction is: *(2005)*

a. c + a – b b. c – a – b
c. a + b + c d. a + b – c

84. For the given reaction, the particle X is: *(2000)*

$_6C^{11} \rightarrow _5B^{11} + β^+ + X$

a. Neutron b. Anti-neutrino
c. Neutrino d. Proton

85. A radioactive elements emits one α and β particle then mass number of daughter element is: *(1999)*

a. Decreased by 4 b. Increased by 4
c. Decreased by 2 d. Increased by 2

86. For nuclear reaction $_{92}U^{235} + _0n^1 \rightarrow _{56}Ba^{144} +$ .... $+ 3_0n^1$: *(1998)*

a. $_{26}Kr^{89}$ b. $_{36}K^{89}$
c. $_{26}Sr^{90}$ d. $_{38}Sr^{89}$

87. Which of the following is used as a moderator in nuclear reaction? *(1997)*

a. Cadmium b. Plutonium
c. Uranium d. Heavy water

88. The binding energies per nucleon for a deuteron and an α-particle are x_1 and x_2 respectively. The energy Q released in the reaction $^2H_1 + ^2H_1 \rightarrow ^4H_1 + Q$, is (1995)

a. $4(x_1 + x_2)$ b. $4(x_2 - x_1)$
c. $2(x_2 - x_1)$ d. $2(x_2 + x_1)$

Nuclear Fission and Nuclear Fusion

89. A certain mass of Hydrogen is changed to Helium by the process of fusion. The mass defect in fusion reaction is 0.02866 u. The energy liberated per u is (given 1 u = 931 MeV): *(2013)*

a. 13.35 MeV b. 2.67 MeV
c. 26.7 MeV d. 6.675 MeV

90. Fusion reaction takes place at high temperature because: *(2011 Pre)*

a. Nuclei break up at high temperature
b. Atoms get ionised at high temperature
c. Kinetic energy is high enough to overcome the coulomb repulsion between nuclei
d. Molecules break up at high temperature

91. In any fission process the ratio (mass of fission products/mass of parent nucleus) is: *(2005)*

a. Less than 1
b. Greater than 1
c. Equal to 1
d. Depends on the mass of parent nucleus

92. Fission of nuclei is possible because the binding energy per nucleon in them: *(2005)*

a. Increases with mass number at high mass numbers
b. Decreases with mass number at high mass numbers
c. Increases with mass number at low mass numbers
d. Decreases with mass number at low mass numbers

93. If in a nuclear fusion process the masses of the fusing nuclei be m_1 and m_2 and the mass of the resultant nucleus be m_3, then: *(2004)*

a. $m_3 = |m_1 - m_2|$ b. $m_3 < (m_1 + m_2)$
c. $m_3 > (m_1 + m_2)$ d. $m_3 = m_1 + m_2$

94. Solar energy is mainly caused due to: *(2003)*

a. Burning of hydrogen in the oxygen
b. Fission of uranium present in the Sun
c. Fusion of protons during synthesis of heavier elements
d. Gravitational contraction

95. Which of the following are suitable for the fusion process: *(2002)*

a. Light nuclei
b. Heavy nuclei
c. Element must be lying in the middle of the periodic table
d. Middle elements, which are lying on binding energy curve

96. Energy is released in nuclear fission due to: *(2001)*

a. Few mass is converted into energy
b. Total binding energy of fragements is more than the B.E. of parental element
c. Total B.E. of fragements is less than the B.E. of parental element
d. Total B.E. of fragements is equals to the B.E. of parental element

97. Nuclear-Fission is best explained by: *(2000)*

a. Liquid droplet theory
b. Yukawa π-meson theory
c. Independent particle model of the nucleus
d. Proton-proton cycle

98. Energy released in the fission of a single $^{235}_{92}U$ nucleus is 200 MeV. The fission rate of $^{235}_{92}U$ filled reactor operating at a power level of 5 W is *(1993)*

a. $1.56 \times 10^{-10}\ s^{-1}$ b. $1.56 \times 10^{11}\ s^{-1}$
c. $1.56 \times 10^{-16}\ s^{-1}$ d. $1.56 \times 10^{-17}\ s^{-1}$

99. Solar energy is due to *(1992)*

a. Fusion reaction
b. Fission reaction
c. Combustion reaction
d. Chemical reaction

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	a	a	c	a	d	d	a	a	c	d	b	d	b	c	a	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
a	a	c	c	a	c	b	a	b	c	a	b	c	a	c	c	a
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
d	c	a	b	c	c	b	a	b	b	b	c	a	d	c	a	c
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
a	c	b	a	a	c	a	a	d	a	c	d	d	a	a	d	c
69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
d	a	a	d	a	b	b	b	d	d	b	d	d	b	b	c	a
86	87	88	89	90	91	92	93	94	95	96	97	98	99			
b	d	b	d	c	a	b	b	c	a	b	a	b	a			

Explanations

1. (d) Nuclear Radius : R is proportional to $R_0(A)^{1/3}$

$$\Rightarrow \frac{R(125)}{R(64)} = \frac{125^{1/3}}{64^{1/3}} = \frac{5}{4}$$

2. (a) From mass energy equivalence

$E = mc^2$

$= 0.5 \times 10^{-3}(3 \times 10^8)^2 = 4.5 \times 10^{13} J$

3. (a) $R \propto A^{1/3}$, $A_{Al} = 27$, $A_{Te} = 125$

$$\frac{R_{Al}}{R_{Te}} = \left(\frac{27}{125}\right)^{1/3} \Rightarrow R_{Te} = \frac{5}{3}R_{Al}$$

4. (c) $R \propto A^{\frac{1}{3}} \Rightarrow \frac{R_{cu}}{R_{Al}} = \left(\frac{64}{27}\right)^{\frac{1}{3}} = \left(\frac{4^3}{3^3}\right)^{\frac{1}{3}} = \frac{4}{3}$

$$\Rightarrow R_{Cu} = \frac{4}{3} \times 3.6 = 4.8 \text{ fermi}$$

5. (a) Nuclear density $= \dfrac{\text{Mass of nucleus}}{\text{Volume of nucleus}}$

Nuclear density $= \dfrac{mA}{\frac{4}{3}\pi R^3} \Rightarrow mA = \text{Mass of nucleus}$

= Mass of 1 nucleon × Mass number

[Considering mass of proton and neutron equal]

Nuclear density $= \dfrac{mA}{\frac{4}{3}\pi\left(R_0A^{1/3}\right)^3} \quad \left\{\because R = R_0A^{1/3}\right\}$

$$= \frac{3mA}{4\pi R_0^3 A} = \frac{3m}{4\pi R_0^3}$$

Hence, Nuclear density is independent of mass number.

6. (d) $\because R = R_0A^{1/3}$ [$\because R_0$ is constant]

A= Mass number

$$\frac{R_{Al}}{R_{Te}} = \left(\frac{A_{Al}}{A_{Te}}\right)^{1/3} \Rightarrow \frac{3.6}{R_{Te}} = \left(\frac{27}{125}\right)^{1/3}$$

$$\Rightarrow \frac{3.6}{R_{Te}} = \frac{3}{5} \Rightarrow R_{Te} = 6.0 \text{ fm}$$

7. (d) As we know that $\left[\text{Mass number}(A)\right]^{1/3}$

$R = R_0A^{1/3}$

Here, $R_{Ge} = 2R_{Be}$

$$\frac{R_{Ge}}{R_{Be}} = \left[\frac{A_{Ge}}{A_{Be}}\right]^{1/3} \Rightarrow \frac{2R_{Be}}{R_{Be}} = \left[\frac{A_{Ge}}{9}\right]^{1/3}$$

$$\Rightarrow (2)^3 = \frac{A_{Ge}}{9}$$

$\Rightarrow A_{Ge} = 72$ So, no. of nucleons in Ge = 72

8. (a) Isotones means nuclei having same number of neutrons i.e. (A–Z) so, in option (a) no. of neutrons in both ${}_{34}Se^{74}$ and ${}_{31}Ga^{71}$ is 40.

9. (a) In ${}^{A}_{Z}X$ Z = Number of protons

A–Z = number of neutrons

10. (c) Mass number = Atomic number + no. of neutrons

For hydrogen, number of neutrons = 0

So, mass number = Atomic number.

Hence mass number is sometimes equal to atomic number.

11. (d) $\dfrac{\text{Volume of atom}}{\text{volume of nucleus}} \sim \left(\dfrac{10^{-10}}{10^{-15}}\right)^3 = 10^{15}$

12. (b) $\dfrac{N_1}{N_2} = \text{ratio}$

Average weight $= \dfrac{N_1W_1 + N_2W_2}{N_1 + N_2}$

$10.81N_1 + 10.81N_2 = 10N_1 + 11N_2$

$0.81N_1 = 0.19N_2 \Rightarrow \boxed{\dfrac{N_1}{N_2} = \dfrac{19}{81}}$

13. (d) Velocity ratio $(v_1 : v_2) = 2 : 1$.

Mass (m) ∝ Volume ∝ r^3.

According to law of conservation of momentum, $m_1v_1 = m_2v_2$

Therefore $\dfrac{v_1}{v_2} = \dfrac{m_2}{m_1} = \dfrac{r_2^3}{r_1^3}$

Or $\dfrac{r_1}{r_2} = \left(\dfrac{v_2}{v_1}\right)^{1/3} = \left(\dfrac{1}{2}\right)^{1/3} = \dfrac{1}{2^{1/3}}$

Or $r_1 : r_2 = 1 : 2^{1/3}$.

14. (b) Mass number of helium $(A_{He}) = 4$ and mass number of sulphur $(A_S) = 32$.

Radius of nucleus, $r = r_0A^{1/3} \propto A^{1/3}$

Therefore $\dfrac{r_s}{r_{He}} = \left(\dfrac{A_s}{A_{He}}\right)^{1/3} = \left(\dfrac{32}{4}\right)^{1/3} = (8)^{1/3} = 2.$

15. (c) The nuclear radius r is related to mass number A as follow

$r = r_0 A^{1/3} \Rightarrow r \propto A^{1/3}$ or $A \propto r^3$

Now density $= \dfrac{\text{Mass}}{\text{Volume}}$

Further mass ∝ A and volume ∝r^3

$\therefore \dfrac{\text{Mass}}{\text{Volume}} = \text{constant}$

16. (a) Electrons are outside the nucleus, nucleus contains only neutrons and protons

17. (a) Z = 11 i.e., number of protons = 11, A = 23

∴ Number of neutrons = A – Z = 12

Number of electron = 0 (No electron in nucleus)

Therefore 11, 12, 0.

18. (a) As $_6C^{13}$ and $_7N^{14}$ have same no. of neutrons (13 – 6 = 7 for C and 14 – 7 = 7 for N), they are isotones.

19. (a) $R \propto A^{1/3}$ from $R = R_0 A^{1/3}$

$\therefore R_{Al} \propto (27)^{1/3}$ and $R_{Te} \propto (125)^{1/3}$

$\therefore \frac{R_{Al}}{R_{Te}} = \frac{3}{5} = \frac{6}{10}$

20. (c) $A^{240} \longrightarrow B^{120} + C^{120}$

Given,

Binding energy per nucleon of A, $E_A = 7.6$ MeV

$E_B = 8.5$ MeV

and $E_C = 8.5$ MeV

Total gain in Binding energy is given as below;

Q = Binding energy of products – Binding energy of rectants

$= [(A_B \times E_B + A_C \times E_C) - (A_A \times E_A)]$

$= [(120 \times 8.5 + 120 \times 8.5) - (240 \times 7.6)]$

$= [2 \times 120 \times 8.5 - 240 \times 7.6]$

$= [2040 - 1824]$

Q = 216 MeV

21. (c) According to Einstein mass - energy equivalence,

$E = mc^2 \Rightarrow m = \frac{E}{c^2} \Rightarrow \Delta m = \frac{\Delta E}{c^2}$

Mass decay per second

$= \frac{\Delta m}{\Delta t} = \frac{\Delta E}{c^2 \Delta t} = \frac{P}{c^2} = \frac{1000 \times 10^3}{(3 \times 10^8)^2}$

$= \frac{1}{9} \times 10^{-10}\,\text{kgs}^{-1}$

Mass decay per hour

$= \frac{1}{9} \times 10^{-10} \times 3600 = 4 \times 10^{-8}\,\text{kg}$

= 40 μg

22. (a) $_1H^2 + _1H^2 = _2He^4 + \Delta E$

$\Delta E = 4(7) - 2(1.1 + 1.1) = 23.6$ eV

23. (c) $\frac{\text{B.E}}{\text{Nucleon}} = \frac{0.042 \times 931}{7} = 5.6$ MeV

24. (b) As we know

$\text{B.E.} = \left[Z_{M_P} + (A - Z)M_n - M(A, Z)\right]c^2$

$= \Delta mc^2$

From here,

$M(A, Z) = Z_{M_P} + (A - Z)M_n - \frac{\text{B.E}}{c^2}$

25. (a) Binding Energy = [Mass of Protons + Mass of neutrons – Mass of nucleus] c^2

Nucleus mass $= M \rightarrow {}^A_Z M$

$\Rightarrow \text{B.E.} = \left[Z_{M_P} + (A - Z)M_n - M(A.Z)c^2\right]$

26. (b) Mass of nucleus is slightly less than sum of masses of its constituents. This mass difference is equivalent to binding energy.

$M(A, Z) = ZM_p + (A - Z)M_n - \frac{B}{c^2}$

$\therefore \Delta m = \frac{E}{c^2}(ZM_p + NM_n) - M(N, Z)$

Hence $M(N, Z) = NM_n + ZM_p - \frac{B}{c^2}$

27. (c) B.E = Δm × 931

= [2(1.0087 + 1.0073) – 4.0015] × 931 = 28.4 MeV

28. (a) When nucleons form a nucleus, the mass of the nucleus is less than the sum of the masses of nucleons this decrease in the mass of nucleons (equal to mass defect) is converted into energy in accordance with the equation $E = mc^2$, which is responsible to hold the nucleons in the nucleus.

29. (b) From binding energy curve, the curve reaches peak for $_{26}Fe^{56}$.

30. (c) 1 a.m.u. – 931 MeV

31. (a) α-particle = $_2He^4$. It contains 2 p and 2n. As some mass is converted into binding energy, therefore, mass of α particle is slightly less than sum of the masses of 2p and 2n.

32. (c) Nuclear force is the same between any two nucleons

33. (c) Nuclear forces are short range forces

34. (a) Average binding energy/nucleon in nuclei is of the order of 8 MeV.

35. (d) R = Activity of the sample = $\lambda N = \frac{0.693}{T} \times N$

$R = \frac{0.693}{1.4 \times 10^{17}} \times 2 \times 10^{21} = 10^4$

36. (c) Half life = 10 minutes,

so $600 \xrightarrow{t_{1/2}} 300 \xrightarrow{t_{1/2}} 150$

$\Rightarrow$ Total time = 2 half life = 20 min.

37. (a) $N = N_0 e^{-\lambda t}$ $\lambda_1 = 8\lambda$ $\lambda_2 = \lambda$

$N_1 = N_0 e^{-\lambda_1 t}$ ...(1)

$N_2 = N_0 e^{-\lambda_2 t}$ ...(2)

Dividing Eq. (1) by Eq. (2), we get

$\frac{N_1}{N_2} = e^{\lambda_2 t - \lambda_1 t}$

$\frac{1}{e} = e^{t(\lambda_2 - \lambda_1)}$

$e^{-1} = e^{t[\lambda - 8\lambda]} \Rightarrow e^{-1} = e^{-7\lambda t}$

$t = \frac{1}{7\lambda}$

38. (b) Using equation of radioactivity

$N_1 = N_o e^{-\lambda t}$

40% decay

$60 = N_o e^{-\lambda t_1}$ .....(1)

85% decay

$15 = N_o e^{-\lambda t_2}$ .....(2)

Dividing Eq. (1) by Eq. (2), we get

$\frac{60}{15} = \frac{e^{-\lambda t_1}}{e^{-\lambda t_2}} \Rightarrow 4 = e^{(\lambda t_2 - t_1 \lambda)}$

Take log both side

$\log 4 = \log e^{\lambda(t_2 - t_1)}$

$2 \ln 2 = \lambda(t_2 - t_1) \qquad \because \lambda = \dfrac{\ln 2}{t_{1/2}}$

$2 \ln 2 = \dfrac{\ln 2}{30}(t_2 - t_1)$

$t_2 - t_1 = 60$ minutes

39. (c) As $\dfrac{N_x}{N_y} = \dfrac{1}{7} \Rightarrow \dfrac{N_x}{N_x + N_y} = \dfrac{1}{8} = \left(\dfrac{1}{2}\right)^3$

So, $t = 3T_{1/2} = 3 \times 1.4 \times 10^9$ yrs. $= 4.2 \times 10^9$ yrs.

40. (c) $\underset{N_x}{X} \rightarrow \underset{N_y}{Y(\text{stable})}$

$\dfrac{N_x}{N_y} = \dfrac{1}{7} \Rightarrow \dfrac{N_x}{N_x + N_y} = \dfrac{N}{N_0} = \dfrac{1}{8}$

By using $N = N_0 e^{-\lambda t}$ we have

$\dfrac{N_0}{8} = N_0 e^{-\lambda t} \Rightarrow t = 3 \times 20 \text{ years} = 60 \text{ years}$

41. (b) Active fraction at instant $t_2, \dfrac{1}{2^{t_2/T_{½}}} = \dfrac{1}{3}$

Active fraction at instant $t_1, \dfrac{1}{2^{t_1/T_{1/2}}} = \dfrac{2}{3}$

$\Rightarrow \dfrac{2^{t_2/T_{1/2}}}{2^{t_1/T_{1/2}}} = 2 \Rightarrow 2^{\frac{t_2 - t_1}{T_{1/2}}} = 2^1$

$\Rightarrow t_2 - t_1 = T_{1/2} = 50$ days

42. (a) $N_A = \dfrac{40}{2^{t/20}}$ and $N_B = \dfrac{160}{2^{t/10}}; \dfrac{N_A}{N_B} = 1$

$\Rightarrow 2^{\frac{t}{10} - \frac{t}{20}} = 4 = 2^2 \Rightarrow 2^{\frac{t}{20}} = 2^2 \Rightarrow t = 40 \text{ s}$

43. (b)

	P	Q
No of nuclei, at t = 0	$4N_0$	N_0
Half Life	1 min	2 min
No. of nuclei after time t	N_p	N_Q

The no. of nuclei of P and Q are equal after time t.

$\because N_P = N_Q$

$\Rightarrow 4N_0\left(\dfrac{1}{2}\right)^{\frac{t}{1}} = N_0\left(\dfrac{1}{2}\right)^{\frac{t}{2}}$

$= \left[\dfrac{1}{2}\right]^{t - \frac{t}{2}} = \dfrac{1}{4} \Rightarrow \left[\dfrac{1}{2}\right]^{t/2} = \left[\dfrac{1}{2}\right]^2$

From here, t = 4 min

So, no. of nuclei of P or Q after 4 min

$\Rightarrow \dfrac{4N_0}{2^4} = \dfrac{N_0}{4}$

Hence, no. of nuclei of $= \left(4N_0 - \dfrac{N_0}{4}\right) + \left(N_0 - \dfrac{N_0}{4}\right)$

$= \dfrac{9N_0}{2}$

44. (b) $\left(\dfrac{N}{N_0}\right) = \left(\dfrac{1}{2}\right)^n$ Where n = No. of half lives

$\Rightarrow \dfrac{1}{16} = \left(\dfrac{1}{2}\right)^n$ N = No. of nuclei left.

$\Rightarrow n = 4$ N_0 = No. of initial nuclei.

Let the age of rock be t years

$\Rightarrow n = \dfrac{t}{T_{½}} \Rightarrow t = n \times T_{½} = 4 \times 50 \text{ years} = 200 \text{ years}$

45. (b) Momentum of emitted photon

$= p_{photon} = \dfrac{h\nu}{c}$

From the law of conservation of linear momentum, Momentum of recoil nucleus

$= p_{nucleus} = P_{photon}$

$\therefore Mv = \dfrac{h\nu}{c}$

where v is the recoil speed of the nucleus

or $v = \dfrac{h\nu}{Mc}$ ....(i)

The recoil energy of the nucleus

$= \dfrac{1}{2}Mv^2 = \dfrac{1}{2}M\left(\dfrac{h\nu}{Mc}\right)^2 = \dfrac{h^2\nu^2}{2Mc^2}$ (Using Eq. (i))

46. (c) $A = \lambda N \Rightarrow A_1 = \lambda N_1 \;\&\; A_2 = \lambda N_2$

$(A_1 - A_2) = \lambda(N_1 - N_2) \Rightarrow (N_1 - N_2) = \dfrac{(A_1 - A_2)}{\lambda}$

47. (a) $N = N_0 e^{-\lambda t} \Rightarrow \dfrac{N_0}{e} = N_0 e^{-\lambda(5)} \Rightarrow \lambda = \dfrac{1}{5}$

Now $\dfrac{N_0}{2} = N_0 e^{-\lambda(t)} \Rightarrow t = \dfrac{1}{\lambda}\log_e 2 = 5\log_e 2$

48. (d) As we know

$N = N_0 e^{-\lambda t}$ N = No. of nuclei

So, $\dfrac{N_1}{N_2} = \dfrac{N_0 e^{-\lambda_1 t}}{N_0 e^{-\lambda_2 t}}$ N_o = Initial no. of nuclei

$\Rightarrow \dfrac{1}{e} = \dfrac{e^{-5\lambda t}}{e^{-\lambda t}}$ λ = Decay constant

$\Rightarrow \dfrac{1}{e} = \dfrac{1}{e^{4\lambda t}}$ t = time

$\Rightarrow 4\lambda t = 1 \Rightarrow t = \dfrac{1}{4\lambda}$

49. (c) Decay constant of A = 5λ

Decay constant of B = λ

As we know,

$N = N_0 e^{-\lambda t}$ N_o = Initial no. of nucleus

Since, N_o is same for both.

$\dfrac{N_A}{N_B} = \dfrac{N_0}{N_0}\dfrac{e^{-\lambda_A t}}{e^{-\lambda_B t}} = \dfrac{e^{-5\lambda t}}{e^{-\lambda t}}$

$\Rightarrow \dfrac{1}{e^2} = \dfrac{1}{e^{4\lambda t}} \Rightarrow 4\lambda t = 2 \Rightarrow t = \dfrac{1}{2\lambda}$

50. (a) Let the initial activity of radioactive material be R_0

Decay constant = λ

Activity at time $t_1 \Rightarrow R_1 = R_0 e^{-\lambda t_1}$

Activity at time $t_2 \Rightarrow R_2 = R_0 e^{-\lambda t_2}$

$\Rightarrow \frac{R_1}{R_2} = \frac{e^{-\lambda t_1}}{e^{-\lambda t_2}}$

$= e^{-\lambda(t_1 - t_2)}$

Hence, $R_1 = R_2 e^{-\lambda(t_1 - t_2)}$

51. (c) $t_{1/2} = 1600$ year

$\frac{N}{N_0} = \frac{25}{100} = \left(\frac{1}{2}\right)^1 \Rightarrow n = 2$

$T = n\, t_{1/2}$

$T = 2 \times 1600 = 3200$ year

52. (a) $N = N_0 e^{-\lambda t} \Rightarrow m = m_0 e^{-\lambda t} = m_0 e^{-\lambda(2/\lambda)}$

$= \frac{10}{e^2} = 1.35\,\text{gm}$

53. (c) $t = nT, X = \frac{X_0}{2^n}, n = \frac{t}{T} = \frac{30}{10} = 3$

Active nuclei $X = \frac{4 \times 10^{16}}{(2)^3}$ and decayed nuclei

$x' = (X_0 - X) = 3.5 \times 10^{16}$

54. (b) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n$ n → no. of decays

$\frac{1}{256} = \left(\frac{1}{2}\right)^8 = \left(\frac{1}{2}\right)^n \quad \therefore n = 8$

Time for 8 half life = 100 hours

55. (a) Half life of radioactive substance is inversely proportional to the decay constant.

$T = \frac{0.693}{\lambda}$

$T = \frac{\ln 2}{\lambda}$

56. (a) The life span of atomic hydrogen is fraction of one second.

57. (c) Decay constant $= \frac{0.693}{T_{1/2}} = \frac{0.693}{77}$

$= 0.009$/days

58. (a) $T_{1/2(A)} = 40$ min, $T_{1/2(B)} = 20$ min, t = 80 min

$n_A = \frac{t}{T_{1/2(A)}} = \frac{80}{40} = 2, \qquad n_B = \frac{t}{T_{1/2(B)}} = \frac{80}{20} = 4$

$\frac{N_A}{N_B} = \frac{N_0/2^2}{N_0/2^4} = \frac{16}{4} = 4:1$

59. (a) Half-life time = 30 minutes; Rate of decrease (N) = 5 per second and total time = 2 hours = 120 minutes. Relation for initial and final count rate

$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\text{time/half-life}} = \left(\frac{1}{2}\right)^{120/30} = \left(\frac{1}{2}\right)^4 = \frac{1}{16}.$

Therefore $N_0 = 16 \times N = 16 \times 5 = 80\ s^{-1}$.

60. (d) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{6400/1600} = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$

61. (a) From equation (i), A loses 2 units of charge by emission of alpha particle.

From equation (ii), B has 2 units of charge more than C. Hence, A and C are isotopes as their charge number is same.

62. (c) Curie is a unit of radioactivity

63. (d) Number of half lives, $n = \frac{t}{T} = \frac{6400}{800} = 8$

$\frac{N}{N_0} = \left(\frac{1}{2}\right)^8 = \frac{1}{256}$

64. (d) After two half lives, the activity becomes one fourth.

Activity on 1 – 8 – 91 was 2 micro curie

∴ Activity before two months,

4 × 2 micro–curie = 8 micro curie

65. (a) After gamma radiation the mass number and atomic number remains unchanged

66. (a) α-particle is nucleus of Helium which has two protons and two neutrons. It is He^{2+} nucleus, it has zero electrons.

67. (d) When α particle is emitted, mass number decreased by 4

Atomic number decreased by 2

When β^- particle is emitted, mass number by remains same

Atomic number increases by one

So, ${}^{m}_{n}X \xrightarrow{\alpha} {}^{m-4}_{n-2}Y \xrightarrow{2\beta^{-1}} {}^{m-4}_{n}X$

68. (c) ${}^{A}_{Z}Y \xrightarrow[X\alpha]{2X\beta} A - 4x$

Here, mass no. change & Atomic no. remains same

z – 2x + 2x

Let no. of β particles emitted be 2x

And no. of α particles emitted be x

Emission of 1 β particle increases atomic no by 1

Emission of 1 α particle decreases atomic no by 2

So, Atomic number remains same

Hence, daughter is an isotope of parent.

69. (d) ${}^{A}_{Z}X \xrightarrow{\beta} {}^{A}_{Z+1}Y \xrightarrow{\alpha} {}^{A-4}_{Z-1}B^* \xrightarrow{\gamma} {}^{A-4}_{Z-1}B$

70. (a) In radioactive decay process, the negatively Charged β-particles are the electrons produced as a result of the decay of neutrons inside the nucleus. ${}^{1}_{0}n \rightarrow {}^{1}_{1}p + {}^{0}_{-1}\beta$

71. (a) Emission of electron (e^-) + antineutrino ($\bar{\gamma}$) ⇒ β-decay.

72. (d) ${}_{8}O^{16} + {}_{1}H^{2} \rightarrow {}_{Z}X^{A} + {}_{2}He^{4}$

use conversation of charge and mass

73. (a) Alpha rays are high speed particles and it consists of two protons and two neutrons, all together by the same strong nuclear force that binds the nucleus of any atom.

74. (b) The most penetrating radiation is γ-rays.

75. (b) On emission of one α-particle, atomic number decreases by 2 units and atomic mass number decrease by 4 units.

Here, decrease in mass number = 200 – 168 = 32.

$\therefore$ Number of α-particles = 32/4 = 8.

While the emission of β-particle does not effect the mass number and atomic number increases by 1 unit.

$\therefore$ Number of p–particles= 16 – 10 = 6.

76. (b) ${}_6C^{12} + {}_0n^1 \rightarrow {}_6C^{13} \rightarrow {}_7N^{13} + {}_{-1}\beta^0$ + Energy

77. (d) Two successive β decays increase the charge no. by 2.

78. (d) ${}^{22}_{11}Na \rightarrow X + e^+ + \nu$

Above equation is of β^+ decay

Therefore the atomic number decreases by 1 and mass number remain unchanged

${}^{22}_{11}Na \rightarrow {}^{22}_{10}Ne + e^+ + \nu$

79. (b) ${}^{A}_{Z}X \xrightarrow{B^+} {}_{Z-1}B \xrightarrow{\alpha} {}_{Z-3}C \xrightarrow{B^-} {}_{Z-2}D$

In B⁺ decay atomic number decreases by 1.

In B⁻ decay atomic number increases by 1.

In α decay atomic number decreases by 2.

80. (d) When a uranium isotope is bombarded with neutron the reaction will take place as:-

$${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{89}_{36}Kr + {}^{144}_{56}Bra + 3{}^{1}_{0}n + Q$$

81. (d) B.E. of ${}_2He^4 = 4 \times 7.06 = 28.24$ MeV

B.E. of ${}^7_3Li = 7 \times 5.60 = 39.20$ MeV

${}^7_3Li + {}^1_1H \rightarrow {}_2He^4 + {}_2He^4 + Q$

39.20 + Q = 28.24 × 2 = 56.48

Q = 56.48 – 39.20 = 17.28 MeV

82. (b) ${}^2_1H + {}^2_1H \rightarrow {}^2_1He$ $\qquad$ B.E.$_{He}$ = 28 MeV

B.E$_{(Deutrons)}$ = 2.2 MeV

Energy released in the reaction

= B.E.$_{(Products)}$ – B.E$_{(Reactants)}$

= 28 – 2 × 2.2 MeV

= 28 – 4.4 = 23.6 MeV

83. (b) ${}^2_1H + {}^3_1H \rightarrow {}^4_2He + {}^0_1\eta$

Q (Energy released in reaction)

= B.E $_{products}$ – B.E $_{reactants}$

= B.E. of ${}^4_2He$ – B.E of ${}^2_1He$ – B.E of ${}^3_1He$

Q = c – a – b

84. (c) ${}_6C^{11} \rightarrow {}_5B^{11} + \beta^+ + X$

$\beta^+ = {}_1e^0$ $\qquad$ neutrino

Note: In β-decay, a positron is always accompanied by a neutrino and an electron is always accompanied by an anti-neutrino.

85. (a) $\alpha + 3\beta$

${}^{A-4}X_{Z-2} + 3e^0_{-1}$

mass number = A – 4 + 3 × 0

= A – 4

mass number of daughter element decreased by 4.

86. (b) ${}_{92}U^{235} + {}_0n^1 \rightarrow {}_{56}Ba^{144} + {}_ZX^A + 3{}_0n^1$

235 + 1 = 144 + A + 3 ⇒ A = 89

92 + 0 = 56 + Z + 0 ⇒ Z = 36

Therefore ${}_{36}X^{89} \rightarrow {}_{36}Kr^{89}$

87. (d) In nuclear fission, the chain; reaction is controlled in such way that only one neutron, produced in each fission, causes further fission, Therefore some moderator is used to slow down the neutrons. Heavy water is used for this purpose.

88. (b) Number of nucleon on reactant side = 4

Binding energy for one nucleon = x_1

Binding energy for 4 nucleons = $4x_1$

Similarly on product side binding energy = $4x_2$

Now, Q = change in binding energy = $4(x_2 - x_1)$.

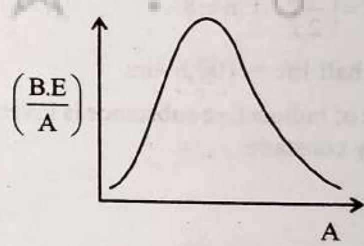
89. (d) Energy released per U

$$= \left(\frac{0.02866}{4}\right)(931\text{ MeV}) = 6.675\text{ MeV}$$

90. (c) Fusion reaction takes place at high temperature because kinetic energy is high enough to overcome the coulomb repulsion between nuclei.[At high temp., K.E. will be more]

91. (a) Ratio of $\dfrac{\text{Mass of fission products}}{\text{Mass of parent nucleus}}$ always less than 1 because some mass gets converted to energy.

92. (b) Fission of nuclei is possible because decreases with mass number at high mass numbers.



93. (b) Product is more stable so that its mass is less than the sum of masses of reactants i.e.

$m_3 < (m_1 + m_2)$

94. (c) Solar energy → fusion of protons into helium.

95. (a) The nuclei of light elements have a lower binding energy than that for the elements of intermediate mass. They are therefore less stable; consequently the fusion of the light elements results in more stable nucleus.

96. (b) The sum of the masses of product nuclei is found to be less than the mass of reactant. This difference (Δm) is converted into energy which is about 200 MeV per fission.

97. (a) The nuclear fission was explained by Bohr and wheeler using liquid drop method.

98. (b) Fission rate $= \dfrac{\text{Total power}}{\text{Energy/fission}} = \dfrac{5}{200 \times 1.6 \times 10^{-13}}$

$= 1.56 \times 10^{11}\ s^{-1}$

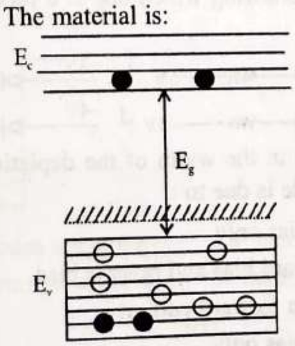
99. (a) Fusion reaction

CHAPTER 14

Semiconductor Electronics

Energy Band in Solids and Basics of Semiconductor

1. The solids which have the negative temperature coefficient of resistance are: (2020)
   a. Insulators only
   b. Semiconductors only
   c. Insulators and semiconductors
   d. Metals
2. A p-n photodiode is fabricated from a semiconductor with a band gap of 2.5 eV. It can detect a signal of wavelength (2009)
   a. 4000 nm b. 6000 nm
   c. 4000Å d. 6000Å
3. A p-n photodiode is made of a material with a band gap of 2.0 eV. The minimum frequency of the radiation that can be absorbed by the material is nearly (2008)
   a. 1×10^{14} Hz b. 20×10^{14} Hz
   c. 10×10^{14} Hz d. 5×10^{14} Hz
4. In the energy band diagram of a material shown below, the open circles and filled circles denote holes and electrons respectively. The material is: (2007)



   a. An insulator b. A metal
   c. An n-type semiconductor d. A p-type semiconductor
5. Carbon, silicon and germanium atoms have four valence electrons each. Their valence and conduction bands are separated by energy band gaps represented by $(E_g)_C$, $(E_g)_{Si}$ and $(E_g)_{Ge}$ respectively. Which one of the following relationships is true in their case? (2005)
   a. $(E_g)_C > (E_g)_{Si}$ b. $(E_g)_C = (E_g)_{Si}$
   c. $(E_g)_C < (E_g)_{Ge}$ d. $(E_g)_C < (E_g)_{Si}$
6. Choose the only false statement from the following (2005)
   a. In conductors, the valence and conduction bands overlap
   b. Substances with energy gap of the order of 10 eV are insulators
   c. The resistivity of a semiconductor increases with increase in temperature
   d. The conductivity of a semiconductor increases with increase in temperature
7. In semiconductors at a room temperature: (2004)
   a. The valence band is completely filled and the conduction band is partially filled
   b. The valence band is completely filled
   c. The conduction band is completely empty
   d. The valence band is partially empty and the conduction band is partially filled
8. A piece of copper and other of germanium are cooled from the room temperature to 80 K, then (1992)
   a. Resistance of each will increase
   b. Resistance of copper will decrease
   c. The resistance of copper will increase while that of germanium will decrease
   d. The resistance of copper will decrease while that of germanium will increase
9. At absolute zero, Si acts as (1988)
   a. Non metal b. Metal
   c. Insulator d. None of these

Intrinsic and Extrinsic Semiconductor (P and N types)

10. The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them. (2021)
    a. current in p-type > current in n-type.
    b. current in n-type > current in p-type.
    c. No current will flow in p-type, current will only flow in n-type.
    d. current in n-type = current in p-type.

11. An intrinsic semiconductor is converted into n-type extrinsic semiconductor by doping it with *(2020-Covid)*
a. Aluminium b. Silver
c. Germanium d. Phosphorous

12. For a p-type semiconductor, which of the following statements is true? *(2019)*
a. Electrons are the majority carriers and trivalent atoms are the dopants.
b. Holes are the majority carriers and trivalent atoms are the dopants.
c. Holes are the majority carriers and pentavalent atoms are the dopants.
d. Electrons are the majority carriers and pentavalent atoms are the dopants.

13. In a n-type semiconductor, which of the following statement is true? *(2013)*
a. Holes are majority carriers and trivalent atoms are dopants
b. Electrons are majority carriers and trivalent atoms are dopants
c. Electron are minority carriers and pentavalent atoms are dopants
d. Holes are minority carriers and pentavalent atoms are dopants

14. C and Si both have same lattice structure, having 4 bonding electrons in each. However, C is insulator whereas Si is intrinsic semiconductor. This is because: *(2012 Pre)*
a. In case of C the valence band is not completely filled at absolute zero temperature.
b. In case of C the conduction band is partly filled even at absolute zero temperature.
c. The four bonding electrons in the case of C lie in the second orbit, whereas in the case of Si they lie in the third.
d. The four bonding electrons in the case of C lie in the third orbit, whereas for Si they lie in the fourth orbit.

15. Pure Si at 500 K has equal number of electron (n_e) and hole (n_h) concentrations of 1.5×10^{16} m^{-3}. Doping by indium increases n_h to 4.5×10^{22} m^{-3}. The doped semiconductor is of: *(2011 Mains)*
a. n-type with electron concentration $n_e = 5 \times 10^{22}$ m^{-3}
b. p-type with electron concentration $n_e = 2.5 \times 10^{10}$ m^{-3}
c. n-type with electron concentration $n_e = 2.5 \times 10^{23}$ m^{-3}
d. p-type having electron concentrations $n_e = 5 \times 10^{9}$ m^{-3}

16. If a small amount of antimony is added to germanium crystal: *(2011 Pre)*
a. It becomes a p-type semiconductor
b. The antimony becomes an acceptor atom
c. There will be more free electrons than holes in the semiconductor
d. Its resistance is increased

17. Which of the following statement is False? *(2010 Pre)*
a. The resistance of intrinsic semiconductor decreases with increase of temperature
b. Pure Si doped with trivalent impurities gives a p-type semiconductor
c. Majority carriers in a n-type semiconductors are holes
d. Minority carriers in a p-type semiconductor are electrons

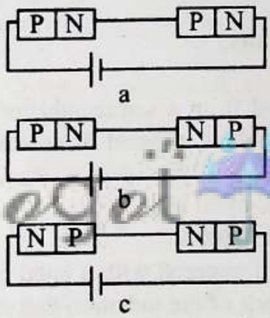
18. To obtain a p-type germanium semiconductor, it must be doped with *(1997)*
a. Indium b. Phosphorus
c. Arsenic d. Antimony

19. When arsenic is added as an impurity to silicon, the resulting material is *(1996)*
a. n-type conductor b. n-type semiconductor
c. p-type semiconductor d. None of these

20. When n-type semiconductor is heated *(1989)*
a. Number of electrons increases while that of holes decreases
b. Number of holes increases while that of electrons decreases
c. Number of electrons and holes remain same
d. Number of electrons and holes increases equally

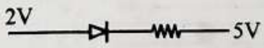
P-N Junction Diode (forward and reverse bias), Diffusion and Drift Current

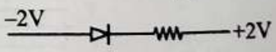
21.

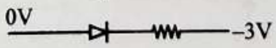


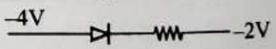
In the given circuits (a), (b) and (c), the potential drop across the two p-n junctions are equal in: *(2022)*
a. Both circuits (a) and (c) b. Circuit (a) only
c. Circuit (b) only d. Circuit (c) only

22. Out of the following which one is a forward biased diode? *(2020-Covid)*

a. 

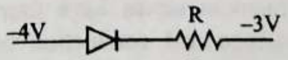
b. 

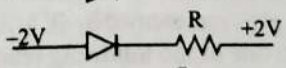
c. 

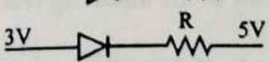
d. 

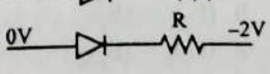
23. The increase in the width of the depletion region in a p-n junction diode is due to : *(2020)*
a. Reverse bias only
b. Both forward bias and reverse bias
c. Increase in forward current
d. Forward bias only

24. Which one of the following represents forward bias diode? *(2017-Delhi, 2006)*

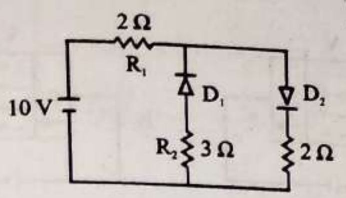
a. 

b. 

c. 

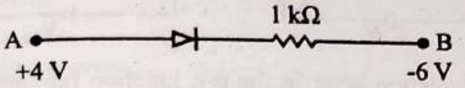
d. 

25. The given circuit has two ideal diodes connected as shown in the figure below. The current flowing through the resistance R_1 will be: *(2016-II)*



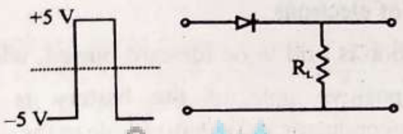
a. 1.43 A
b. 3.13 A
c. 2.5 A
d. 10.0 A

26. Consider the junction diode as ideal. The value of current flowing through AB is: *(2016-I)*

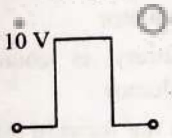


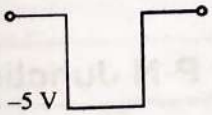
a. 0 A
b. 10^{-2} A
c. 10^{-1} A
d. 10^{-3} A

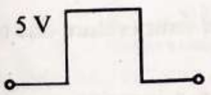
27. If in a p-n junction, a square input signal of 10 V is applied, as shown

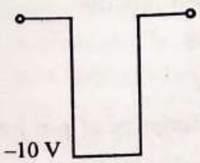


then the output across R_L will be: *(2015)*

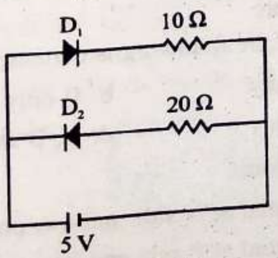
a. 

b. 

c. 

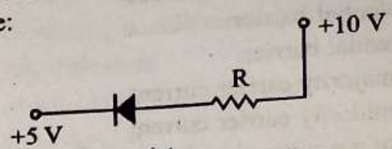
d. 

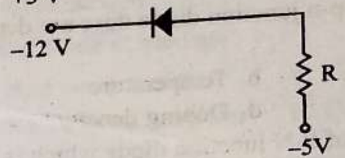
28. Two ideal diodes are connected to a battery as shown in the circuit. The current supplied by the battery is: *(2012 Pre)*

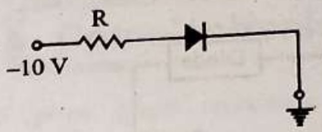


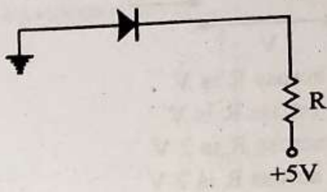
a. 0.75 A
b. Zero
c. 0.25 A
d. 0.5 A

29. In the following figure, the diodes which are forward biased, are: *(2011 Mains)*

a. 

b. 

c. 

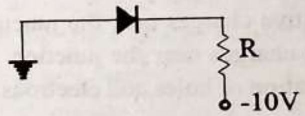
d. 

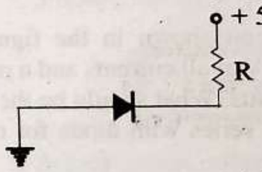
a. (c) only
b. (b) and (a)
c. (b) and (d)
d. (a), (b) and (d)

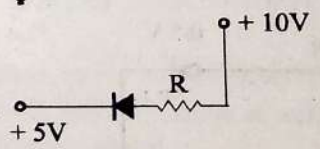
30. In forward biasing of the p-n junction: *(2011 Pre)*
   a. The positive terminal of the battery is connected to p-side and the depletion region becomes thick
   b. The positive terminal of the battery is connected to n-side and the depletion region becomes thin
   c. The positive terminal of the battery is connected to n-side and the depletion region becomes thick
   d. The positive terminal of the battery is connected to p-side and the depletion region becomes thin

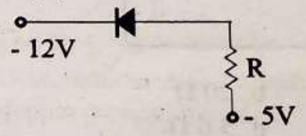
31. Application of a forward bias to a p-n junction *(2005)*
   a. Widens the depletion zone
   b. Increases the potential difference across the depletion zone
   c. Increases the number of donors on the n side
   d. Decreases the electric field in the depletion zone

32. Of the diodes shown in the following diagrams, which one of the diode is reverse biased? *(2004)*

a. 

b. 

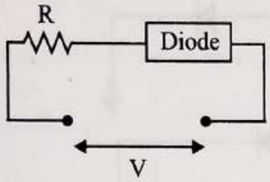
c. 

d. 

33. Reverse bias applied to a junction diode: *(2003)*
a. Lowers the potential barrier
b. Raises the potential barrier
c. Increases the majority carrier current
d. Increases the minority carrier current

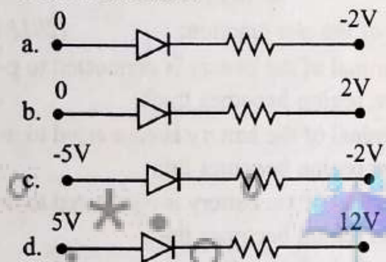
34. Barrier potential of a p-n junction diode does not depend on: *(2003)*
a. Diode design
b. Temperature
c. Forward bias
d. Doping density

35. For the given circuit of P-N junction diode which is correct: *(2002)*



a. In F.B. the voltage across R is V
b. In R.B. the voltage across R is V
c. In F.B. the voltage across R is 2 V
d. In R.B. the voltage across R is 2 V

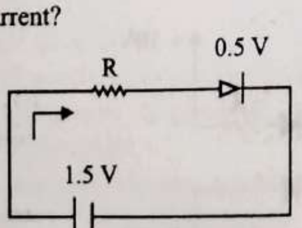
36. From the following diode circuit. Which diode is in forward biased condition: *(2000)*



37. One part of a device is connected with the negative terminal of a battery and another part is connected with the positive terminal of a battery. If their ends now altered, current does not flow in circuit, then the device will be: *(1998)*
a. P-N Junction
b. Transistor
c. Zener diode
d. Triode

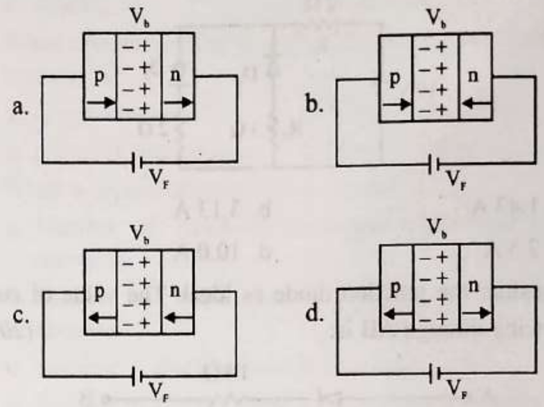
38. The cause of potential barrier in a P-N junction diode is: *(1998)*
a. Concentration of positive and negative ions near the junction
b. Concentration of positive charges near the junction
c. Depletion of negative charges near the junction
d. Increment in concentration of holes and electrons near the junction

39. The diode used in the circuit shown in the figure has a constant voltage drop at 0.5 V at all currents and a maximum power rating of 100 milli watts. What should be the value of the resistor R, connected in series with diode for obtaining maximum current? *(1997)*



a. 6.76 Ω
b. 20 Ω
c. 5 Ω
d. 5.6 Ω

40. In the case of forward biasing of p-n junction, which one of the following figures correctly depicts the direction of flow of carriers? *(1995)*



41. The depletion layer in the p-n junction region is caused by *(1991)*
a. Drift of holes
b. Diffusion of charge carriers
c. Migration of impurity ions
d. Drift of electrons

42. p-n junction is said to be forward biased, when *(1988)*
a. The positive pole of the battery is joined to the p-semiconductor and negative pole to the n-semiconductor
b. The positive pole of the battery is joined to the n-semiconductor and p-semiconductor
c. The positive pole of the battery is connected to n-semiconductor and p-semiconductor
d. A mechanical force is applied in the forward direction.

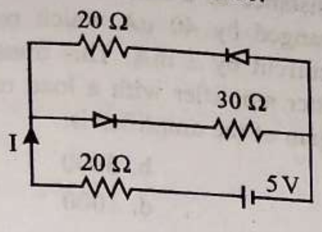
Characteristics of P-N Junction

43. In a p-n junction diode, change in temperature due to heating *(2018)*
a. Does not affect resistance of p-n junction
b. Affects only forward resistance
c. Affects only reverse resistance
d. Affects the overall V–I characteristics of p-n junction

44. The barrier potential of a p-n junction depends on: *(2014)*
A. Type of semiconductor material
B. Amount of doping
C. Temperature
Which one of the following is correct?
a. A and B only
b. B only
c. B and C only
d. A, B and C

45. In a PN junction: *(2002)*
a. High potential at N side and low potential at P side
b. High potential at P side and low potential at N side
c. P and N both are at same potential
d. Undetermined

46. The current (I) in the circuit will be: (2001)



a. $\frac{5}{40}$ A
b. $\frac{5}{50}$ A
c. $\frac{5}{10}$ A
d. $\frac{5}{20}$ A

47. Depletion layer has (for an unbiased PN junction): (1999)

a. Electrons
b. Holes
c. Static Ions
d. Neutral atoms

Diode as a Rectifier

48. In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be: (2022)

a. 120 Hz
b. 0
c. 30 Hz
d. 60 Hz

49. The peak voltage in the output of a half wave diode rectifier fed with a sinusoidal signal without filter is 10 V. The d.c. component of the output voltage is: (2004)

a. $\frac{10}{\pi}$ V
b. 10 V
c. $\frac{20}{\pi}$ V
d. $\frac{10}{\sqrt{2}}$ V

50. If a full wave rectifier circuit is operating from 50 Hz mains, the fundamental frequency in the ripple will be: (2003)

a. 25 Hz
b. 50 Hz
c. 70.7 Hz
d. 100 Hz

Zener Diode

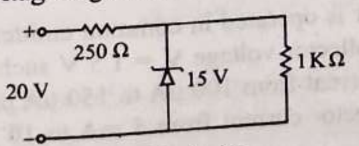
51. Consider the following statements (A) and (B) and identify the correct answer. (2021)

(A) A zener diode is Connected in reverse bias, when used as a voltage regulator.

(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.

a. (A) and (B) both are incorrect.
b. (A) is correct and (B) is incorrect.
c. (A) is incorrect but (B) is correct.
d. (A) and (B) both are correct.

52. A Zener diode, having breakdown voltage equal to 15 V, is used in a voltage regulator circuit shown in figure.



(2011 Mains)

The current through the diode is:

a. 10 mA
b. 15 mA
c. 20 mA
d. 5 mA

53. Zener diode is used for: (2005)

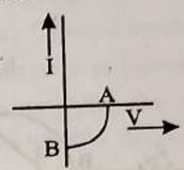
a. Producing oscillations in an oscillator
b. Amplification
c. Stabilisation
d. Rectification

54. Zener diode is used as: (1999)

a. Half wave rectifier
b. Full wave rectifier
c. A.C. voltage stablizer
d. D.C. voltage stablizer

Optoelectronic Devices

55. The given graph represents V-I characteristic for a semiconductor device. (2014)



Which of the following statement is correct?

a. It is V–I characteristic for solar cell where point A represents open circuit voltage and point B short circuit current
b. It is for a solar cell and points A and B represent open circuit voltage and current, respectively
c. It is for a photo diode and points A and B represent open circuit voltage and current, respectively
d. It is for a LED and points A and B represents open circuit voltage and short circuit current respectively

56. In a p-n junction photo cell, the value of the photo electromotive force produced by monochromatic light is proportional to: (2005, 2004)

a. The intensity of the light falling on the cell
b. The frequency of the light falling on the cell
c. The voltage applied at the p-n junction
d. The barrier voltage at the p-n junction

57. For an electronic valve, the plate current I and plate voltage V in the space charge limited region are related as (1992)

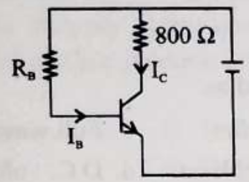
a. I is proportional to $V^{3/2}$
b. I is proportional to $V^{2/3}$
c. I is proportional to V
d. I is proportional to V^2

Transistor

58. For transistor action, which of the following statements is correct? (2020)

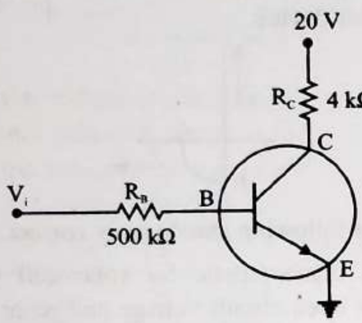
a. Base, emitter and collector regions should have same size.
b. Both emitter junction as well as the collector junction are forward biased.
c. The base region must be very thin and lightly doped.
d. Base, emitter and collector regions should have same doping concentrations.

59. A n-p-n transistor is connected in common emitter configuration (see figure) in which collector voltage drop across load resistance (800 Ω) connected to the collector circuit is 0.8 V. The collector current is, *(2020-Covid)*



a. 0.1 mA b. 1 mA
c. 0.2 mA d. 2 mA

60. In the circuit shown in the figure, the input voltage V_i is 20 V, $V_{BE} = 0$ and $V_{CE} = 0$. The values of I_B, I_C and β are given by *(2018)*



a. $I_B = 20\ \mu A$, $I_C = 5 mA$, $\beta = 250$
b. $I_B = 25\ \mu A$, $I_C = 5 mA$, $\beta = 200$
c. $I_B = 40\ \mu A$, $I_C = 10\ mA$, $\beta = 250$
d. $I_B = 40\ \mu A$, $I_C = 5 mA$, $\beta = 125$

61. In a common emitter transistor amplifier the audio signal voltage across the collector is 3 V. The resistance of collector is 3 kΩ. If current gain is 100 and the base resistance is 2 kΩ, the voltage and power gain of the amplifier is: *(2017-Delhi)*

a. 15 and 200 b. 150 and 15000
c. 20 and 2000 d. 200 and 1000

62. A npn transistor is connected in common emitter configuration in a given amplifier. A load resistance of 800 Ω is connected in the collector circuit and the voltage drop across it is 0.8 V. If the current amplification factor is 0.96 and the input resistance of the circuit is 192 Ω, the voltage gain and the power gain of the amplifier will respectively be: *(2016-I)*

a. 4, 3.84 b. 3.69, 3.84
c. 4, 4 d. 4, 3.69

63. The input signal given to a CE amplifier having a voltage gain of 150 is $V_i = 2\cos\left(15t + \frac{\pi}{3}\right)$. The corresponding output signal will be: *(2015 Pre)*

a. $300\cos\left(15t + \frac{4\pi}{3}\right)$ b. $300\cos\left(15t + \frac{\pi}{3}\right)$
c. $75\cos\left(15t + \frac{2\pi}{3}\right)$ d. $2\cos\left(15t + \frac{5\pi}{6}\right)$

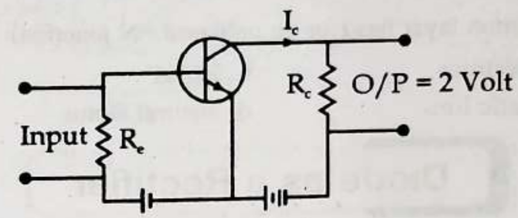
64. In a common emitter (CE) amplifier having a voltage Gain G, the transistor used has transconductance 0.03 mho and current gain 25. If the above transistor is replaced with another one with transconductance 0.02 mho and current gain 20, the voltage gain will be: *(2013)*

a. 5/4 G b. 2/3 G
c. 1.5 G d. 1/3 G

65. The input resistance of a silicon transistor is 100 Ω. Base current is changed by 40 μA which results in a change in collector current by 2 mA. This transistor is used as a common emitter amplifier with a load resistance of 4 kΩ. The voltage gain of the amplifier is: *(2012 Mains)*

a. 2000 b. 3000
c. 4000 d. 1000

66. In a CE transistor amplifier, the audio signal voltage across the collector resistance of 2 kΩ is 2 V. If the base resistance is 1 kΩ and the current amplification of the transistor is 100, the input signal voltage is: *(2012 Pre)*



a. 0.1 V b. 1.0 V
c. 1 mV d. 10 mV

67. A transistor is operated in common emitter configuration at $V_C = 2$ V such that a change in the base current from 100 μA to 300 μA produces a change in the collector current from 10 mA to 20 mA. The current gain is: *(2011 Pre, 2009)*

a. 50 b. 75
c. 100 d. 25

68. For transistor action *(2010 Mains)*

1. Base, emitter and collector regions should have similar size and doping concentrations.
2. The base region must be very thin and lightly doped.
3. The emitter-base junction is forward biased and base-collector junction is reverse biased.
4. Both the emitter-base junction as well as the base collector junction are forward biased.

Which one of the following pairs of statements is correct?

a. 4, 1 b. 1, 2
c. 2, 3 d. 3, 4

69. A common emitter amplifier has a voltage gain of 50, an input impedance of 100 Ω and an output impedance of 200 Ω. The power gain of the amplifier is: *(2010 Pre, 2007)*

a. 50 b. 500
c. 1000 d. 1250

70. The voltage gain of an amplifier with 9% negative feedback is 10. The voltage gain without feedback will be: *(2008)*

a. 100 b. 90
c. 10 d. 1.25

71. A transistor is operated in common emitter configuration at constant collector voltage $V_c = 1.5$ V such that a change in the base current from 100 μA to 150 μA produces a change in the collector current from 5 mA to 10 mA. The current gain (β) is: *(2006)*

a. 50 b. 75
c. 100 d. 125

72. A n-p-n transistor conducts when: *(2003)*

a. Both collector and emitter are positive with respect to the base

b. Collector is positive and emitter is negative with respect to the base

c. Collector is positive and emitter is at same potential as the base

d. Both collector and emitter are negative with respect to the base

73. For a transistor $\frac{I_C}{I_E} = 0.96$, then current gain for common emitter configuration: *(2002)*

a. 12 b. 6

c. 48 d. 24

74. For a common emitter circuit if $\frac{I_C}{I_E} = 0.98$ then current gain for common emitter circuit will be: *(2001)*

a. 49 b. 98

c. 4.9 d. 25.5

75. The correct relation for α and β for a transistor: *(2000)*

a. $\beta = \frac{1-\alpha}{\alpha}$ b. $\beta = \frac{\alpha}{1-\alpha}$

c. $\alpha = \frac{\beta-1}{\beta}$ d. $\alpha\beta = 1$

76. Common emitter circuit is used as amplifier, its current gain is 50. If input resistance is 1 kΩ and input voltage is 5 volt then output current will be: *(1998)*

a. 250 mA b. 30 mA

c. 50 mA d. 100 mA

77. The correct relationship between the two current gains α and β in a transistor is *(1997)*

a. $\alpha = \frac{\beta}{1+\beta}$ b. $\alpha = \frac{1+\beta}{\beta}$

c. $\beta = \frac{\alpha}{1+\alpha}$ d. $\beta = \frac{\alpha}{\alpha-1}$

78. When *npn* transistor is used as an amplifier, then *(1996)*

a. Electrons move from collector to base

b. Holes move from base to emitter

c. Electrons move from emitter to collector

d. Hole move from emitter to base

79. When using a triode, as an amplifier, the electrons are emitted by *(1996)*

a. Grid and collected by cathode only

b. Cathode and collected by the anode only

c. Anode and collected by cathode only

d. Anode and collected by the grid and by cathode.

80. The part of the transistor which is heavily doped to produce large number of majority carriers is: *(1993)*

a. Emitter

b. Base

c. Collector

d. Any of the above depending upon the nature of transistor

81. For amplification by a triode, the signal to be amplified is given to *(1992)*

a. The cathode b. The grid

c. The glass envelope d. The anode

82. To use a transistor as an amplifier *(1991)*

a. The emitter base junction is forward biased and the base collector junction is reversed biased

b. No bias voltage is required

c. Both junction are forward biased

d. Both junction are reverse biased

83. In a common base amplifier the phase difference between the input signal voltage and the output voltage is: *(1990)*

a. 0 b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$ d. π

84. When a triode is used as an amplifier the phase difference between the input signal voltage and the output is *(1990)*

a. 0 b. π

c. $\pi/2$ d. $\pi/4$

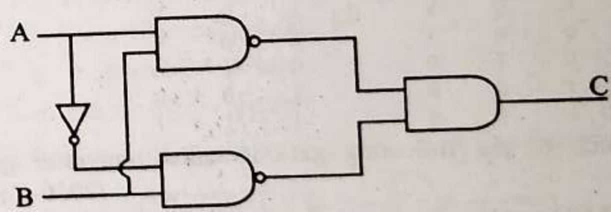
85. Radiowaves of constant amplitude can be generated with *(1989)*

a. FET b. Filter

c. Rectifier d. Oscillator

Digital Electronics and Logic Gates

86.



The truth table for the given logic circuit is: *(2022)*

a.

A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

b.

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

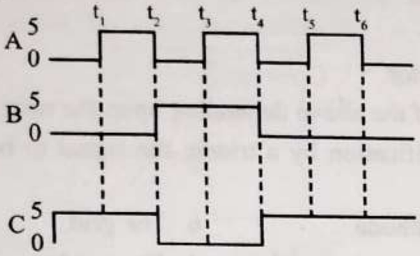
c.

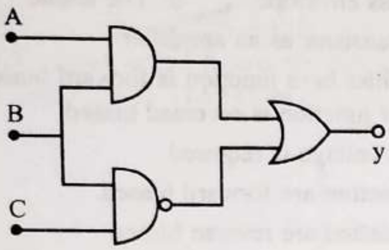
A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

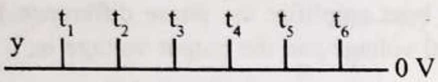
d.

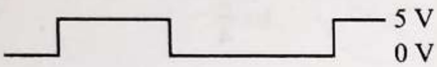
A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

87. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y? *(2021)*

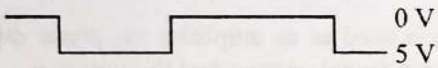




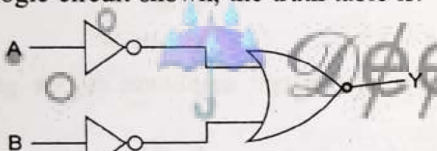
a. 

b. 

c. 

d. 

88. For the logic circuit shown, the truth table is: *(2020)*



a.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

b.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

c.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

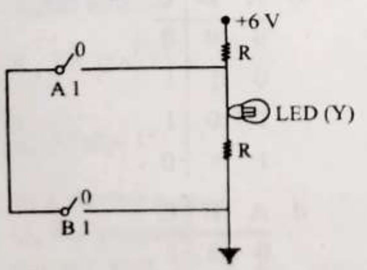
d.

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

89. Which of the following gate is called universal gate? *(2020-Covid)*

a. AND gate b. NAND gate
c. NOT gate d. OR gate

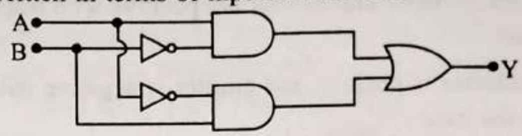
90.



The correct Boolean operation represented by the circuit diagram drawn is: *(2019)*

a. AND b. OR
c. NAND d. NOR

91. In the combination of the following gates the output Y can be written in terms of inputs A and B as *(2018)*

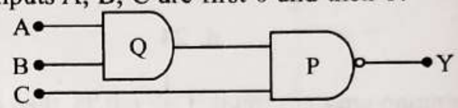


a. $\overline{A.B} + A.B$ b. $A.\overline{B} + \overline{A}.B$
c. $\overline{A.B}$ d. $\overline{A+B}$

92. The given electrical network is equivalent to: *(2017-Delhi)*

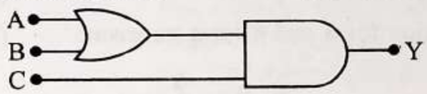
a. OR gate b. NOR gate
c. NOT gate d. AND gate

93. What is the output Y in the following circuit, when all the three inputs A, B, C are first 0 and then 1? *(2016 - II)*



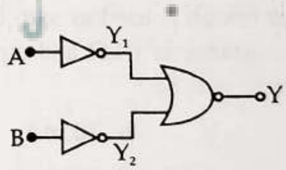
a. 1, 0 b. 1, 1
c. 0, 1 d. 0, 0

94. To get output 1 for the following circuit, the correct choice for the input is: *(2016-I, 2012 Mains, 2010 Pre)*



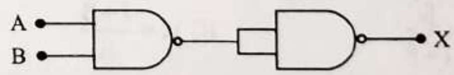
a. A = 0, B = 1, C = 0 b. A = 1, B = 0, C = 0
c. A = 1, B = 1, C = 0 d. A = 1, B = 0, C = 1

95. Which logic gate is represented by the following combination of logic gates? *(2015)*



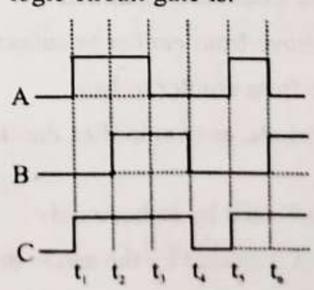
a. NAND b. AND
c. NOR d. OR

96. The output (*X*) of the logic circuit shown in figure will be: *(2013)*



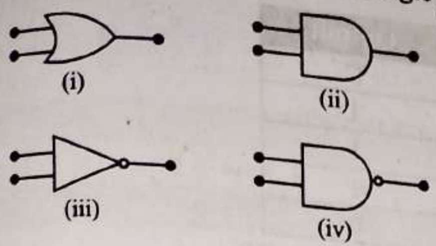
a. $X = \overline{A+B}$ b. $X = \overline{\overline{A}.\overline{B}}$
c. $X = \overline{A.B}$ d. $X = A.B$

97. The figure shows a logic circuit with two inputs A and B and the output C. The voltage wave forms across A, B and C are as given. The logic circuit gate is: *(2012 Pre)*



a. NAND gate b. OR gate
c. NOR gate d. AND gate

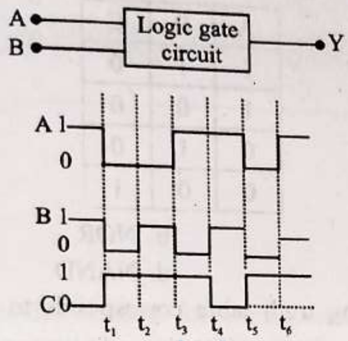
98. Symbolic representation of four logic gates are shown as



Pick out which ones are for AND, NAND and NOT gates, respectively (2011 Pre)

a. (ii), (iii) and (iv)
b. (iii), (ii) and (i)
c. (iii), (ii) and (iv)
d. (ii), (iv) and (iii)

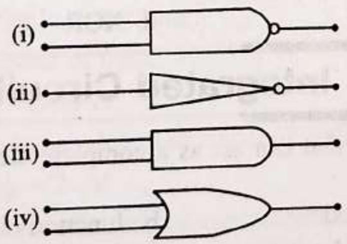
99. The following figure shows a logic gate circuit with two inputs A and B and the output Y. The voltage waveforms of A, B and Y are as shown below



The logic gate is: (2010 Mains)

a. NOR gate
b. OR gate
c. AND gate
d. NAND gate

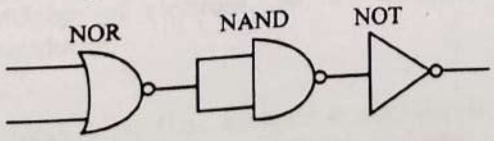
100. The symbolic representation of four logic gates are given below:



The logic symbols for OR, NOT and NAND gates are respectively: (2009)

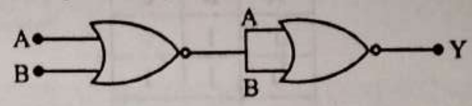
a. (iv), (i), (iii)
b. (iv), (ii), (i)
c. (i), (iii), (iv)
d. (iii), (iv), (ii)

101. The following circuit is equivalent to: (2008)



a. OR gate
b. AND gate
c. NAND gate
d. NOR gate

102. In the following circuit, the output Y for all possible inputs A and B is expressed by the truth table: (2007)



a.

A	B	Y
0	1	1
0	1	1
1	0	1
1	1	0

b.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

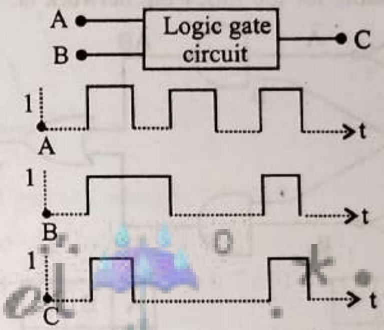
c.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

d.

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

103. The following figure shows a logic gate circuit with two inputs A and B and the output C. The voltage waveforms of A, B and C are as shown below: (2006)



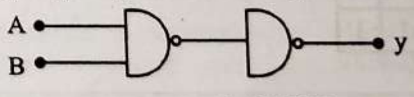
The logic circuit gate is :

a. AND gate
b. NAND gate
c. NOR gate
d. OR gate

104. The output of OR gate is 1: (2004)

a. If either or both inputs are 1
b. Only if both inputs are 1
c. If either input is zero
d. If both inputs are zero

105. Following diagram performs the logic function of: (2003)



a. AND gate
b. NAND gate
c. OR gate
d. XOR gate

106. The given truth table is for which logic gate: (2002)

A	B	Y
1	1	0
0	1	1
1	0	1
0	0	1

a. NAND
b. XOR
c. NOR
d. OR

107. Following truth table represent which logic gate: *(2001)*

A	B	C
1	1	0
0	1	1
1	0	1
0	0	1

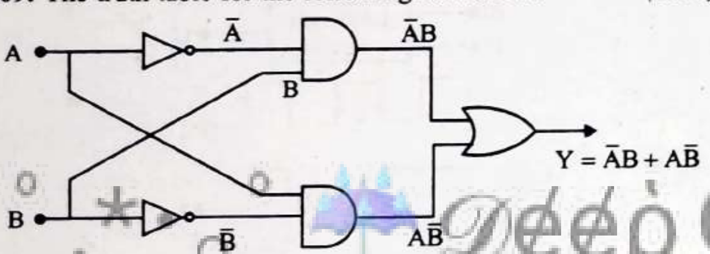
a. XOR b. NOT
c. NAND d. AND

108. Given Truth table is correct for: *(2000)*

A	B	Y
1	1	1
1	0	0
0	1	0
0	0	0

a. NAND b. AND
c. NOR d. OR

109. The truth table for the following network is: *(1999)*



a.

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

b.

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

c.

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

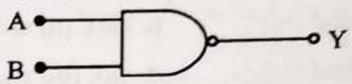
d. None of the above

110. Following table is for which logic gate: *(1998)*

Input		Output
A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

a. AND b. OR
c. NAND d. NOT

111. Following logic gate is: *(1998, 1996)*



a. AND b. NAND
c. EX-OR d. OR

112. The following truth table belongs to which one of the following four gates? *(1997,95)*

A	B	C
1	1	0
1	0	0
0	1	0
0	0	1

a. XOR b. NOR
c. OR d. NAND

113. The following truth table corresponds to the logical gate *(1991)*

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

a. NAND b. OR
c. AND d. XOR

Integrated Circuit

114. The device that can act as a complete electronic circuit is: *(2010 Pre)*

a. Zener diode b. Junction diode
c. Integrated circuit d. Junction transistor

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	c	d	d	a	c	d	d	c	b	d	b	d	c	d	c	c
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
a	b	d	a	c	a	d	c	b	c	d	b	d	c	b	b	a
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
a	a	a	a	c	b	b	a	d	d	a	b	c	d	a	d	b
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
d	c	d	a	a	a	c	b	d	b	a	a	b	a	d	a	c
69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
d	a	c	b	d	a	b	a	a	c	b	a	b	a	a	b	d
86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102
d	c	d	b	c	b	b	a	d	b	d	b	d	d	b	d	c
103	104	105	106	107	108	109	110	111	112	113	114					
a	a	a	a	c	b	b	c	b	b	b	c					

Explanations

1. **(c)** For insulators and for semiconductors, temperature coefficient of resistance is negative but for metals temperature coefficient of resistance is positive.

2. **(c)** Band gap = 2.5 eV

   The wavelength corresponding to 2.5 eV is-

   $$\lambda = \frac{hc}{E} = \frac{12400\ \text{Å}}{2.5} = 4960\text{Å}$$

   $\therefore$ 4000Å can excite this.

   Where: h: Planks's constant

   c: Speed of light

   hc = 12400 Å

   So photodiode can detect of signal of 4000 Å wavelength.

3. **(d)** Band gap = 2 eV

   Wavelength of radiation corresponding to this energy,

   $$\lambda = \frac{hc}{E} = \frac{12400\ \text{Å}}{2} = 6200\ \text{Å}$$

   The frequency of this radiation is-

   $$\nu = \frac{c}{\lambda} = \frac{3\times10^{8}\,\text{ms}^{-1}}{6200\times10^{-10}\,\text{m}} = 5\times10^{14}\,\text{Hz}$$

4. **(d)** In the given figure, the number of holes is more than the number of electrons. So, the material is a p-type semiconductor.

   Or

   An energy level (i.e., acceptor energy level) is near the valance band. So it is p-type of semiconductor.

5. **(a)** Band gap of carbon is 5.5 eV.

   Band gap of silicon is 1.1 eV.

   Band gap of germanium is 0.7 eV.

   $(E_g)_C > (E_g)_{si} > (E_g)_{Ge}$

   Hence, $(E_g)_C > (E_g)_{Si}$

6. **(c)** Resistivity of a semiconductor decreased with increase in the temperature.

   Semiconductor shows negative temperature of coefficient of resistivity at room temperature.

7. **(d)** In semiconductors at room temperature the electrons get enough energy so that they able to over come the forbidden gap. Thus at room temperature the valence band is partially empty and conduction band is partially filled. Conduction band in semiconductor is completely empty only at 0 K.

8. **(d)** Germanium is semiconductor which on decreasing temperature resistance increases, whereas Copper is a conductor so its resistance decreases on decreasing temperature as thermal agitation decreases.

9. **(c)** At absolute zero (0 K), all semiconductors act as insulators.

   At T = 0K, movement of charge carriers are not present hence conductivity become zero.

10. **(b)** Current can be given as,

    $I = neAv_d$

    $I = neA(\mu E)$ $\quad (\because v_d$ is drift velocity, $v_d = \mu E)$

    $\because$ In N-type, e^- are majority charge carriers whereas in P-type, hole (h) are majority charge carriers.

$$\Rightarrow \frac{I_n}{I_p} = \frac{n_e e A \mu_e E}{n_r e A \mu_r E}$$

$$\frac{I_n}{I_p} = \frac{\mu_e}{\mu_r} \qquad (\because \text{ Concentration is same i.e., } n_e = n_h)$$

Since, mobility of electrons is greater than mobility of holes.

i.e., $\mu_e > \mu_h$

$$\Rightarrow \frac{I_n}{I_p} > 1 \Rightarrow I_n > I_p$$

11. (d) Phosphorous is a pentavalent element. For n-type semiconductor intrinsic semiconductor is doped by pentavalent impurity.

12. (b) In p-type semiconductor, an intrinsic semiconductor is doped with trivalent impurities, that creates deficiencies of valence electrons called holes which are majority charge carriers.

For p-type: Majority ⇒ Holes
Minority ⇒ Electrons
Dopants ⇒ Trivalent atoms.

13. (d) The n-type semiconductor, can be produced by doping an impurity atom of valence 5 so, electrons are majority charge carriers and holes are minority charge carrier.

For n-type: Majority ⇒ Electrons
Minority ⇒ Holes
Dopants ⇒ Pentavalent atoms

14. (c) In the case of silicon electrons are in the third orbit, therefore it is easy to remove electrons from silicon than carbon.

Energy band gap of C (5.5 eV) > Energy band of Si (1.1 eV)

15. (d) P-type semiconductor is obtained when Si or Ge is doped with trivalent element like Al, B, In, etc., i.e. Group 14 with Group 13.

$\because$ Intrinsic concentration $(n_i) = 1.5 \times 10^{16} m^{-3}$

Hole concentration $(n_h) = 4.5 \times 10^{22}\ m^{-3}$

Using $n_e n_h = n_i^2$

$$n_e = \frac{n_i^2}{n_h} = \frac{(1.5\times10^{16})^2}{4.5\times10^{22}}$$

$n_e = 5 \times 10^9\ m^{-3}$

16. (c) Antimony is element of group 15 whereas Germanium is element of group 14 so, Antimony has one more valence electron than Germanium.

Hence, the crystal has more free electrons than holes in the semi conductor (i.e., n-type semiconductor).

17. (c) Majority carriers in a n-type semi conductors are electrons.

Minority carriers in a n-type semiconductor are holes.

Pure Si doped with trivalent impurities gives a p-type whereas doped with pentavalend impurities gives a n-type semiconductors.

18. (a) In p type germanium semiconductor, it must be doped with a trivalent impurity atom. Since indium is a third group member, therefore germanium must be doped in indium.

19. (b) Arsenic is pentavalent, therefore when added with silicon it leaves one electron as a free electron. In this case the conduction of electricity is due to motion of electrons, so the resulting material is n-type semiconductor.

20. (d) When n-type semiconductor is heated then covalent bonds start to break and electron-hole pair is generated. It results in equal number of electron and holes.

21. (a) In (a) & (c) circuits, both the junctions are in similar biasing conditions so they equal resistances. Since both are in series, therefore equal potential will drop across them.

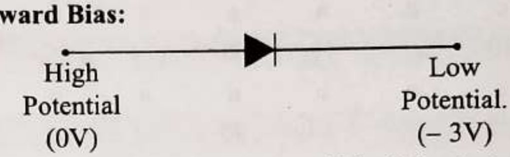
Depletion layer → static ions

22. (c) For a forward bias p-n junction diode:

- p-type should be connected with high potential.
- n-type should be connected with low potential.

So, in given options-c, p-side is connected with high potential (0V) and n-side is connected with low potential (– 3V).

Forward Bias:



23. (a) In a p-n junction diode the width of the depletion region increases due to reverse biasing.

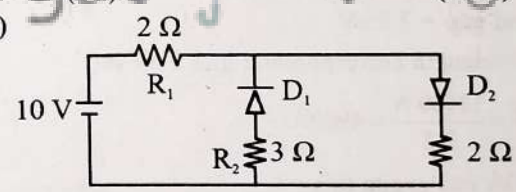
24. (d) For a forward bias p-n junction diode:

- p-type should be connected with high potential.
- n-type should be connected with low potential.

So, in given options-d, p-side is connected with high potential (0V) and n-side is connected with low potential (– 2V).

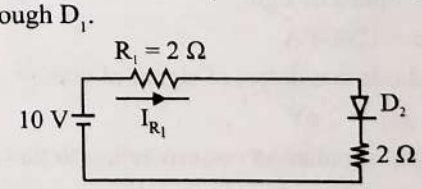
Forward Bias:

25. (c)



D_1 and D_2 are ideal diodes.

D_2 is forward biased and D_1 is reverse biased, current do not flow through D_1.

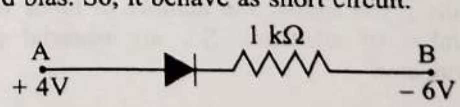


As D_2 is ideal so behave as short circuit.

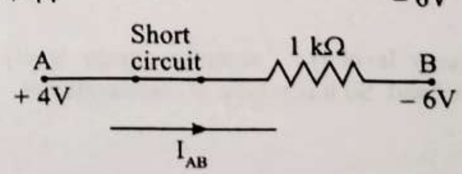
$$\therefore\ I_{R_1} = \frac{10}{2+2}$$

$I_{R_1} = 2.5$ A

26. (b) The junction diode in the given arrangement is ideal and forward bias. So, it behave as short circuit.



⇒

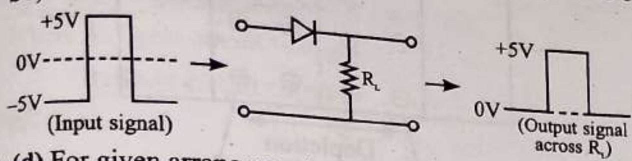


$\therefore I_{AB} = \frac{\Delta V}{R} = \frac{4-(-6)}{1\times10^{3}}$

$I_{AB} = \frac{10}{1000} = 0.01\ A = 10^{-2} A$

27. **(c)** Diode is reverse biased for negative voltage i.e., – 5V to 0V. Diode is forward biased for positive voltage i.e., 0V to 5V.

So,

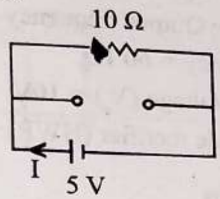


28. **(d)** For given arrangement

D_1 (Forward Biased) → Short circuit

D_2 (Reverse Biased) → Open circuit

So, circuit becomes



$\therefore I = \frac{5}{10} = 0.5\ A$

29. **(b)** For a forward bias p-n junction diode:

- p-type should be connected with high potential.
- n-type should be connected with low potential.

So, in given options-b, p-side is connected with high potential (– 5V) and n-side is connected with low potential (– 12V).

Forward Bias:

30. **(d)** For forward biased p-n junction diodes:

- Positive terminal of battery is connected to p-side.
- Negative terminal of battery is connected to n-side.
- Depletion region becomes thin.
- Barrier voltage decreases.
- Forward current increases.

31. **(c)** In forward biasing:

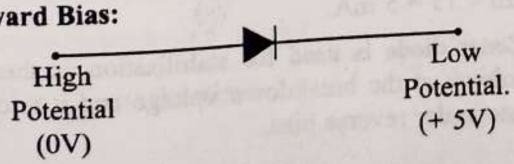
- Potential barrier reduces
- Depletion layer reduces
- Increases the number of donor on the n-side

32. **(b)** For a reverse bias p-n junction diode:

- p-type should be connected with low potential.
- n-type should be connected with high potential.

So, in given options-b, p-side is connected with low potential (0V) and n-side is connected with high potential (+ 5V).

Forward Bias:



33. **(b)** In reverse biasing:

- Potential barrier increases
- Depletion layer widens
- Decreases the number of donor on the n-side

34. **(a)** The potential difference required to move the charge carriers through the electric field is called the barrier potential. It is also called barrier hill. It depends on the type of semiconductor material, amount of doping, temperature and biasing of diode.

- It does not depends on diode design.
- Barrier potential for Si at room temperature = 1.1eV
- Barrier potential for Ge at room temperature = 0.7 eV

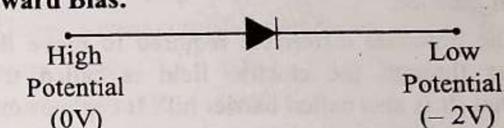
35. **(a)** In forward biasing, the resistance of *p-n* junction diode is very low to the flow of current. So practically all the voltage will be dropped across the resistance R, i.e., voltage across R will be V. In reverse biasing, the resistance of *p-n* junction diode is very high. So the voltage drop across R is zero.

36. **(a)** For a forward bias p-n junction diode:

- p-type should be connected with high potential.
- n-type should be connected with low potential.

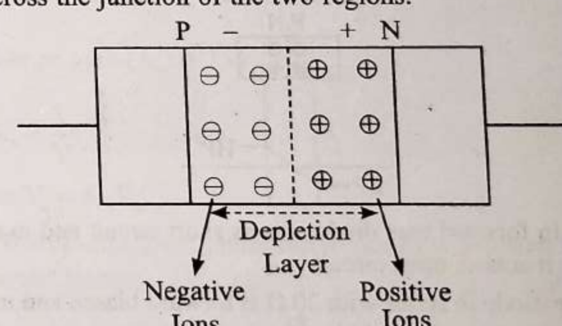
So, in given options-a, p-side is connected with high potential (0V) and n-side is connected with low potential (– 2V).

Forward Bias:



37. **(a)** In p-n junction diodes, In first case diode is forward biased, while after altering the ends, diode is reverse biased that results in very small current flow through the diode. It is unidirectional device (only conduct in forward bias).

38. **(a)** The loss of electrons from the n-region and the gain of electron by the p-region causes a difference of potential across the junction of the two regions.



39. **(c)** Voltage drop across diode (V_D) = 0.5 V;

Maximum power rating of diode (P) = 100 mW = 100×10^{-3} W

And source voltage (V_s) = 1.5 v.

The resistance of diode (R_D) = $\frac{V_D^2}{P} = \frac{(0.5)^2}{100\times10^{-3}} = 2.5\ \Omega$

Diode current (I_D) = $\frac{V_D}{R_D} = \frac{0.5}{2.5} = 0.2\,A$

Total resistance in circuit (R) = $\frac{V_s}{I_D} = \frac{1.5}{0.2} = 7.5\ \Omega$

And the value of the series resistor = Total resistance of the circuit – Resistance of diode = 7.5 – 2.5 = 5Ω.

40. **(b)** As soon as the p–n junction is formed, there is an immediate diffusion of the charge carrier across the junction due to thermal agitation. After diffusion, these charge carriers combine with their counterparts and neutralise each other.

So, charge carriers of n-side (i.e., electrons) move towards p-side and charge carrier of p-side (i.e., holes) move towards n-side.

Therefore correct direction of flow carriers is depicted in figure b.

41. **(b)** As p-n junction is formed, diffusion of charge carrier take place (i.e., majority flows from high concentration to low concentration). As carrier flows from high to low concentration, diffusion is performed near the junction and depletion layer is formed.

So, the depletion layer in the p-n junction region is caused by diffusion of charge carriers.

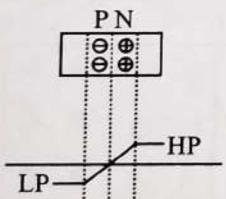
42. **(a)** In p-n junction, the positive terminal of external battery is to be connected to p semiconductor and negative terminal of battery to the n semiconductor for forward biasing.

43. **(d)** It affects all the p–n diode characteristics. Because, temperature increase, increases reverse saturation current, decreases barrier potential, and knee voltage also decreases as we increase temperature affects overall V-I characteristics of p–n junction.

44. **(d)** The potential difference required to move the charge carriers through the electric field is called the barrier potential. It is also called barrier hill. It depends on the type of semiconductor material, amount of doping, temperature and biasing of diode.

45. **(a)** As depletion layer is formed, the accumulation of electric electric charges of opposite polarities in the two regions of junction shows rise to an electric field between these regions.

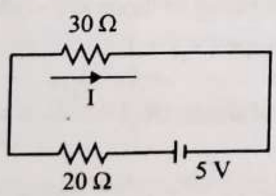
Therefore, in p-n junction high potential (HP) is at n-side and low potential (LP) is at p-side.



46. **(b)** In forward bias diode acts as short circuit and in reverse bias it acts as open circuit.

Here diode in series with 30 Ω is forward biased and in series with 20 Ω is reversed biased.

So, circuit becomes:

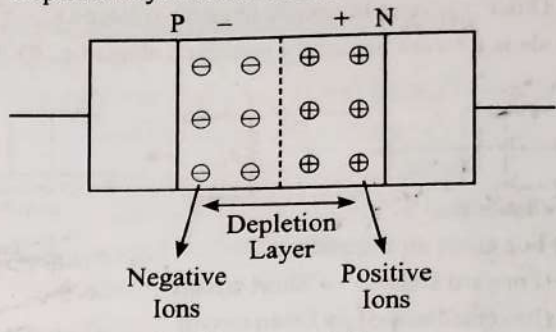


$$\therefore\ I = \frac{5\text{V}}{(30+20)\Omega}$$

$$I = \frac{5}{50}\text{A}$$

47. **(c)** For an unbiased p-n junction diode:

- Depletion layer has only static ions (Positive ions on n-side and negative ions on p-side).
- Depletion layer has no free electrons.
- Depletion layer has no free holes.



48. **(d)** In the case of half wave rectifier

Input frequency = Output frequency

⇒ Output frequency = 60 Hz

49. **(a)** Given, Peak Voltage (V_o) = 10V

For half wave diode rectifier (HWR):

$$V_{dc} = \frac{V_o}{\pi} = \frac{10}{\pi}\text{ volts}$$

50. **(d)** For full wave rectifier (FWR):

Ripple frequency = 2 × Source frequency

$\therefore\ f_r = 2f_s$

$f_r = 2 \times 50\text{ Hz} = 100\text{ Hz}$

51. **(b)** When a zener diode is connected in reverse bias, then there is leakage of current and it keeps continue until it reaches breakdown voltage. After this point, voltage across zener diode becomes constant which helps it to use as voltage regulator.

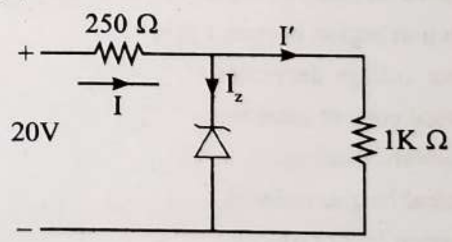
Potential barrier of Ge is 0.3 V.

Potential barrier of Si is 0.7 V.

Here, statement (A) is correct and (B) is incorrect.

52. **(d)** Voltage drop across 1Ω resistance is:

$V_z = 15$ V



$$I' = \frac{15\text{ V}}{1\text{ k}\Omega} = 15\text{ mA}$$

Voltage drop across 250 Ω = 20 – 15 = 5 V

So, Current through $250\,\Omega,\ I = \frac{5}{250} = 20\text{ mA}$

$\therefore$ Current through the Zener diode $I_z = I - I'$

$I_z = 20 - 15 = 5$ mA

53. **(c)** Zener diode is used for stabilisation as there is zener breakdown at the breakdown voltage and it is designed to operate under reverse bias.

54. (d) When a zener diode is connected in reverse bias, then there is a leakage current untill is reaches to breakdown. At the breakdown, current may increase but voltage across it will remain same.

Thus the output voltage across the zener diode is a fixed voltage. Hence, zener diode can be used as DC voltage stablizer.

55. (a) It is V – I characteristic for solar cell.

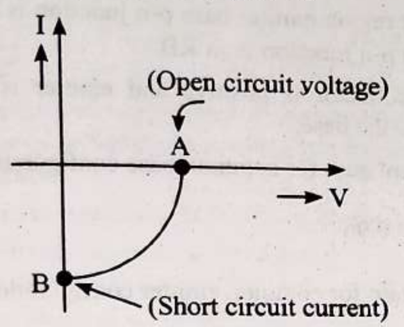
Where A: Open circuit voltage

B: Short circuit current

For Solar Cell:

Open circuit i.e., $I = 0 \Rightarrow$ Potential = V (Point A)

Short circuit i.e., $I = I \Rightarrow$ Potential = 0



56. (a) In photocell, photoelectromotive force is the force that stimulates the emission of an electric current when photovoltaic action creates a potential difference between two points and the electric current depends on the intensity of incident light.

Photo emf $\propto$ Current through cell

Current $\propto$ Intensity

So, photo emf $\propto$ Intensity of light falling on the cell.

57. (a) Child's equation

$i_p = kV^{3/2}$

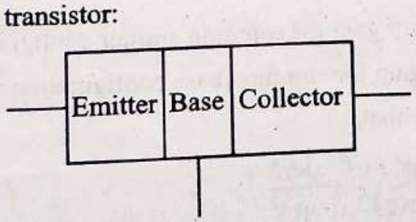
Where i_p : Plate current

V : Plate voltage

k : Constant

$\therefore$ Current $\propto V^{3/2}$

58. (c) For a transistor:



Sizer order: Collector > Emitter > Base

Doping order: Emitter > Collector > Base

Hence, base region must be vary thin and lightly doped.

59. (b) Across load resistance voltage drop = 0.8V

$\therefore$ Current through load resistor:

$$I_c = \frac{V_c}{R_c} = \frac{0.8}{800} = 0.001 = 1\text{mA}$$

60. (d) From KVL at input side.

$$V_i - R_B I_B - V_{BE} = 0$$

$$20 - (500)\, I_B - 0 = 0$$

$$\therefore\ I_B = \frac{20}{500} = 40\ \mu A$$

from KVL at output side

$$20 - R_C I_C - V_{CE} = 0$$

$$20 - (4)\, I_C - 0 = 0$$

$$\therefore\ I_C = \frac{20}{4} = 5\text{ mA}$$

$$\text{and}\quad \beta = \frac{I_C}{I_B} = \frac{5\text{ mA}}{40\ \mu A} = 125$$

61. (b) Given, Current gain $(\beta) = 100$

Base resistance $(R_B) = 2k\Omega$

Collector resistance $(R_C) = 3k\Omega$

So, voltage gain:

$$A_v = \beta \frac{R_C}{R_B}$$

$$A_v = 100 \times \frac{3}{2} = 150$$

$\therefore$ Power gain = βA_v = 100 × 150 =15000

62. (a) Given current amplification factor, $\propto = 0.36$

Load resistance $(R_L) = 800\Omega$

Input resistance $(R_i) = 192\Omega$

$\therefore$ Voltage gain for common base configuration

$$A_V = \propto \frac{R_L}{R_i} = 0.96 \times \frac{800}{192} = 4$$

$\therefore$ Power gain for common base configuration

$$P_V = A_V \propto = 4 \times 0.96 = 3.84$$

63. (a) Input signal $V_{in} = 2\cos\left(15t + \frac{\pi}{3}\right)$

Voltage gain $(A_V) = 150$

$$A_V = \frac{V_o}{V_{in}}$$

$$\text{so } V_o = A_V V_{in}$$

CE amplifier gives phase difference of π between input and output signals.

$$\text{So,}\quad V_o = 150 \times 2\cos\left(15t + \frac{\pi}{3} + \pi\right)$$

$$V_o = 300\cos\left(15t + \frac{4\pi}{3}\right)$$

64. (b) $\because$ Voltage gain $A_V = \frac{\Delta V_C}{\Delta V_B} = \frac{R_L \Delta I_C}{\Delta V_B} = g_m R_L$

$\therefore A_V \propto g_m$ (where g_m is trans conductance)

$$\therefore\ \frac{A_{V_1}}{A_{V_2}} = \frac{g_{m_1}}{g_{m_2}}$$

$$\frac{G}{A_{V_2}} = \frac{0.03}{0.02} \Rightarrow A_{V_2} = \frac{2}{3}G$$

65. (a) Voltage gain = $\frac{V_o}{V_{in}} = \beta\frac{R_o}{R_{in}}$

$$A_V = \frac{\Delta I_C}{\Delta I_B}\frac{R_o}{R_{in}} = \frac{2\times10^{-3}}{40\times10^{-6}}\times\frac{4\times10^{3}}{100}$$

$A_V = 2000$

66. (d) $\beta = 100$, $R_L = 2\ k\Omega$, $R_i = 1\ k\Omega$

Voltage magnification = $\frac{V_{out}}{V_{in}} = \frac{\beta R_L}{R_i}$

$$\Rightarrow \frac{2}{V_{in}} = \frac{(100)(2\times10^{3})}{1\times10^{3}} \Rightarrow V_{in} = 0.01\ V = 10\ mV$$

67. (a) Change in collector current;

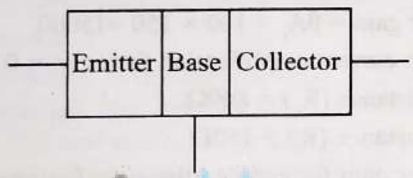
$\Delta I_C = 20 - 10 = 10mA = 10 \times 10^{-3}A$

Change in base current:

$\Delta I_B = 300 - 100 = 200\mu A = 200 \times 10^{-6}A$

$$\therefore \text{ Current gain } (\beta) = \frac{\Delta I_C}{\Delta I_B} = \frac{10\times10^{-3}}{200\times10^{-6}} = 50$$

68. (c) For a transistor:



Sizer order: Collector > Emitter > Base

Doping order: Emitter > Collector > Base

For transistor action the base region must be very thin and lightly doped and the emitter-base junction is forward biased and base-collector junction is reverse biased.

69. (d) Given, voltage gain (A_v) = 50

Input resistance (R_i) = 100Ω

Output resistance (R_{out}) = 200Ω

$$\text{Voltage gain} = \beta\left(\frac{R_{out}}{R_{in}}\right)$$

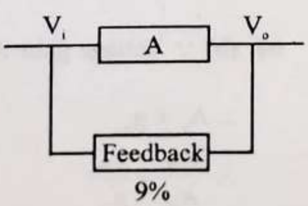
$$\therefore \text{ Current gain } (\beta) = \beta = \frac{50\times100}{200} = 25$$

Power gain = (current gain) × (Voltage gain) = βA_V

Power gain = 25 × 50 = 1250

70. (a) Voltage gain of an amplifier with 9% negative feedback is 10.

$$\Rightarrow V_o = A\left(V_i - \frac{9}{100}\text{ of } V_o\right)V$$



Since, $V_o = 10\ V_i$

$$\Rightarrow 10V_i = A\left(V_i - \frac{9}{100}\times10V_i\right)$$

$\Rightarrow A = 100$

So, the voltage gain without feedback = 100.

71. (c) ΔI_C = Change in collector current

= (10 – 5) mA

= 5 mA = 5×10^{-3} A

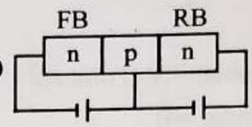
ΔI_b = Change in base current

= (150 – 100)μA

= 50μA = 50×10^{-6}A

$$\text{Current gain } (\beta) = \frac{\Delta I_C}{\Delta I_B} = \frac{5\times10^{-3}}{50\times10^{-6}} = 100$$

72. (b)



In active region emitter base p-n junction is in FB and base collector p-n junction is in RB.

Hence, collector is positive and emitter is negative with respect to the base.

73. (d) Current gain for common base configuration:

$$\alpha = \frac{I_C}{I_E} = 0.96$$

Current gain for common emitter configuration:

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\therefore \beta = \frac{0.96}{1 - 0.96}$$

$\Rightarrow \beta = 24$

74. (a) Current gain for common base configuration:

$$\alpha = \frac{I_C}{I_E} = 0.98$$

Current gain for common emitter configuration:

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\therefore \beta = \frac{0.98}{1-0.98}$$

$\Rightarrow \beta = 49$

75. (b) Current gain for common emitter configuration = β

Current gain for common base configuration = α

We know that,

$$\beta = \frac{\text{Collector Current}}{\text{Base Current}}$$

$$\beta = \frac{I_C}{I_B} = \frac{I_C}{I_E - I_C} \qquad \{\because I_E = I_B + I_C\}$$

$$\beta = \frac{I_C / I_E}{1 - \frac{I_C}{I_E}} = \frac{\alpha}{1-\alpha} \qquad \left\{\because \alpha = \frac{I_C}{I_E}\right\}$$

$$\beta = \frac{\alpha}{1-\alpha}$$

76. (a) Given, Input voltage (V) = 1V

Input resistance (R) = 1kΩ

Input current, $I_B = \frac{V}{R} = \frac{5}{10^3} = 5 \times 10^{-3}$

Current gain, $\beta = \frac{I_C}{I_B} = 50 = \frac{\text{output current } (I_C)}{\text{input current } (I_B)}$

$50 = \frac{I_C}{5 \times 10^{-3}} \quad \Rightarrow I_C = 250 \text{ mA}$

77. (a) $\because$ Current gain $(\beta) = \frac{\alpha}{1-\alpha}$

$\beta - \beta\alpha = \alpha$

$\therefore \beta = \alpha + \beta\alpha = \alpha(1+\beta)$

$\therefore \alpha = \frac{\beta}{+\beta}$

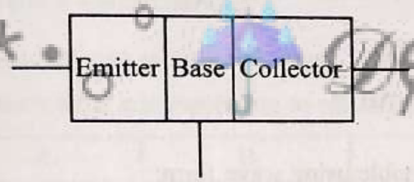
78. (c) The electrons are majority carriers in emitter, which move from emitter to collector via base while using n-p-n transistor as an amplifier.

In active mode, transistor works as an amplifier.

79. (b) When triode is used as an amplifier, the electron are emitted by cathode and collected by the anode.

Anode has positive potential while cathode has negative potential

80. (a) For a transistor:



Sizer order: Collector > Emitter > Base

Doping order: Emitter > Collector > Base

The function of emitter is to supply the majority carriers. So, it is heavily doped.

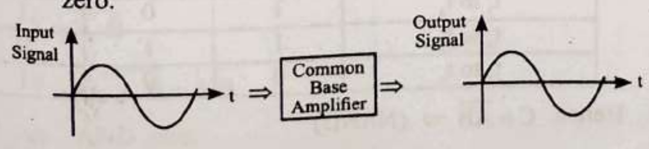
81. (b) The amplifying action of a triode is based on the fact that a small change in grid voltage causes a large change in plate current. The AC input signal which is to be amplified is superimposed on the grid potential.

82. (a) For transistor to be used as an amplifier the emitter base junction is forward bias while the collector base junction is reverse biased.

This is known as active mode of transistor.

Emitter Base Junction	Collector Base Junction	Mode	Application
Forward Bias	Reverse Bias	Active mode	As an Amplifier

83. (a) The phase difference between output voltage and input signal voltage in common base transistor or circuit is zero.



84. (b) Voltage gain of an amplifier

$A_V = \frac{\text{Output voltage}}{\text{Input voltage}} = -\frac{\mu R_L}{R_L + r_P}$

Where, R_L: Load resistance

r_p: Dynamic plate

μ: Amplification factor

The negative sign indicates that the output voltage differs in phase from the input voltage by 180° or π. This holds for a pure resistive load.

85. (d) For the given options:

- FET (Field Effect Transistor) is a voltage controlled device. It uses voltage supplied to its input terminal to control the current flow.
- Filter is used to pass the right signal and cancel out the undesired signals.
- Rectifier is used convert to convert AC into DC.
- An Oscillator operates on the oscillator principles. Radio waves are electromagnetic radiation. An oscillator can be used to produce electromagnetic waves.

86. (d) $C = \overline{A \cdot B}.\overline{\bar{A} \cdot B}$

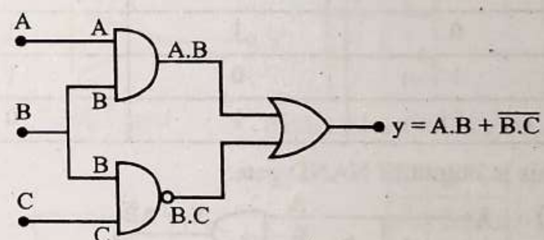
using De-Morgan Theorem

$C = \overline{A \cdot B + \bar{A} \cdot B} \Rightarrow C = \overline{B(A + \bar{A})} = \bar{B}$

Therefore

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

87. (c)

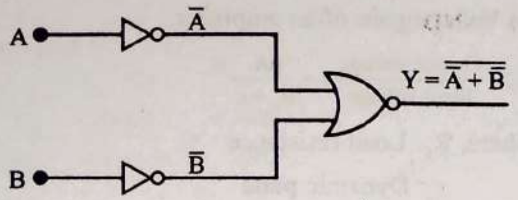


Truth Table (According to the given wave form)

Clk	A	B	C	$y_{O/P} = A.B + \overline{B.C}$
0 to t_1	0	0	1	1
t_1 to t_2	1	0	1	1
t_2 to t_3	0	1	0	1
t_3 to t_4	1	1	0	1
t_4 to t_5	0	0	1	1
t_5 to t_6	1	0	1	1
t_6 to t_7	0	0	1	1

Here, output function is, $y = A.B + \overline{B.C}$ and it gives high output in all clock pulse. So, output voltage is, $V_0 = 5V$.

88. (d)



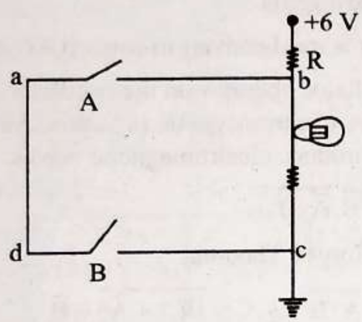
$Y=\overline{\bar{A}+\bar{B}}=\overline{\overline{A\cdot B}}=A\cdot B$

Therefore the given logic circuit is AND gate.

Option d is truth table for AND gate.

89. (b) NOR gate and NAND gate are considered as universal logic gates because we can design any gate using NOR and NAND.

90. (c) From the given logic circuit LED will glow, when voltage across LED is high.



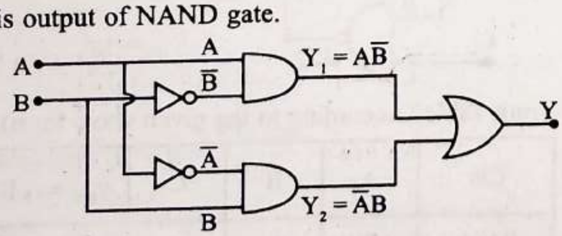
It will be high when circuit abcda remains open. Therefore truth table will be as follows.

Truth table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

This is output of NAND gate.

91. (b)

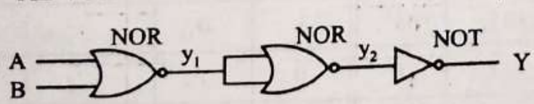


$Y_1 = A\bar{B}$ and $Y_2 = \bar{A}B$

Now, $Y = Y_1 + Y_2$

$Y = A\bar{B} + \bar{A}B$

92. (b)



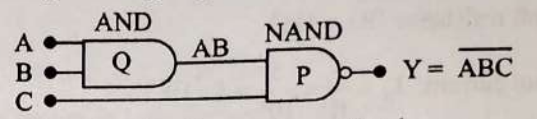
$y_1 = \overline{A+B}$

$y_2 = \overline{y_1 + y_1} = \overline{y_1} = \overline{\overline{A+B}} = A+B$

$y = \overline{y_2} = \overline{A+B}$

Hence, circuit is equivalent to NOR gate.

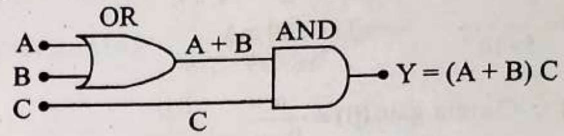
93. (a) Using working of gates:



When $A = B = C = 0 \Rightarrow Y = 1$

$A = B = C = 1 \Rightarrow Y = 0$

94. (d) Using working of gates:

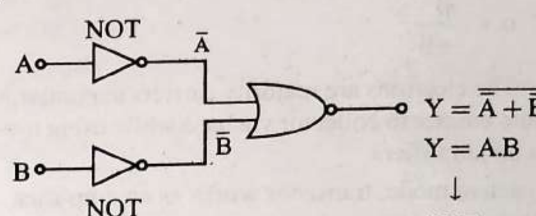


Option (d) satisfies the given condition, Y = 1

$Y = (1 + 0)1$

$Y = 1$

95. (b)



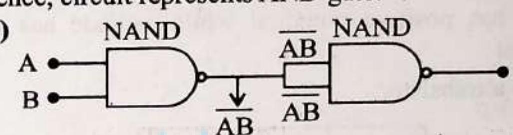
$Y = A.B$

↓

AND gate

Hence, circuit represents AND gate.

96. (d)



$X = \overline{\overline{AB}.\overline{AB}}$

$X = AB + AB$

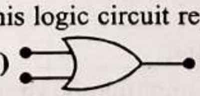
$X = AB$

97. (b) Truth table using wave form:

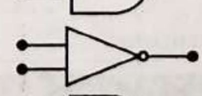
Clk	A	B	C
t_1 to t_2	1	0	1
t_2 to t_3	1	1	1
t_3 to t_4	0	1	1
t_4 to t_5	0	0	0
t_5 to t_6	1	0	1

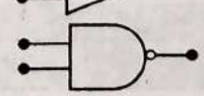
This logic circuit represents the truth table for OR gate.

98. (d)


⇒ OR gate


⇒ AND gate


⇒ NOT gate

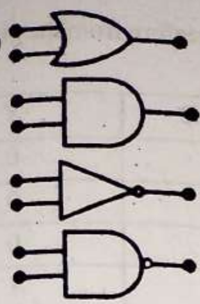

⇒ NAND gate

99. (d) Truth table using wave form:

Clk	A	B	C
t_1 to t_2	0	0	1
t_2 to t_3	0	1	1
t_3 to t_4	1	0	1
t_4 to t_5	1	1	0
t_5 to t_6	0	0	1

Hence, $C = \overline{AB} \Rightarrow$ (NAND)

100. (b)



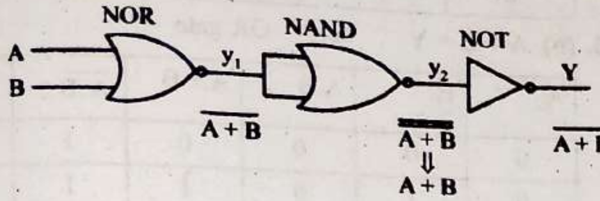
⇒ OR gate

⇒ AND gate

⇒ NOT gate

⇒ NAND gate

101. (d)



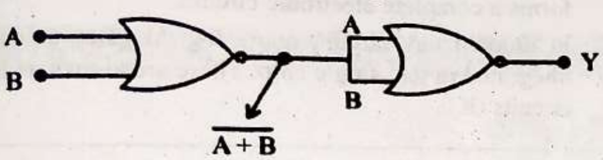
$y_1 = \overline{A+B}$

$y_2 = \overline{y_1 + y_1} = \overline{y_1} = \overline{\overline{A+B}} = A+B$

$y = \overline{y_2} = \overline{A+B}$

Hence circuit is equivalent to NOR gate.

102. (c)



$Y = \overline{\overline{A+B}} = A+B \Rightarrow$ OR Gate

Option c is the truth table for OR gate.

103. (a) The truth table corresponding to waveform is given by

A	B	C
1	1	1
0	1	0
1	0	0
0	0	0

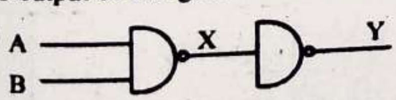
∴ The given logic circuit gate is AND gate.

104. (a) For OR gate truth table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

The output of OR gate is 1 if either or both inputs are 1.

105. (a)

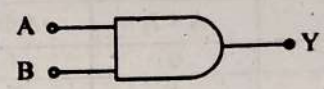


$X = \overline{A.B}$

$Y = \overline{X} = \overline{\overline{A.B}} = A.B$

⇒ AND gate.

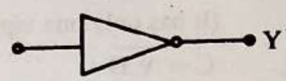
106. (a) NAND gate is a combination of AND and NOT gate.



AND gate

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

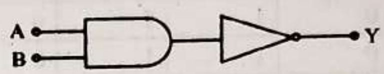
Truth table of AND gate



NOT gate

A	Y
0	1
1	0

Truth table of NOT gate



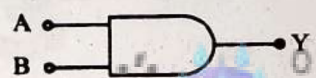
NAND gate

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

Truth table of NAND gate

Hence the given truth table is of a NAND gate.

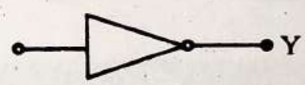
107. (c) NAND gate is a combination of AND and NOT gate.



AND gate

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

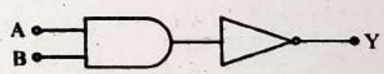
Truth table of AND gate



NOT gate

A	Y
0	1
1	0

Truth table of NOT gate



NAND gate

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

Truth table of NAND gate

Hence the given truth table is of a NAND gate.

108. (b)

A	B	A.B	A + B	$\overline{A.B}$	$\overline{A+B}$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

∴ C = A.B

Hence, the truth table is correct for AND gate

109. (b) $Y = \bar{A}B + A\bar{B} = A \oplus B$

Ex – OR Gate

A	B	A + B	A⊕B
0	0	0	0
1	0	1	1
0	1	1	1
1	1	1	0

110. (c) AND gate → $C = A \cdot B$

OR gate → $C = A + B$

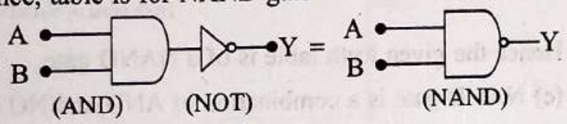
NOT gate → (It has only one input)

NAND → $C = \overline{A.B}$

A	B	A.B	A + B	$\overline{A.B}$	$\overline{A+B}$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

Hence, table is for NAND gate

111. (b)



112. (b) For NOR gate, $Y = \overline{A+B}$. Therefore from the given truth table

A	B	A + B	Y
1	1	1	0
1	0	1	0
0	1	1	0
0	0	0	1

113. (b) $A + B = Y$ ⇒ OR gate

A	B	A.B	A + B	$\overline{A.B}$	$\overline{A+B}$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

114. (c) Integrated circuit is a set of electronic circuits on a small piece of semiconductor material (normally used silicon). It forms a complete electronic circuits.

In modern day circuit, many logical gates or circuits are integrated in one single chip. These are known as Integrated circuits (IC).

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